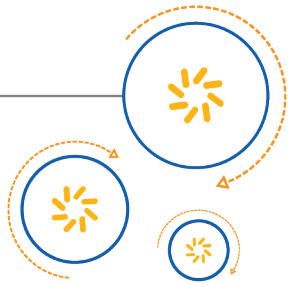
Qualcomm Technologies, Inc.



QMI UIM 1.54 for MPSS.JO.1.0

QMI User Identity Module Spec

80-NV300-12 D

September 18, 2015

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Revision History

Revision	Date	Description
A	Oct 2014	Initial release. Created from 80-NH952-12 AH. Updates for this revision include minor version 42 through minor version 44. Updated: • Mandatory Event Mask TLV (Sections 3.18.1 and 3.18.2) • Section 3.18.3 Updated: • Mandatory TLVs: – Event Mask (Section 3.18.1) – Capabilities Mask (Section 3.26.1) • Optional TLVs: – Event Registration Result (Section 3.18.2) – Service Number (Section 3.26.1) – Service Status (Section 3.26.2) • Sections 3.13.3, 3.30.3, and 3.48.3 Added new messages: • QMI_UIM_GET_SIM_PROFILE (Section 3.54) • QMI_UIM_SET_SIM_PROFILE (Section 3.55) • QMI_UIM_SUPPLY_VOLTAGE (Section 3.56) • QMI_UIM_CARD_ACTIVATION_STATUS_IND (Section 3.57) Added: • Section B.10 Reduced Card Indications • Section B.11 Vcc Power Down and Recovery • Section B.12 Multiple Profiles

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Revision	Date	Description
B	Feb 2015	Updates for this revision include minor version 45. Moved reference documents and acronyms to an appendix. Updated: • Mandatory TLV Event Mask (Section 3.27.1) • Optional TLV Event Registration Result (Section 3.27.2) • Sections 3.4.3, 3.22.3, and 3.35.3 Added new TLVs: • Encrypt Data Request (Section 3.4.1) • Read Result With Request Information (Sections 3.4.2 and 3.5.1) Added QMI_ERR_OP_PARTIAL_FAILURE error code to the following: • QMI_UIM_PERSONALIZATION_RESP (Section 3.59.2) • QMI_UIM_PERSONALIZATION_SECURE_RESP (Section 3.72.2) Added new messages: • QMI_UIM_DEPERSONALIZATION_SECURE (Section 3.71) • QMI_UIM_PERSONALIZATION_SECURE (Section 3.72) • QMI_UIM_EMERGENCY_ONLY (Section 3.73) • QMI_UIM_SIMLOCK_CONFIGURATION (Section 3.74) • QMI_UIM_SIMLOCK_CONFIGURATION_IND (Section 3.75) Added: • Section B.13 Enable Remote Simlock Configuration • Section B.14 Read or Write Simlock Data • Appendix D Encrypted Requests and Responses
C	Apr 2015	Updates for this revision include minor version 46 through minor version 49. Updated: • Optional TLV Configuration Mask (Section 3.40.1) Added new TLVs: • USB UICC supported flag (Section 3.40.2) • Support for SAP in client mode (Section 3.40.2) • Search Data (Section 3.40.2) • Response in Indication (Section 3.40.2) • Reactivate Personalization feature (Section 3.59.1) Added new messages: • QMI_UIM_GBA (Section 3.76) • QMI_UIM_GBA_IND (Section 3.77) • QMI_UIM_GET_GBA_IMPI (Section 3.78) • QMI_UIM_SEARCH_RECORD (Section 3.79) • QMI_UIM_SEARCH_RECORD_IND (Section 3.80)

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Revision	Date	Description
D	Sep 2015	Updates for this revision include minor version 50 through minor version 54. Updated: • Mandatory TLV: Event Mask (Section 3.27.1) • Optional TLVs: – Event Registration Result (Section 3.27.2) – Configuration Mask (Section 3.40.1) • Sections 3.40.3, and 3.76.3. Added new TLVs: • Termination of Application (Section 3.35.1) • Temporary Depersonalization Status (Section 3.40.2) • Personalization Reactivation Status (Section 3.40.2) • APN Name (Section 3.76.1) Added new messages: • QMI_UIM_REMOTE_UNLOCK (Section 3.81) • QMI_UIM_VERIFY_IMSI (Section 3.82) • QMI_UIM_TEMPORARY_UNLOCK_STATUS_IND (Section 3.83)

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MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

1 Introduction

1.1 Purpose

This specification documents Major Version 1 of the Qualcomm Messaging Interface (QMI) for the User Identity Module (QMI_UIM).

1.2 Scope

This document is intended for QMI clients to perform operations and to access the SIM card on Qualcomm MSM™ devices via the QMI_UIM.

This document provides the following details about QMI_UIM:

- Theory of operation – Chapter 2 provides the theory of operation of QMI_UIM. The chapter includes messaging conventions, assigned QMI service type, fundamental service concepts, and state variables related to the service.
- Message formats, syntax, and semantics – Chapter 3 provides the specific syntax and semantics of messages included in this version of the QMI_UIM specification.
- Additional information – Appendix A through Appendix D provide information on sessions, call flows, and refresh indication behaviors.

1.3 Conventions

Function declarations, function names, type declarations, and code samples appear in a different font, for example, `#include`.

1.4 Technical Assistance

For assistance or clarification on information in this document, submit a case to Qualcomm Technologies, Inc. (QTI) at https://createpoint.qti.qualcomm.com/.

If you do not have access to the CDMATech Support website, register for access or send email to support.cdmatech@qti.qualcomm.com.

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2 Theory of Operation

2.1 Generalized QMI Service Compliance

The QMI_UIM service complies with the generalized QMI service specification, including the rules for messages, indications and responses, byte ordering, arbitration, constants, result, and error code values described in 80-VB816-1. Extensions to the generalized QMI service theory of operation are noted in subsequent sections of this chapter.

2.2 UIM Service Type

The UIM is assigned QMI service type 0x0B.

2.3 Message Definition Template

2.3.1 Response Message Result TLV

This Type-Length-Value (TLV) is present in all Response messages defined in this document. It is not present in the Indication messages.

Name	Version introduced	Version last modified
Result Code	Corresponding response's *Version introduced*	Corresponding response's *Version last modified*

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x02			1	Result Code
Length	4			2	
Value	→	uint16	qmi_result	2	Result code • QMI_RESULT_SUCCESS • QMI_RESULT_FAILURE
		uint16	qmi_error	2	Error code – Possible error code values are described in the error codes section of each message definition

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2.4 QMI_UIM Fundamental Concepts

2.4.1 UIM Overview

The UIM module is used to access the card available on the device. The implementation supports SIM and USIM cards used for GSM/WCDMA devices, as well as RUIM and CSIM cards used for CDMA devices.

2.4.2 Multiple Applications on the Same Card

Multimode UICC cards allow true multimode operation with a single smart card. Previously, only one application was accessed. Up to two active provisioning applications are now possible, and application-specific requests are required to specify the application to which the request pertains.

2.4.3 Sessions

To access each application available on the card, sessions are used, as described in 80-VT475-1. The service automatically opens six sessions to the provisioning applications for both 1X and GSM/WCDMA and to the card in each slot for the files under MF. Depending on the capabilities of the target, some of these sessions might be invalid. The same notion of sessions is also used in the case of ICC cards, even though they do not support logical channels or applications.

In addition to providing access to files in the provisioning applications, the service also makes it possible to access other files, specifying the Application Identifier (AID) or the slot of the card; in these cases, an additional session is opened on-the-fly and maintained by the QMI service.

See Appendix A for more information, including a list of session types.

2.4.4 Commands Supported via QMI_UIM

The QMI_UIM module enables access to several features of the card.

- Access to files on the card
  - Read transparent and records
  - Write transparent and records
  - Get file attributes
- PIN operation
  - Enable and disable
  - Verify
  - Unblock
  - Change

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- Other tasks
  - Support for file refresh operations
  - Support for personalization
  - Power up/power down the card
  - Authentication
  - Select provisioning applications from the card
  - Get modem configuration
  - Send raw APDUs to the card
  - SIM Access Profile

A control point can register for notification of card events. The QMI layer maintains a cached status of the card and updates the status with notifications from the modem.

2.4.5 Refresh Procedure via QMI_UIM

The card can automatically start a refresh procedure. Refresh is one of the proactive commands; however, it must be handled by the modem in a special way, because it might impact many other modules (e.g., call manager, phonebook).

During the refresh procedure, the control point receives multiple events at different stages.

Details about the events and the call flows for the various refresh scenarios are described in 80-VM566-1.

2.5 Service State Variables

2.5.1 Shared State Variables

No QMI_UIM state variables are shared across control points.

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3 QMI_UIM Messages

Table 3-1 QMI_UIM messages

Command	ID	Description
QMI_UIM_RESET	0x0000	Resets the state of the issuing control point kept by the service.
QMI_UIM_GET_SUPPORTED_MSGS	0x001E	Queries the set of messages implemented by the currently running software.
QMI_UIM_GET_SUPPORTED_FIELDS	0x001F	Queries the fields supported for a single command as implemented by the currently running software.
QMI_UIM_READ_TRANSPARENT	0x0020	Provides read access to any transparent file in the card and provides access by the path.
QMI_UIM_READ_TRANSPARENT_IND	0x0020 indication	Indication with a read transparent command result.
QMI_UIM_READ_RECORD	0x0021	Provides read access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.
QMI_UIM_READ_RECORD_IND	0x0021 indication	Indication with a read record command result.
QMI_UIM_WRITE_TRANSPARENT	0x0022	Provides write access to any transparent file in the card and provides access by the path.
QMI_UIM_WRITE_TRANSPARENT_IND	0x0022 indication	Indication with a write transparent command result.
QMI_UIM_WRITE_RECORD	0x0023	Provides write access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.
QMI_UIM_WRITE_RECORD_IND	0x0023 indication	Indication with a write record command result.
QMI_UIM_GET_FILE_ATTRIBUTES	0x0024	Retrieves the file attributes for any EF or DF in the card and provides access by the path.
QMI_UIM_GET_FILE_ATTRIBUTES_IND	0x0024 indication	Indication with a get file attributes command result.
QMI_UIM_SET_PIN_PROTECTION	0x0025	Enables or disables the protection of the UIM contents by a specific PIN.
QMI_UIM_SET_PIN_PROTECTION_IND	0x0025 indication	Indication with a set PIN protection command result.

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Table 3-1 QMI_UIM messages (cont.)

Command	ID	Description
QMI_UIM_VERIFY_PIN	0x0026	Verifies the PIN before the card content is accessed.
QMI_UIM_VERIFY_PIN_IND	0x0026 indication	Indication with a verify PIN command result.
QMI_UIM_UNBLOCK_PIN	0x0027	Unblocks a blocked PIN using the PUK code.
QMI_UIM_UNBLOCK_PIN_IND	0x0027 indication	Indication with an unblock PIN command result.
QMI_UIM_CHANGE_PIN	0x0028	Changes the value of the specified PIN.
QMI_UIM_CHANGE_PIN_IND	0x0028 indication	Indication with a change PIN command result.
QMI_UIM_DEPERSONALIZATION	0x0029	Deactivates or unblocks the personalization on the phone.
QMI_UIM_REFRESH_REGISTER	0x002A	Registers for file change notifications triggered by the card.
QMI_UIM_REFRESH_OK	0x002B	Enables the client to indicate whether it is OK to start the Refresh procedure.
QMI_UIM_REFRESH_COMPLETE	0x002C	Invoked when the client has finished the Refresh procedure.
QMI_UIM_REFRESH_GET_LAST_EVENT	0x002D	Provides the ability to retrieve the last refresh event.
QMI_UIM_EVENT_REG	0x002E	Registers for event notifications from the card.
QMI_UIM_GET_CARD_STATUS	0x002F	Retrieves the current status of the card.
QMI_UIM_POWER_DOWN	0x0030	Powers down the card.
QMI_UIM_POWER_UP	0x0031	Powers up the card.
QMI_UIM_STATUS_CHANGE_IND	0x0032 indication	Indicates that the status of the card has changed.
QMI_UIM_REFRESH_IND	0x0033 indication	Received when a refresh command comes from the card.
QMI_UIM_AUTHENTICATE	0x0034	Executes the authentication algorithm on the card.
QMI_UIM_AUTHENTICATE_IND	0x0034 indication	Indication with an authenticate command result.
QMI_UIM_CLOSE_SESSION	0x0035	Forcibly closes a session to a nonprovisioning application.
QMI_UIM_GET_SERVICE_STATUS	0x0036	Retrieves the status of a service on the card.
QMI_UIM_SET_SERVICE_STATUS	0x0037	Changes the status of a service on the card.
QMI_UIM_CHANGE_PROVISIONING_ SESSION	0x0038	Changes the provisioning session.

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Table 3-1 QMI_UIM messages (cont.)

Command	ID	Description
QMI_UIM_GET_LABEL	0x0039	Gets the label of an application on a UICC card.
QMI_UIM_GET_CONFIGURATION	0x003A	Gets the modem configuration for the UIM module.
QMI_UIM_SEND_APDU	0x003B	Sends an APDU to the card.
QMI_UIM_SEND_APDU_IND	0x003B indication	Indication with a Send APDU command result.
QMI_UIM_SAP_CONNECTION	0x003C	Establishes and releases a connection to the UIM module as a SAP client.
QMI_UIM_SAP_REQUEST	0x003D	Executes a SAP request.
QMI_UIM_SAP_CONNECTION_IND	0x003E indication	Indicates a change in the status of the SAP connection.
QMI_UIM_LOGICAL_CHANNEL	0x003F	Opens and closes a logical channel to an application on a UICC card.
QMI_UIM_SUBSCRIPTION_OK	0x0040	Enables the client to indicate whether it is OK to proceed with publishing the subscription.
QMI_UIM_GET_ATR	0x0041	Retrieves the Answer to Reset of a specific card.
QMI_UIM_OPEN_LOGICAL_CHANNEL	0x0042	Opens a logical channel on a UICC card.
QMI_UIM_SESSION_CLOSED_IND	0x0043 indication	Indicates either a provisioning or a nonprovisioning session is closed by the modem.
QMI_UIM_REFRESH_REGISTER_ALL	0x0044	Registers for file change notifications triggered by the card for all files.
QMI_UIM_SET_FILE_STATUS	0x0045	Changes the activation status of the file on the card.
QMI_UIM_SWITCH_SLOT	0x0046	Switches the binding between a logical slot and a physical slot.
QMI_UIM_GET_SLOTS_STATUS	0x0047	Retrieves the current status of the physical and logical slots.
QMI_UIM_SLOT_STATUS_CHANGE_IND	0x0048 indication	Indicates that the status of the physical slot has changed.
QMI_UIM_READ_TRANSPARENT_LONG_IND	0x0049 indication	Indication with a read transparent command result.
QMI_UIM_SIM_BUSY_STATUS_IND	0x004A indication	Indicates the busy status of a SIM card.
QMI_UIM_GET_PLMN_NAME_TABLE_INFO	0x004B	Retrieves information about the SE.13 table used by the modem.
QMI_UIM_PERSONALIZATION	0x004C	Activates and sets the personalization data on the phone.

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Table 3-1 QMI_UIM messages (cont.)

Command	ID	Description
QMI_UIM_INCREASE	0x004D	Performs an increase operation for any file on the card and provides access by the path.
QMI_UIM_INCREASE_IND	0x004D indication	Indicates the clients with an increase confirmation.
QMI_UIM_RECOVERY	0x004E	Performs recovery on an identified slot.
QMI_UIM_RESELECT	0x004F	Performs a SELECT of an application on the specified logical channel.
QMI_UIM_RECOVERY_IND	0x0050 indication	Indicates that a SIM recovery completed successfully.
QMI_UIM_SEND_STATUS	0x0051	Sends a STATUS command.
QMI_UIM_GET_SIM_PROFILE	0x0052	Queries the profile information from the card.
QMI_UIM_SET_SIM_PROFILE	0x0053	Switches the profile on the SIM card.
QMI_UIM_SUPPLY_VOLTAGE	0x0054	Informs the modem to proceed with the supply voltage Vcc deactivation.
QMI_UIM_SUPPLY_VOLTAGE_IND	0x0054 indication	Indicates that the modem must deactivate or activate the supply voltage Vcc line of the UICC.
QMI_UIM_CARD_ACTIVATION_STATUS_IND	0x0055 indication	Indicates the card activation status.
QMI_UIM_DEPERSONALIZATION_SECURE	0x0056	Performs a temporary unlock or unlock without the control key.
QMI_UIM_PERSONALIZATION_SECURE	0x0057	Activates and locks the device to simlock codes.
QMI_UIM_EMERGENCY_ONLY	0x0058	Sets the phone in emergency-only mode.
QMI_UIM_SIMLOCK_CONFIGURATION	0x0059	Provides access to the simlock configuration data.
QMI_UIM_SIMLOCK_CONFIGURATION_IND	0x0059 indication	Indication with the simlock configuration data access request.
QMI_UIM_GBA	0x005A	Initializes the GBA and generates a key that is shared with a NAF.
QMI_UIM_GBA_IND	0x005A indication	Delayed response indication containing status and shared key related information.
QMI_UIM_GET_GBA_IMPI	0x005B	Retrieves the IMPI used for the GBA corresponding to the session.
QMI_UIM_SEARCH_RECORD	0x005C	Provides access to search for records in the card using a given pattern.
QMI_UIM_SEARCH_RECORD_IND	0x005C indication	Indication with a search record command result.

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Table 3-1 QMI_UIM messages (cont.)

Command	ID	Description
QMI_UIM_REMOTE_UNLOCK	0x005D	Performs operations related to remote unlock.
QMI_UIM_VERIFY_IMSI	0x005E	Verifies whether a subscription related to IMSI is validated by simlock.
QMI_UIM_TEMPORARY_UNLOCK_ STATUS_IND	0x005F indication	Indicates the temporary unlock status of the modem.

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3.1 QMI_UIM_RESET

Resets the state of the issuing control point kept by the service.

UIM message ID

0x0000

Version introduced

Major - 1, Minor - 0

3.1.1 Request - QMI_UIM_RESET_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

None

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3.1.2 Response - QMI_UIM_RESET_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

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Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.

3.1.3 Description of QMI_UIM_RESET REQ/RESP

This command resets the issuing control points state kept by the service. This is the equivalent to closing and reopening the service, only performed as one operation, so the client ID of the requesting control point does not change.

The state variables for the control point change to their default values before the response is issued. Among other possible operations, the following actions are performed:

- The list of events to which the client is registered is emptied. The client must register again using the QMI_UIM_EVENT_REG request to start receiving the events again.
- The list of files to be refreshed is deleted, and the client is deregistered from refresh indications. The client must register for refresh again to receive refresh indications.

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3.2 QMI_UIM_GET_SUPPORTED_MSGS

Queries the set of messages implemented by the currently running software.

UIM message ID

0x001E

Version introduced

Major - 1, Minor - 28

3.2.1 Request - QMI_UIM_GET_SUPPORTED_MSGS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

None

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3.2.2 Response - QMI_UIM_GET_SUPPORTED_MSGS_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Common version introduced	Common version last modified
Result Code	1.6	1.7

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Optional TLVs

Name	Common version introduced	Common version last modified
List of Supported Messages	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	List of Supported Messages
Length	Var			2	
Value	→	uint16	supported_msgs_len	2	Number of sets of the following elements: • supported_msgs
		uint8	supported_msgs	Var	This array of uint8 is a bitmask where each bit represents a message ID, i.e., starting with the LSB, bit 0 represents message ID 0, bit 1 represents message ID 1, etc. The bit is set to 1 if the message is supported; otherwise, it is set to zero. For example, if a service supports exactly four messages with IDs 0, 1, 30, and 31 (decimal), the array (in hexadecimal) is 4 bytes [03 00 00 c0].

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate a response.
QMI_ERR_INFO_UNAVAILABLE	Information is not available.

3.2.3 Description of QMI_UIM_GET_SUPPORTED_MSGS REQ/RESP

This command queries the set of messages implemented by the currently running software. This may be a subset of the messages defined in this revision of the service.

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3.3 QMI_UIM_GET_SUPPORTED_FIELDS

Queries the fields supported for a single command as implemented by the currently running software.

UIM message ID

0x001F

Version introduced

Major - 1, Minor - 28

3.3.1 Request - QMI_UIM_GET_SUPPORTED_FIELDS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Common version introduced	Common version last modified
Service Message ID	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Service Message ID
Length	2			2	
Value	→	uint16	msg_id	2	ID of the command for which the supported fields are requested.

Optional TLVs

None

3.3.2 Response - QMI_UIM_GET_SUPPORTED_FIELDS_RESP

Message type

Response

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Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Common version introduced	Common version last modified
Result Code	1.6	1.7

Optional TLVs

Name	Common version introduced	Common version last modified
List of Supported Request Fields	1.6	1.6
List of Supported Response Fields	1.6	1.6
List of Supported Indication Fields	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	List of Supported Request Fields
Length	Var			2	
Value	→	uint8	request_fields_len	1	Number of sets of the following elements: • request_fields
		uint8	request_fields	Var	This field describes which optional field IDs are supported in the QMI request. The array of uint8 is a bitmask where each bit represents a field (TLV) ID. Because fields 0 to 15 (decimal) are mandatory by definition, the first bit represents field ID 16. Starting with the LSB, bit 0 represents field ID 16, bit 1 represents field ID 17, etc. The bit is set to 1 if the field ID is supported; otherwise, it is set to zero. For example, if a service supports exactly four fields with IDs 16, 17, 30, and 31 (decimal), the array (in hexadecimal) is 2 bytes [03 c0].
Type	0x11			1	List of Supported Response Fields
Length	Var			2	
Value	→	uint8	response_fields_len	1	Number of sets of the following elements: • response_fields

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	response_fields	Var	This field describes which optional field IDs are supported in the QMI response. Its format is the same as request_fields.
Type	0x12			1	List of Supported Indication Fields
Length	Var			2	
Value	→	uint8	indication_fields_len	1	Number of sets of the following elements: • indication_fields
		uint8	indication_fields	Var	This field describes which optional field IDs are supported in the QMI indication. Its format is the same as request_fields.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate a response.
QMI_ERR_REQUESTED_NUM_ UNSUPPORTED	Requested message ID is not supported by the currently running software.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_INFO_UNAVAILABLE	Information is not available.

3.3.3 Description of QMI_UIM_GET_SUPPORTED_FIELDS REQ/RESP

This command queries the fields supported for a single command as implemented by the currently running software.

If the request, response, or indication is supported for the given message ID, the corresponding optional array is included in QMI_UIM_GET_SUPPORTED_FIELDS_RESP, even if the message does not contain any optional fields. This enables the client to distinguish this case from one where the service does not support the request, response, or indication.

Examples are:

- If the specified message ID is not supported by the service, the response has qmi_result = QMI_RESULT_FAILURE and qmi_error = QMI_ERR_REQUESTED_NUM_UNSUPPORTED.
- If the specified message ID is an empty message, the response has qmi_result = QMI_RESULT_SUCCESS and qmi_error = QMI_ERR_NONE. None of the optional arrays are included.
- If the specified message ID supports the request with 0 optional fields, the response with 3 optional fields (16, 17, and 18 decimal), and does not support an indication, the response has the following:
  - qmi_result = QMI_RESULT_SUCCESS
  - qmi_error = QMI_ERR_NONE
  - request_fields array is included with length zero

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– response_fields array is included with length 1 value [07]

– indication_fields array is not included

Trailing zero bytes are omitted from the response. For example, if the message defines 20 different fields but the response only contains 16 bits, the client is to assume the last four fields are not supported.

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3.4 QMI_UIM_READ_TRANSPARENT

Provides read access to any transparent file in the card and provides access by the path.

UIM message ID

0x0020

Version introduced

Major - 1, Minor - 0

3.4.1 Request - QMI_UIM_READ_TRANSPARENT_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
File ID	1.0	1.0
Read Transparent	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Read Transparent
Length	4			2	
Value	→	uint16	offset	2	Offset for the Read operation.
		uint16	length	2	Length of the content to be read. The value 0 is used to read the complete file.

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.12	1.12
Encrypt Data	1.18	1.18
Encrypt Request Data	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.
Type	0x11			1	Encrypt Data
Length	1			2	
Value	→	boolean	encryption	1	Indicates whether the data read from the card is to be encrypted.
Type	0x12			1	Encrypt Request Data
Length	1			2	
Value	→	boolean	encryption_ext	1	Indicates whether the session information, file ID, and data read from the card are to be encrypted.

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3.4.2 Response - QMI_UIM_READ_TRANSPARENT_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.29

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.0	1.0
Read Result	1.0	1.0
Response in Indication	1.12	1.12
Encrypted Data	1.18	1.18
Requested Length	1.20	1.20
Long Data Token	1.35	1.35
Read Result With Request Information	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Read Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Read content; sequence of bytes as read from the card.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.
Type	0x13			1	Encrypted Data
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	boolean	encryption	1	Indicates whether the data from the card passed in read_result is encrypted.
Type	0x14			1	Requested Length
Length	2			2	
Value	→	uint16	file_length	2	When this TLV is present, the requested length exceeds the maximum size supported by the QMI UIM. QMI_ERR_INSUFFICIENT_ RESOURCES is returned to the client and this value indicates the total length.
Type	0x15			1	Long Data Token
Length	4			2	
Value	→	uint32	long_data_token	4	When this TLV is present, the requested length exceeds the maximum size supported by QMI UIM. QMI_ERR_INSUFFICIENT_ RESOURCES is returned to the client, and the result is provided in subsequent indications. Long data token is used to reconstruct the entire read response, possibly spanning multiple indications.
Type	0x16			1	Read Result With Request Information
Length	Var			2	
Value	→	uint16	read_result_ext_len	2	Number of sets of the following elements: • read_result_ext
		opaque	read_result_ext	Var	Read request information and content. Includes session information, file ID as provided in the read request, and the sequence of bytes as read from the card. See Section D.5 for the result format.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when there is an attempt to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

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QMI_ERR_INSUFFICIENT_ RESOURCES	Response is longer than the maximum supported size.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

3.4.3 Description of QMI_UIM_READ_TRANSPARENT REQ/RESP

This function provides read access to any transparent file in the card and access by the path.

The client using this function is only able to access files that have a flat data structure. An error is reported for all other cases.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the read request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_READ_TRANSPARENT_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

A client can request to encrypt the data read from the card. This procedure guarantees protection against a man-in-the-middle attack between the service and the client. The security algorithm and keys used for the encryption are not part of this document. The service might not support the encryption and provide a non-encrypted response.

A client can request to encrypt the session and file information in addition to the data read from the card.

A client can request either to encrypt the data read from the card or to encrypt session and file information along with the data read from the card.

The service returns QMI_ERR_INSUFFICIENT_RESOURCES when the file being read is too large, in which case the file_length field contains the total length of the file. The client can use this information to request the data with multiple operations that read smaller blocks. Additionally, the Long Data Token TLV is sent. The service then sends one or more QMI_UIM_READ_TRANSPARENT_LONG_IND indications to the client. The client can use the long_data_token and file_length fields to reconstruct the final data for the read operation.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.5 QMI_UIM_READ_TRANSPARENT_IND

Indication with a read transparent command result.

UIM message ID

0x0020

Version introduced

Major - 1, Minor - 12

3.5.1 Indication - QMI_UIM_READ_TRANSPARENT_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12
Read Result	1.12	1.12
Encrypted Data	1.18	1.18
Requested Length	1.20	1.20
Long Data Token	1.35	1.35

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Name	Version introduced	Version last modified
Read Result With Request Information	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Read Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Read content; sequence of bytes as read from the card.
Type	0x12			1	Encrypted Data
Length	1			2	
Value	→	boolean	encryption	1	Indicates whether the data from the card passed in the read_result is encrypted.
Type	0x13			1	Requested Length
Length	2			2	
Value	→	uint16	file_length	2	When this TLV is present, the requested length exceeds the maximum size supported by the QMI UIM. QMI_ERR_INSUFFICIENT_ RESOURCES is returned to the client, and this value indicates the total length.
Type	0x14			1	Long Data Token
Length	4			2	
Value	→	uint32	long_data_token	4	When this TLV is present, the read result exceeds the maximum size supported by Read result TLV, and the result is provided in subsequent QMI_UIM_READ_TRANSPARENT_ LONG_IND messages. Long data token is used to reconstruct the entire read response, possibly spanning multiple indications.
Type	0x15			1	Read Result With Request Information
Length	Var			2	
Value	→	uint16	read_result_ext_len	2	Number of sets of the following elements: • read_result_ext
		opaque	read_result_ext	Var	Read request information and content. Includes session information, file ID as provided in the read request, and the sequence of bytes as read from the card. See Section D.5 for the result format.

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3.5.2 Description of QMI_UIM_READ_TRANSPARENT_IND

This indication is received when a read transparent request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_READ_TRANSPARENT_RESP.

Even when the client requests that the data be passed in a subsequent indication, the client must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

When QMI_ERR_INSUFFICIENT_RESOURCES is returned because the requested length is too long, the File Length TLV and the Long Data Token TLV are also sent. The service then sends one or more QMI_UIM_READ_TRANSPARENT_LONG_IND messages to the client. The client can use file_length and long_data_token fields to reconstruct the final data for the read operation.

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3.6 QMI_UIM_READ_RECORD

Provides read access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.

UIM message ID

0x0021

Version introduced

Major - 1, Minor - 0

3.6.1 Request - QMI_UIM_READ_RECORD_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
File ID	1.0	1.0
Read Record	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Read Record
Length	4			2	
Value	→	uint16	record	2	Record number (starting from 1).
		uint16	length	2	Length of the content to be read. The value 0 is used to read the complete record.

Optional TLVs

Name	Version introduced	Version last modified
Last Record	1.8	1.8
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Last Record
Length	2			2	
Value	→	uint16	last_record	2	Last record. This value is used to read multiple records at the same time.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

3.6.2 Response - QMI_UIM_READ_RECORD_RESP

Message type

Response

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Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.29

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.0	1.0
Read Result	1.0	1.0
Additional Read Result	1.8	1.8
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Read Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Read content. The content is the sequence of bytes as read from the card.
Type	0x12			1	Additional Read Result
Length	Var			2	
Value	→	uint16	additional_record_len	2	Number of sets of the following elements: • additional_record
		uint8	additional_record	Var	Read content of all the additional records. Each record has the same size as the first record.
Type	0x13			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

3.6.3 Description of QMI_UIM_READ_RECORD REQ/RESP

This function provides read access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the read request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_READ_RECORD_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

When multiple records are requested, the QMI UIM stops reading additional records from the file as soon as a Read operation returns an error. The QMI UIM returns only the records that are read successfully from the card without any indication of the specific error for the additional records.

When the records are large, the client must avoid reading too many records at the same time.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.7 QMI_UIM_READ_RECORD_IND

Indication with a read record command result.

UIM message ID

0x0021

Version introduced

Major - 1, Minor - 12

3.7.1 Indication - QMI_UIM_READ_RECORD_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12
Read Result	1.12	1.12
Additional Read Result	1.12	1.12

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Read Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Read content. The content is the sequence of bytes as read from the card.
Type	0x12			1	Additional Read Result
Length	Var			2	
Value	→	uint16	additional_record_len	2	Number of sets of the following elements: • additional_record
		uint8	additional_record	Var	Read content of all the additional records. Each record has the same size as the first record.

3.7.2 Description of QMI_UIM_READ_RECORD_IND

This indication is received when a read record request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_READ_RECORD_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.8 QMI_UIM_WRITE_TRANSPARENT

Provides write access to any transparent file in the card and provides access by the path.

UIM message ID

0x0022

Version introduced

Major - 1, Minor - 0

3.8.1 Request - QMI_UIM_WRITE_TRANSPARENT_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
File ID	1.0	1.0
Write Transparent	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Write Transparent
Length	Var			2	
Value	→	uint16	offset	2	Offset for the Write operation.
		uint16	data_len	2	Number of sets of the following elements: • data
		uint8	data	Var	Content to write.

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

3.8.2 Response - QMI_UIM_WRITE_TRANSPARENT_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.29

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Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.0	1.0
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

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3.8.3 Description of QMI_UIM_WRITE_TRANSPARENT REQ/RESP

This function provides write access to any transparent file in the card and provides access by the path.

The client using this function is able to access only files that have a flat data structure. An error is reported for all other cases.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the write request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_WRITE_TRANSPARENT_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.9 QMI_UIM_WRITE_TRANSPARENT_IND

Indication with a write transparent command result.

UIM message ID

0x0022

Version introduced

Major - 1, Minor - 12

3.9.1 Indication - QMI_UIM_WRITE_TRANSPARENT_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

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Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.9.2 Description of QMI_UIM_WRITE_TRANSPARENT_IND

This indication is received when a write transparent request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_WRITE_TRANSPARENT_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.10 QMI_UIM_WRITE_RECORD

Provides write access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.

UIM message ID

0x0023

Version introduced

Major - 1, Minor - 0

3.10.1 Request - QMI_UIM_WRITE_RECORD_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
File ID	1.0	1.0
Write Record	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Write Record
Length	Var			2	
Value	→	uint16	record	2	Record number (starting from 1). This field is ignored for cyclic files.
		uint16	data_len	2	Number of sets of the following elements: • data
		uint8	data	Var	Content to write.

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

3.10.2 Response - QMI_UIM_WRITE_RECORD_RESP

Message type

Response

Sender

Service

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Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.29

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.0	1.0
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

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3.10.3 Description of QMI_UIM_WRITE_RECORD REQ/RESP

This function provides write access to a specific record in a linear fixed or cyclic file in the card and provides access by the path.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the write request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_WRITE_RECORD_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.11 QMI_UIM_WRITE_RECORD_IND

Indication with a write record command result.

UIM message ID

0x0023

Version introduced

Major - 1, Minor - 12

3.11.1 Indication - QMI_UIM_WRITE_RECORD_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

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Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.11.2 Description of QMI_UIM_WRITE_RECORD_IND

This indication is received when a write record request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_WRITE_RECORD_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.12 QMI_UIM_GET_FILE_ATTRIBUTES

Retrieves the file attributes for any EF or DF in the card and provides access by the path.

UIM message ID

0x0024

Version introduced

Major - 1, Minor - 0

3.12.1 Request - QMI_UIM_GET_FILE_ATTRIBUTES_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
File ID	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.
		uint8	path_len	1	Number of sets of the following elements: • path

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.12	1.12
UICC Security Attributes	1.30	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.
Type	0x11			1	UICC Security Attributes
Length	1			2	
Value	→	boolean	uicc_security_attributes	1	Indicates whether the security attributes are required by the client. Valid values: • 0 – Skip security attributes of the file for the UICC card • 1 – Return security attributes of the file for the UICC card If the UICC Security Attributes TLV is missing, security attributes are returned. This TLV is ignored for ICC cards.

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3.12.2 Response - QMI_UIM_GET_FILE_ATTRIBUTES_RESP

Message type

Response

Sender

Service

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Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.29

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.0	1.0
File Attributes	1.0	1.0
Response in Indication	1.12	1.12
File Status	1.32	1.32

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	File Attributes
Length	Var			2	
Value	→	uint16	file_size	2	File size.
		uint16	file_id	2	File ID.
		enum8	file_type	1	File type. Valid values: • UIM_FILE_TYPE_TRANSPARENT (0x00) – Transparent • UIM_FILE_TYPE_CYCLIC (0x01) – Cyclic • UIM_FILE_TYPE_LINEAR_ FIXED (0x02) – Linear fixed • UIM_FILE_TYPE_DEDICATED_ FILE (0x03) – Dedicated file • UIM_FILE_TYPE_MASTER_ FILE (0x04) – Master file
		uint16	rec_size	2	Size of the records (only for cyclic and linear fixed files).
		uint16	rec_count	2	Total number of records (only for linear fixed files).

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	sec_read	1	Read security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_read_mask	2	Mask with read security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_write	1	Write security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_write_mask	2	Mask with write security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_increase	1	Increase security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	sec_increase_mask	2	Mask with increase security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_deactivate	1	Deactivate security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_deactivate_mask	2	Mask with deactivate security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_activate	1	Activate security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_activate_mask	2	Mask with Activate security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		uint16	raw_len	2	Number of sets of the following elements: • raw_value

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	raw_value	Var	Raw value of file attributes.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.
Type	0x13			1	File Status
Length	4			2	
Value	→	enum	file_status	4	Indicates the status of the file. Valid values: • UIM_FILE_STATUS_DEACTIVATED (0x00) – File is deactivated. • UIM_FILE_STATUS_ACTIVATED (0x01) – File is activated. Other values are reserved for future use and are to be ignored by the client.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

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3.12.3 Description of QMI_UIM_GET_FILE_ATTRIBUTES REQ/RESP

This function retrieves the file attributes for any EF or DF in the card and provides access by the path.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the select request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_GET_FILE_ATTRIBUTES_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

To retrieve the attributes for a DF or for the MF, the special value 0xFFFF is used for the file ID in the request.

On a UICC card, the security attributes are normally stored in the Access Rule Reference file (EF-ARR). If a client does not require the security attributes, the client can request to skip them. This improves the performance of the request because the device is not required to read the EF-ARR. In this case, the security attributes in the response indicate that all operations are always permitted.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.13 QMI_UIM_GET_FILE_ATTRIBUTES_IND

Indication with a get file attributes command result.

UIM message ID

0x0024

Version introduced

Major - 1, Minor - 12

3.13.1 Indication - QMI_UIM_GET_FILE_ATTRIBUTES_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12
File Attributes	1.12	1.12
File Status	1.32	1.32

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	File Attributes
Length	Var			2	
Value	→	uint16	file_size	2	File size.
		uint16	file_id	2	File ID.
		enum8	file_type	1	File type. Valid values: • UIM_FILE_TYPE_TRANSPARENT (0x00) – Transparent • UIM_FILE_TYPE_CYCLIC (0x01) – Cyclic • UIM_FILE_TYPE_LINEAR_ FIXED (0x02) – Linear fixed • UIM_FILE_TYPE_DEDICATED_ FILE (0x03) – Dedicated file • UIM_FILE_TYPE_MASTER_ FILE (0x04) – Master file
		uint16	rec_size	2	Size of the records (only for cyclic and linear fixed files).
		uint16	rec_count	2	Total number of records (only for linear fixed files).
		enum8	sec_read	1	Read security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_read_mask	2	Mask with read security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	sec_write	1	Write security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_write_mask	2	Mask with write security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_increase	1	Increase security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_increase_mask	2	Mask with increase security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_deactivate	1	Deactivate security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	sec_deactivate_mask	2	Mask with deactivate security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		enum8	sec_activate	1	Activate security attributes. Valid values: • UIM_SEC_CONDITION_ALWAYS (0x00) – Always • UIM_SEC_CONDITION_NEVER (0x01) – Never • UIM_SEC_CONDITION_AND_ COND (0x02) – AND condition • UIM_SEC_CONDITION_OR_ COND (0x03) – OR condition • UIM_SEC_CONDITION_SINGLE_ COND (0x04) – Single condition
		uint16	sec_activate_mask	2	Mask with Activate security attributes. This field is valid only when required by security attributes. Valid values: • Bit 0 – PIN1 • Bit 1 – PIN2 • Bit 2 – UPIN • Bit 3 – ADM
		uint16	raw_len	2	Number of sets of the following elements: • raw_value
		uint8	raw_value	Var	Raw value of file attributes.
Type	0x12			1	File Status
Length	4			2	
Value	→	enum	file_status	4	Indicates the status of the file. Valid values: • UIM_FILE_STATUS_DEACTIVATED (0x00) – File is deactivated. • UIM_FILE_STATUS_ACTIVATED (0x01) – File is activated. Other values are reserved for future use and are to be ignored by the client.

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3.13.2 Description of QMI_UIM_GET_FILE_ATTRIBUTES_IND

This indication is received when a get file attributes request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_GET_FILE_ATTRIBUTES_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.14 QMI_UIM_SET_PIN_PROTECTION

Enables or disables the protection of the UIM contents by a specific PIN.

UIM message ID

0x0025

Version introduced

Major - 1, Minor - 0

3.14.1 Request - QMI_UIM_SET_PIN_PROTECTION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Set PIN Protection	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Set PIN Protection
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	pin_id	1	Indicates the PIN ID to be enabled or disabled. Valid values: • UIM_PIN_ID_PIN_1 (0x01) – PIN1 (also called PIN) • UIM_PIN_ID_PIN_2 (0x02) – PIN2 • UIM_PIN_ID_UNIVERSAL (0x03) – Universal PIN • UIM_PIN_ID_HIDDEN_KEY (0x04) – Hidden key
		enum8	pin_operation	1	Indicates whether the PIN is enabled or disabled. Valid values: • UIM_PIN_OPERATION_DISABLE (0x00) – Disable the PIN • UIM_PIN_OPERATION_ENABLE (0x01) – Enable the PIN
		uint8	pin_len	1	Number of sets of the following elements: • pin_value
		char	pin_value	Var	PIN value. This value is a sequence of ASCII characters.

Optional TLVs

Name	Version introduced	Version last modified
Key Reference ID	1.10	1.10
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Key Reference ID
Length	1			2	
Value	→	enum8	pin_key_ref_id	1	Indicates the PIN key reference ID. Valid values are from 1 to 8, respectively, for application 1 to application 8, as specified in ETSI TS 102 221. This TLV is used only for PIN1 and PIN2 and is ignored in all other cases.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.14.2 Response - QMI_UIM_SET_PIN_PROTECTION_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.0	1.0
Encrypted PIN1	1.7	1.7
Response in Indication	1.12	1.12
Card Result	1.29	1.29

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the enable/disable operation has failed.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 is enabled successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x13			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	PIN in the request is incorrect.
QMI_ERR_PIN_BLOCKED	PIN is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	PIN is permanently blocked. The SIM is unusable.
QMI_ERR_SIM_NOT_INITIALIZED	PIN is not yet initialized because the SIM initialization has not finished. Try the PIN operation later.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device nor by the SIM card.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_INVALID_PINID	PIN in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.14.3 Description of QMI_UIM_SET_PIN_PROTECTION REQ/RESP

This message enables or disables the protection of UIM contents by a specific PIN.

The same PIN can be used by multiple sessions (i.e., the PIN is shared between GSM and RUIM in an ICC card). The PIN is automatically set for all the sessions when the command is executed.

The hidden key PIN type does not refer to a real PIN of the card, but is used to manipulate the EF-HIDDENKEY file present on the card, as specified in 3GPP TS 31.102. When the PIN is enabled, the data in the file has a valid value and the contacts that are marked as hidden are not displayed. When the PIN is disabled, the data in the file is invalid and all contacts are displayed.

Note: The hidden key PIN type is applicable only to provisioning sessions.

The operation to enable the hidden key PIN also sets its value and can be used only when the hidden key PIN is disabled.

The optional TLV with the number of retries is not sent for a hidden key PIN type.

The optional TLV with an encrypted PIN1 is only sent after PIN1 is successfully enabled and the modem supports the feature.

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The optional TLV with the PIN key reference ID is used by the client to force a specific key reference to be used in the PIN operation. This TLV is not used in normal conditions, so the appropriate key reference can be selected by the modem.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_SET_PIN_PROTECTION_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.15 QMI_UIM_SET_PIN_PROTECTION_IND

Indication with a set PIN protection command result.

UIM message ID

0x0025

Version introduced

Major - 1, Minor - 12

3.15.1 Indication - QMI_UIM_SET_PIN_PROTECTION_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.12	1.12
Encrypted PIN1	1.12	1.12
Card Result	1.29	1.29

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the PIN operation has failed.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 operation is completed successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.15.2 Description of QMI_UIM_SET_PIN_PROTECTION_IND

This indication is received when a set PIN protection request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_SET_PIN_PROTECTION_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.16 QMI_UIM_VERIFY_PIN

Verifies the PIN before the card content is accessed.

UIM message ID

0x0026

Version introduced

Major - 1, Minor - 0

3.16.1 Request - QMI_UIM_VERIFY_PIN_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Verify PIN	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Verify PIN
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	pin_id	1	Indicates the PIN ID to be verified. Valid values: • UIM_PIN_ID_PIN_1 (0x01) – PIN1 (also called PIN) • UIM_PIN_ID_PIN_2 (0x02) – PIN2 • UIM_PIN_ID_UNIVERSAL (0x03) – Universal PIN • UIM_PIN_ID_HIDDEN_KEY (0x04) – Hidden key
		uint8	pin_len	1	Number of sets of the following elements: • pin_value
		char	pin_value	Var	PIN value. This value is a sequence of ASCII characters.

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Value of PIN1	1.7	1.7
Key Reference ID	1.10	1.10
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Value of PIN1 (This field can be used only when the PIN type is PIN1.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x11			1	Key Reference ID
Length	1			2	
Value	→	enum8	pin_key_ref_id	1	Indicates the PIN key reference ID. Valid values are from 1 to 8, respectively, for application 1 to application 8, as specified in ETSI TS 102 221. This TLV is used only for PIN1 and PIN2 and is ignored in all other cases.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.16.2 Response - QMI_UIM_VERIFY_PIN_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.0	1.0
Encrypted PIN1	1.7	1.7
Response in Indication	1.12	1.12
Card Result	1.29	1.29

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the verify operation fails.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 is verified successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x13			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	PIN in the request is incorrect.
QMI_ERR_PIN_BLOCKED	PIN is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	PIN is permanently blocked. The SIM is unusable.
QMI_ERR_SIM_NOT_INITIALIZED	PIN is not yet initialized because the SIM initialization has not finished. Try the PIN operation later.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device nor by the SIM card.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_INVALID_PINID	PIN in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.16.3 Description of QMI_UIM_VERIFY_PIN REQ/RESP

This function verifies the PIN before the card content is accessed.

The same PIN can be used by multiple sessions (i.e., the PIN is shared between GSM and RUIM in an ICC card). The PIN is automatically verified for all the sessions when the command is executed.

The hidden key PIN type does not refer to a real PIN of the card, but is used to verify the value entered by the user against the content of the file EF-HIDDENKEY present on the card, as specified in 3GPP TS 31.102. If the hidden key is enabled (file EF-HIDDENKEY has a valid value), the hidden contacts are shown only after the value of this PIN has been successfully verified.

Note: The hidden key PIN type is applicable only to provisioning sessions.

When the optional TLV with an encrypted PIN1 value is present in the request, the modem ignores the clear PIN1 value present in the request and uses the encrypted value only.

The optional TLV with number of retries is not sent for a hidden key PIN type.

The optional TLV with an encrypted PIN1 is only sent after the successful verification of PIN1, when the modem supports the feature and the verification was done using a clear PIN1.

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The optional TLV with the PIN key reference ID is used by the client to force a specific key reference to be used in the PIN operation. This TLV is not used in normal conditions, so the appropriate key reference can be selected by the modem.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_VERIFY_PIN_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.17 QMI_UIM_VERIFY_PIN_IND

Indication with a verify PIN command result.

UIM message ID

0x0026

Version introduced

Major - 1, Minor - 12

3.17.1 Indication - QMI_UIM_VERIFY_PIN_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.12	1.12
Encrypted PIN1	1.12	1.12
Card Result	1.29	1.29

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the PIN operation has failed.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 operation is completed successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.17.2 Description of QMI_UIM_VERIFY_PIN_IND

This indication is received when a verify PIN request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_VERIFY_PIN_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.18 QMI_UIM_UNBLOCK_PIN

Unblocks a blocked PIN using the PUK code.

UIM message ID

0x0027

Version introduced

Major - 1, Minor - 0

3.18.1 Request - QMI_UIM_UNBLOCK_PIN_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Unblock PIN	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Unblock PIN
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	pin_id	1	Indicates the PIN ID to be unblocked. Valid values: • UIM_PIN_ID_PIN_1 (0x01) – PIN1 (also called PIN) • UIM_PIN_ID_PIN_2 (0x02) – PIN2 • UIM_PIN_ID_UNIVERSAL (0x03) – Universal PIN
		uint8	puk_len	1	Number of sets of the following elements: • puk_value
		char	puk_value	Var	PIN unlock key value. This value is a sequence of ASCII characters.
		uint8	new_pin_len	1	Number of sets of the following elements: • new_pin_value
		char	new_pin_value	Var	New PIN value. This value is a sequence of ASCII characters.

Optional TLVs

Name	Version introduced	Version last modified
Key Reference ID	1.10	1.10
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Key Reference ID
Length	1			2	
Value	→	enum8	pin_key_ref_id	1	Indicates the PIN key reference ID. Valid values are from 1 to 8, respectively, for application 1 to application 8, as specified in ETSI TS 102 221. This TLV is used only for PIN1 and PIN2 and is ignored in all other cases.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.18.2 Response - QMI_UIM_UNBLOCK_PIN_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.0	1.0
Encrypted PIN1	1.7	1.7
Response in Indication	1.12	1.12
Card Result	1.29	1.29

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the unblock operation fails.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 is unblocked successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x13			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	PIN in the request is incorrect.
QMI_ERR_PIN_BLOCKED	PIN is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	PIN is permanently blocked. The SIM is unusable.
QMI_ERR_SIM_NOT_INITIALIZED	PIN is not yet initialized because the SIM initialization has not finished. Try the PIN operation later.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device nor by the SIM card.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_INVALID_PINID	PIN in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.18.3 Description of QMI_UIM_UNBLOCK_PIN REQ/RESP

This function unblocks a blocked PIN using the PUK code. The client must pass PUK1 to unblock PIN1 or PUK2 to unblock PIN2.

The same PIN can be used by multiple sessions (i.e., the PIN is shared between GSM and RUIM in an ICC card). The PIN is automatically unblocked for all the sessions when the command is executed.

When the card is blocked because the PUK was entered incorrectly too many times, the card cannot be restored and must be replaced by the operator.

The optional TLV with an encrypted PIN1 is only sent when PIN1 is successfully unblocked and the modem supports the feature.

The optional TLV with the PIN key reference ID is used by the client to force a specific key reference to be used in the PIN operation. This TLV is not used in normal conditions, so the appropriate key reference can be selected by the modem.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_UNBLOCK_PIN_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

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A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.19 QMI_UIM_UNBLOCK_PIN_IND

Indication with an unblock PIN command result.

UIM message ID

0x0027

Version introduced

Major - 1, Minor - 12

3.19.1 Indication - QMI_UIM_UNBLOCK_PIN_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.12	1.12
Encrypted PIN1	1.12	1.12
Card Result	1.29	1.29

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the PIN operation has failed.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 operation is completed successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.19.2 Description of QMI_UIM_UNBLOCK_PIN_IND

This indication is received when an unblock PIN request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_UNBLOCK_PIN_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.20 QMI_UIM_CHANGE_PIN

Changes the value of the specified PIN.

UIM message ID

0x0028

Version introduced

Major - 1, Minor - 0

3.20.1 Request - QMI_UIM_CHANGE_PIN_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Change PIN	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Change PIN
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	pin_id	1	Indicates the PIN ID to be changed. Valid values: • UIM_PIN_ID_PIN_1 (0x01) – PIN1 (also called PIN) • UIM_PIN_ID_PIN_2 (0x02) – PIN2 • UIM_PIN_ID_UNIVERSAL (0x03) – Universal PIN • UIM_PIN_ID_HIDDEN_KEY (0x04) – Hidden key
		uint8	old_pin_len	1	Number of sets of the following elements: • old_pin_value
		char	old_pin_value	Var	Old PIN value. The value is a sequence of ASCII characters.
		uint8	new_pin_len	1	Number of sets of the following elements: • new_pin_value
		char	new_pin_value	Var	New PIN value. This value is a sequence of ASCII characters.

Optional TLVs

Name	Version introduced	Version last modified
Key Reference ID	1.10	1.10
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Key Reference ID
Length	1			2	
Value	→	enum8	pin_key_ref_id	1	Indicates the PIN key reference ID. Valid values are from 1 to 8, respectively, for application 1 to application 8, as specified in ETSI TS 102 221. This TLV is used only for PIN1 and PIN2 and ignored in all other cases.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.20.2 Response - QMI_UIM_CHANGE_PIN_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.0	1.0
Encrypted PIN1	1.7	1.7
Response in Indication	1.12	1.12
Card Result	1.29	1.29

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the change PIN operation fails.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 is changed successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x13			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	PIN in the request is incorrect.
QMI_ERR_PIN_BLOCKED	PIN is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	PIN is permanently blocked. The SIM is unusable.
QMI_ERR_SIM_NOT_INITIALIZED	PIN is not yet initialized because the SIM initialization has not finished. Try the PIN operation later.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device nor by the SIM card.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_INVALID_PINID	PIN in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.20.3 Description of QMI_UIM_CHANGE_PIN REQ/RESP

This function changes the value of the specified PIN. The application must pass both the new and the old values of the PIN to complete the operation.

The same PIN can be used by multiple sessions (i.e., the PIN is shared between GSM and RUIM in an ICC card). The PIN is automatically changed for all the sessions when the command is executed.

The hidden key PIN type does not refer to a real PIN of the card, but is used to change the value stored in the file EF-HIDDENKEY present on the card, as specified in 3GPP TS 31.102. This operation can be performed only when the hidden key PIN is enabled and was previously verified successfully. If it is disabled, the new value can be set using the QMI_UIM_SET_PIN_PROTECTION command.

Note: The hidden key PIN type is applicable only to provisioning sessions.

The optional TLV with number of retries is not sent for hidden key PIN types.

The optional TLV with an encrypted PIN1 is only sent when the PIN1 value has been changed successfully and the modem supports the feature.

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The optional TLV with the PIN key reference ID is used by the client to force a specific key reference to be used in the PIN operation. This TLV is not used in normal conditions, so the appropriate key reference can be selected by the modem.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_CHANGE_PIN_IND indication. In this case, the immediate response indicates only when the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.21 QMI_UIM_CHANGE_PIN_IND

Indication with a change PIN command result.

UIM message ID

0x0028

Version introduced

Major - 1, Minor - 12

3.21.1 Indication - QMI_UIM_CHANGE_PIN_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.12	1.12
Encrypted PIN1	1.12	1.12
Card Result	1.29	1.29

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining (This value is returned only when the PIN operation has failed.)
Length	2			2	
Value	→	uint8	verify_left	1	Number of remaining attempts to verify the PIN.
		uint8	unblock_left	1	Number of remaining attempts to unblock the PIN.
Type	0x11			1	Encrypted PIN1 (This value is returned only when PIN1 operation is completed successfully and the feature is supported.)
Length	Var			2	
Value	→	uint8	enc_pin1_len	1	Number of sets of the following elements: • enc_pin1_value
		uint8	enc_pin1_value	Var	Encrypted PIN1 value.
Type	0x12			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

3.21.2 Description of QMI_UIM_CHANGE_PIN_IND

This indication is received when a change PIN request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_CHANGE_PIN_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.22 QMI_UIM_DEPERSONALIZATION

Deactivates or unblocks the personalization on the phone.

UIM message ID

0x0029

Version introduced

Major - 1, Minor - 0

3.22.1 Request - QMI_UIM_DEPERSONALIZATION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Depersonalization	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Depersonalization
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	feature	1	Indicates the personalization feature to deactivate or unblock. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		enum8	operation	1	Indicates the operation to perform. Valid values: • UIM_PERSO_OPERATION_ DEACTIVATE (0x00) – Deactivate personalization • UIM_PERSO_OPERATION_ UNBLOCK (0x01) – Unblock personalization
		uint8	ck_len	1	Number of sets of the following elements: • ck_value
		char	ck_value	Var	Control key value. This value is a sequence of ASCII characters.

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Optional TLVs

Name	Version introduced	Version last modified
Slot	1.40	1.40

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

3.22.2 Response - QMI_UIM_DEPERSONALIZATION_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining This value is returned only when deactivation or unblock fails.
Length	2			2	
Value	→	uint8	verify_left	1	Number of the remaining attempts to verify the personalization.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	unblock_left	1	Number of the remaining attempts to unblock the personalization.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	Control key in the request is incorrect.
QMI_ERR_PIN_BLOCKED	Simlock feature is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	Simlock feature is permanently blocked. The UE is unusable.
QMI_ERR_SIM_NOT_INITIALIZED	PIN is not yet initialized because the SIM initialization has not finished. Try the PIN operation later.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.22.3 Description of QMI_UIM_DEPERSONALIZATION REQ/RESP

This function deactivates or unblocks the personalization on the phone, as described in 3GPP TS 22.022. Each feature can be deactivated/unblocked independently of the other features.

Slot 1 is the default slot. The appropriate slot information is necessary in case personalization deactivation is requested on the other slots.

The modem might be configured to have an unlimited number of retries. In this case, the value 0xFF is passed in the response to indicate the configuration.

If the modem does not support the Unblock operation, the QMI_ERR_OP_DEVICE_UNSUPPORTED error is returned.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.23 QMI_UIM_REFRESH_REGISTER

Registers for file change notifications triggered by the card.

UIM message ID

0x002A

Version introduced

Major - 1, Minor - 0

3.23.1 Request - QMI_UIM_REFRESH_REGISTER_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Register Refresh	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Register Refresh
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	boolean	register_flag	1	Flag that indicates whether to register or deregister for refresh indications. Valid values: • 0 – Deregister • 1 – Register
		boolean	vote_for_init	1	Flag that indicates whether to vote for the init when there is a refresh. Valid values: • 0 – Client does not vote for initialization • 1 – Client votes for initialization
		uint16	num_files	2	Number of sets of the following elements: • file_id • path_len • path
		uint16	file_id	2	File ID.
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	Path value. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).

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Optional TLVs

None

3.23.2 Response - QMI_UIM_REFRESH_REGISTER_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

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Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.23.3 Description of QMI_UIM_REFRESH_REGISTER REQ/RESP

This function registers for file change notifications triggered by the card.

The client can specify a list of files. The client is notified only when one of the files is modified by the Refresh procedure.

This function can be invoked multiple times for each session type. If the function is invoked twice with the same session type, the new values overwrite the previous values. The client can also use this function to stop receiving indications of the refresh.

The client can specify whether it wants to vote for the refresh. If the client indicates that it wants to vote, it receives a QMI_UIM_REFRESH_IND indication before the FCN Refresh procedure is started and is requested to indicate whether it is OK to proceed with the refresh. For more information on the Refresh procedure, refer to 80-VM566-1.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.24 QMI_UIM_REFRESH_OK

Enables the client to indicate whether it is OK to start the Refresh procedure.

UIM message ID

0x002B

Version introduced

Major - 1, Minor - 0

3.24.1 Request - QMI_UIM_REFRESH_OK_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
OK for Refresh	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	OK for Refresh
Length	1			2	
Value	→	boolean	ok_to_refresh	1	Indicates whether a refresh is OK. Valid values: • 0 – Not OK to refresh • 1 – OK to refresh

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Optional TLVs

None

3.24.2 Response - QMI_UIM_REFRESH_OK_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

3.24.3 Description of QMI_UIM_REFRESH_OK REQ/RESP

This function enables the client to indicate whether it is OK to start the Refresh procedure. This command is used only after a refresh event is received, which indicates the need to vote.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.25 QMI_UIM_REFRESH_COMPLETE

Invoked when the client has finished the Refresh procedure.

UIM message ID

0x002C

Version introduced

Major - 1, Minor - 0

3.25.1 Request - QMI_UIM_REFRESH_COMPLETE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30
Refresh Complete	1.0	1.0

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Refresh Complete
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	boolean	refresh_success	1	Indicates whether the refresh was successful. Valid values: • 0 – Refresh was not completed successfully • 1 – Refresh was completed successfully

Optional TLVs

None

3.25.2 Response - QMI_UIM_REFRESH_COMPLETE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

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3.25.3 Description of QMI_UIM_REFRESH_COMPLETE REQ/RESP

This function is invoked when the client has finished the Refresh procedure (has reread all the cached files) and communicates this to the modem. This function enables the terminal response to be sent to the card.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.26 QMI_UIM_REFRESH_GET_LAST_EVENT

Provides the ability to retrieve the last refresh event.

UIM message ID

0x002D

Version introduced

Major - 1, Minor - 0

3.26.1 Request - QMI_UIM_REFRESH_GET_LAST_EVENT_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.0	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.

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Optional TLVs

None

3.26.2 Response - QMI_UIM_REFRESH_GET_LAST_EVENT_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Refresh Event	1.0	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Refresh Event
Length	Var			2	
Value	→	enum8	stage	1	Indicates the stage of the Refresh procedure. Valid values: • UIM_REFRESH_STAGE_WAIT_FOR_OK (0x00) – Waiting for OK to refresh • UIM_REFRESH_STAGE_START (0x01) – Refresh started • UIM_REFRESH_STAGE_END_WITH_SUCCESS (0x02) – Refresh ended successfully • UIM_REFRESH_STAGE_END_WITH_FAILURE (0x03) – Refresh failed

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	mode	1	Indicates the Refresh mode. Valid values: • UIM_REFRESH_MODE_RESET (0x00) – Reset • UIM_REFRESH_MODE_INIT (0x01) – Init • UIM_REFRESH_MODE_INIT_ FCN (0x02) – Init and FCN • UIM_REFRESH_MODE_FCN (0x03) – FCN • UIM_REFRESH_MODE_INIT_ FULL_FCN (0x04) – Init and Full FCN • UIM_REFRESH_MODE_APP_ RESET (0x05) – Application reset • UIM_REFRESH_MODE_3G_RESET (0x06) – 3G session reset
		enum8	session_type	1	Indicates the session type. Valid values: 0 through 7 and 10 through 21. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value. This value is required only for a nonprovisioning session type and is ignored in all other cases.
		uint16	num_of_files	2	Number of sets of the following elements: • file_id • path_len • path
		uint16	file_id	2	File ID.
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	Path value. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

3.26.3 Description of QMI_UIM_REFRESH_GET_LAST_EVENT REQ/RESP

This function provides the ability to retrieve the last refresh event. The event information is usually passed in as an indication from the QMI to the application and is saved by the application at that time. If the event information is not saved, the client can retrieve the last refresh event.

Details regarding the Refresh procedure (i.e., the stages and actions that an application must complete) are described in 80-VM566-1.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.27 QMI_UIM_EVENT_REG

Registers for event notifications from the card.

UIM message ID

0x002E

Version introduced

Major - 1, Minor - 0

3.27.1 Request - QMI_UIM_EVENT_REG_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Event Mask	1.0	1.54

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Event Mask
Length	4			2	
Value	→	uint32	event_mask	4	Bitmask of the events to be received by the client. Additional bits are reserved for future use and must be set to zero by the caller. • Bit 0 – Card status • Bit 1 – SAP connection • Bit 2 – Extended card status • Bit 3 – Close of provisioning sessions • Bit 4 – Physical slot status • Bit 5 – SIM busy status • Bit 6 – Reduced card status • Bit 7 – Recovery complete • Bit 8 – Supply voltage Vcc status • Bit 9 – Card activation status • Bit 10 – Remote simlock configuration • Bit 11 – Simlock temporary unlock status

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Optional TLVs

None

3.27.2 Response - QMI_UIM_EVENT_REG_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Event Registration Result	1.0	1.54

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Event Registration Result
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	uint32	event_mask	4	Bitmask of the events that were successfully enabled. This result can be different from the mask used in the request when notifications are not supported. Additional bits are reserved for future use and can be ignored by the caller. • Bit 0 – Card status • Bit 1 – SAP connection • Bit 2 – Extended card status • Bit 3 – Close of provisioning sessions • Bit 4 – Physical slot status • Bit 5 – SIM busy status • Bit 6 – Reduced card status • Bit 7 – Recovery complete • Bit 8 – Supply voltage Vcc status • Bit 9 – Card activation status • Bit 10 – Remote simlock configuration • Bit 11 – Simlock temporary unlock status

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

3.27.3 Description of QMI_UIM_EVENT_REG REQ/RESP

This function registers for event notifications from the card. The client must verify the mask in the response to determine which events were registered successfully. Events not supported correctly are not registered.

The client can deregister from all event notifications by passing an empty bitmask in the request.

A client can register for both the legacy and the extended card status. In this case, the subsequent indication contains both.

A client can also register for reduced legacy and/or extended card status indications to prevent receiving transient state indications during card initialization.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.28 QMI_UIM_GET_CARD_STATUS

Retrieves the current status of the card.

UIM message ID

0x002F

Version introduced

Major - 1, Minor - 0

3.28.1 Request - QMI_UIM_GET_CARD_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Extended Card Status	1.30	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Extended Card Status
Length	1			2	
Value	→	boolean	extended_card_status	1	Indicates whether the extended card status is requested. Valid values: • 0 – Legacy card status (up to 2 cards) • 1 – Extended card status If the Extended Card Status TLV is missing, the legacy card status with support for only 2 cards is returned.

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3.28.2 Response - QMI_UIM_GET_CARD_STATUS_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

Name	Version introduced	Version last modified
Card Status	1.0	1.37
Hot-Swap Status	1.13	1.14
Valid Card Status	1.23	1.23
Extended Card Status	1.30	1.37
Extended Hot-Swap Status	1.30	1.30
SIM Busy Status	1.35	1.35

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Status
Length	Var			2	
Value	→	uint16	index_gw_pri	2	Index of the primary GW provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint16	index_1x_pri	2	Index of the primary 1X provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	index_gw_sec	2	Index of the secondary GW provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint16	index_1x_sec	2	Index of the secondary 1X provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint8	num_slot	1	Indicates the number of slots available on the device. The following block is repeated for each slot.
		enum8	card_state	1	Indicates the state of the card. Valid values: • UIM_CARD_STATE_ABSENT (0x00) – Absent • UIM_CARD_STATE_PRESENT (0x01) – Present • UIM_CARD_STATE_ERROR (0x02) – Error
		enum8	upin_state	1	Indicates the state of UPIN. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	upin_retries	1	Indicates the number of retries remaining to verify the UPIN.
		uint8	upuk_retries	1	Indicates the number of retries remaining to unblock the UPIN.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	error_code	1	Indicates the reason for the card error, and is valid only when the card state is Error. Valid values: • UIM_CARD_ERROR_CODE_ UNKNOWN (0x00) – Unknown • UIM_CARD_ERROR_CODE_ POWER_DOWN (0x01) – Power down • UIM_CARD_ERROR_CODE_POLL_ ERROR (0x02) – Poll error • UIM_CARD_ERROR_CODE_NO_ ATR_RECEIVED (0x03) – No ATR received • UIM_CARD_ERROR_CODE_VOLT_ MISMATCH (0x04) – Volt mismatch • UIM_CARD_ERROR_CODE_ PARITY_ERROR (0x05) – Parity error • UIM_CARD_ERROR_CODE_ POSSIBLY_REMOVED (0x06) – Unknown, possibly removed • UIM_CARD_ERROR_CODE_SIM_ TECHNICAL_PROBLEMS (0x07) – Card returned technical problems • UIM_CARD_ERROR_CODE_NULL_ BYTES (0x08) – Card returned NULL bytes • UIM_CARD_ERROR_CODE_SAP_ CONNECTED (0x09) – Terminal in SAP mode Other values are possible and reserved for future use. When an unknown value is received, it is to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	num_app	1	Number of sets of the following elements: • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	app_type	1	Indicates the type of the application. Valid values: • UIM_APP_TYPE_UNKNOWN (0x00) – Unknown • UIM_APP_TYPE_SIM (0x01) – SIM card • UIM_APP_TYPE_USIM (0x02) – USIM application • UIM_APP_TYPE_RUIM (0x03) – RUIM card • UIM_APP_TYPE_CSIM (0x04) – CSIM application • UIM_APP_TYPE_ISIM (0x05) – ISIM application Other values are reserved for the future and are to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	app_state	1	Indicates the state of the application. Valid values: • UIM_APP_STATE_UNKNOWN (0x00) – Unknown • UIM_APP_STATE_DETECTED (0x01) – Detected • UIM_APP_STATE_PIN1_OR_ UPIN_REQ (0x02) – PIN1 or UPIN is required • UIM_APP_STATE_PUK1_OR_ PUK_REQ (0x03) – PUK1 or PUK for UPIN is required • UIM_APP_STATE_PERSON_ CHECK_REQ (0x04) – Personalization state must be checked • UIM_APP_STATE_PIN1_PERM_ BLOCKED (0x05) – PIN1 is blocked • UIM_APP_STATE_ILLEGAL (0x06) – Illegal • UIM_APP_STATE_READY (0x07) – Ready
		enum8	perso_state	1	Indicates the state of the personalization for the application. Valid values: • UIM_PERSO_STATE_UNKNOWN (0x00) – Unknown • UIM_PERSO_STATE_IN_PROGRESS (0x01) – Personalization operation is in progress • UIM_PERSO_STATE_READY (0x02) – Ready • UIM_PERSO_STATE_CODE_ REQ (0x03) – Personalization code is required • UIM_PERSO_STATE_PUK_REQ (0x04) – PUK for personalization code is required • UIM_PERSO_STATE_ PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	perso_feature	1	Indicates the personalization feature. This applies only when a personalization code is required to deactivate or unblock personalization. Valid values: • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_STATUS_ GW_SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_STATUS_ GW_CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_STATUS_ GW_UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_STATUS_ 1X_SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_STATUS_ 1X_CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_STATUS_ 1X_RUIM (0x0A) – 1X RUIM personalization • UIM_PERSO_FEATURE_STATUS_ UNKNOWN (0x0B) – Unknown
		uint8	perso_retries	1	Indicates the number of retries remaining to disable the personalization. This value is set to 0xFF if the modem configuration allows unlimited retries.
		uint8	perso_unblock_retries	1	Indicates the number of retries remaining to unblock the personalization.
		uint8	aid_len	1	Number of sets of the following elements: • aid_value

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	aid_value	Var	Application identifier value.
		enum8	univ_pin	1	Indicates whether UPIN replaces PIN1. Valid values: • UIM_UNIV_PIN_PIN1_USED (0x00) – PIN1 is used • UIM_UNIV_PIN_UPIN_REPLACES_PIN1 (0x01) – UPIN replaces PIN1
		enum8	pin1_state	1	Indicates the state of PIN1. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin1_retries	1	Indicates the number of retries remaining to verify PIN1.
		uint8	puk1_retries	1	Indicates the number of retries remaining to unblock PIN1.
		enum8	pin2_state	1	Indicates the state of PIN2. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin2_retries	1	Indicates the number of retries remaining to verify PIN2.
		uint8	puk2_retries	1	Indicates the number of retries remaining to unblock PIN2.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x11			1	Hot-Swap Status
Length	Var			2	
Value	→	uint8	hot_swap_len	1	Number of sets of the following elements: • hot_swap
		enum8	hot_swap	Var	Indicates the status of the hot-swap switch. Valid values: • UIM_HOT_SWAP_NOT_SUPPORTED (0x00) – Hot-swap is not supported • UIM_HOT_SWAP_STATUS_NOT_SUPPORTED (0x01) – Hot-swap is supported, but the status of the switch is not supported • UIM_HOT_SWAP_STATUS_PRESENT (0x02) – Switch indicates that the card is present • UIM_HOT_SWAP_STATUS_NOT_PRESENT (0x03) – Switch indicates that the card is not present
Type	0x12			1	Valid Card Status
Length	Var			2	
Value	→	uint8	card_status_validity_len	1	Number of sets of the following elements: • card_status_validity
		boolean	card_status_validity	Var	Indicates whether the reported status of the card on the corresponding slot is valid. Valid values: • 0 – Status of the card is unknown (even when reported as absent) • 1 – Status of the card is valid
Type	0x13			1	Extended Card Status
Length	Var			2	
Value	→	uint8	index_gw_len	1	Number of sets of the following elements: • index_gw
		uint16	index_gw	Var	Index of the GW provisioning applications. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint8	index_1x_len	1	Number of sets of the following elements: • index_1x

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	index_1x	Var	Index of the 1X provisioning applications. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint8	num_slot	1	Number of sets of the following elements: • card_state • pin_state • pin_retries • puk_retries • error_code • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	card_state	1	Indicates the state of the card. Valid values: • UIM_EXTENDED_CARD_STATE_ABSENT (0x00) – Absent • UIM_EXTENDED_CARD_STATE_PRESENT (0x01) – Present • UIM_EXTENDED_CARD_STATE_ERROR (0x02) – Error • UIM_EXTENDED_CARD_STATE_UNKNOWN (0x03) – Unknown

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	upin_state	1	Indicates the state of UPIN. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	upin_retries	1	Indicates the number of retries remaining to verify this UPIN.
		uint8	upuk_retries	1	Indicates the number of retries remaining to unblock UPIN.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	error_code	1	Indicates the reason for the card error, and is valid only when the card state is Error. Valid values: • UIM_CARD_ERROR_CODE_UNKNOWN (0x00) – Unknown • UIM_CARD_ERROR_CODE_POWER_DOWN (0x01) – Power down • UIM_CARD_ERROR_CODE_POLL_ERROR (0x02) – Poll error • UIM_CARD_ERROR_CODE_NO_ATR_RECEIVED (0x03) – No ATR received • UIM_CARD_ERROR_CODE_VOLT_MISMATCH (0x04) – Volt mismatch • UIM_CARD_ERROR_CODE_PARITY_ERROR (0x05) – Parity error • UIM_CARD_ERROR_CODE_POSSIBLY_REMOVED (0x06) – Unknown, possibly removed • UIM_CARD_ERROR_CODE_SIM_TECHNICAL_PROBLEMS (0x07) – Card returned technical problems • UIM_CARD_ERROR_CODE_NULL_BYTES (0x08) – Card returned NULL bytes • UIM_CARD_ERROR_CODE_SAP_CONNECTED (0x09) – Terminal in SAP mode Other values are possible and reserved for future use. When an unknown value is received, it is to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	num_app	1	Number of sets of the following elements: • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	app_type	1	Indicates the type of the application. Valid values: • UIM_APP_TYPE_UNKNOWN (0x00) – Unknown • UIM_APP_TYPE_SIM (0x01) – SIM card • UIM_APP_TYPE_USIM (0x02) – USIM application • UIM_APP_TYPE_RUIM (0x03) – RUIM card • UIM_APP_TYPE_CSIM (0x04) – CSIM application • UIM_APP_TYPE_ISIM (0x05) – ISIM application Other values are reserved for the future and are to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	app_state	1	Indicates the state of the application. Valid values: • UIM_APP_STATE_UNKNOWN (0x00) – Unknown • UIM_APP_STATE_DETECTED (0x01) – Detected • UIM_APP_STATE_PIN1_OR_ UPIN_REQ (0x02) – PIN1 or UPIN is required • UIM_APP_STATE_PUK1_OR_ PUK_REQ (0x03) – PUK1 or PUK for UPIN is required • UIM_APP_STATE_PERSON_ CHECK_REQ (0x04) – Personalization state must be checked • UIM_APP_STATE_PIN1_PERM_ BLOCKED (0x05) – PIN1 is blocked • UIM_APP_STATE_ILLEGAL (0x06) – Illegal • UIM_APP_STATE_READY (0x07) – Ready
		enum8	perso_state	1	Indicates the state of the personalization for the application. Valid values: • UIM_PERSO_STATE_UNKNOWN (0x00) – Unknown • UIM_PERSO_STATE_IN_PROGRESS (0x01) – Personalization operation is in progress • UIM_PERSO_STATE_READY (0x02) – Ready • UIM_PERSO_STATE_CODE_ REQ (0x03) – Personalization code is required • UIM_PERSO_STATE_PUK_REQ (0x04) – PUK for personalization code is required • UIM_PERSO_STATE_ PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	perso_feature	1	Indicates the personalization feature. This applies only when a personalization code is required to deactivate or unblock personalization. Valid values: • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_STATUS_ GW_SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_STATUS_ GW_CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_STATUS_ GW_UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_STATUS_ 1X_SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_STATUS_ 1X_CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_STATUS_ 1X_RUIM (0x0A) – 1X RUIM personalization • UIM_PERSO_FEATURE_STATUS_ UNKNOWN (0x0B) – Unknown
		uint8	perso_retries	1	Indicates the number of retries remaining to disable the personalization. This value is set to 0xFF if the modem configuration allows unlimited retries.
		uint8	perso_unblock_retries	1	Indicates the number of retries remaining to unblock the personalization.
		uint8	aid_len	1	Number of sets of the following elements: • aid_value

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	aid_value	Var	Application identifier value.
		enum8	univ_pin	1	Indicates whether UPIN replaces PIN1. Valid values: • UIM_UNIV_PIN_PIN1_USED (0x00) – PIN1 is used • UIM_UNIV_PIN_UPIN_REPLACES_PIN1 (0x01) – UPIN replaces PIN1
		enum8	pin1_state	1	Indicates the state of PIN1. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin1_retries	1	Indicates the number of retries remaining to verify PIN1.
		uint8	puk1_retries	1	Indicates the number of retries remaining to unblock PIN1.
		enum8	pin2_state	1	Indicates the state of PIN2. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin2_retries	1	Indicates the number of retries remaining to verify PIN2.
		uint8	puk2_retries	1	Indicates the number of retries remaining to unblock PIN2.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x14			1	Extended Hot-Swap Status
Length	Var			2	
Value	→	uint8	extended_hot_swap_len	1	Number of sets of the following elements: • extended_hot_swap
		enum8	extended_hot_swap	Var	Indicates the status of the hot-swap switch. Valid values: • UIM_HOT_SWAP_NOT_SUPPORTED (0x00) – Hot-swap is not supported • UIM_HOT_SWAP_STATUS_NOT_SUPPORTED (0x01) – Hot-swap is supported, but the status of the switch is not supported • UIM_HOT_SWAP_STATUS_PRESENT (0x02) – Switch indicates that the card is present • UIM_HOT_SWAP_STATUS_NOT_PRESENT (0x03) – Switch indicates that the card is not present
Type	0x15			1	SIM Busy Status
Length	Var			2	
Value	→	uint8	sim_busy_status_len	1	Number of sets of the following elements: • sim_busy_status
		boolean	sim_busy_status	Var	Indicates whether the card on the corresponding slot is busy. Valid values: • 0 – SIM card is not busy • 1 – SIM card is busy

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.

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3.28.3 Description of QMI_UIM_GET_CARD_STATUS REQ/RESP

This function retrieves the current status of the cards and the status of all applications available on the cards. The function also returns support information for the hot-swap feature and the status of the switch used to detect a card removal/insertion.

Support of the hot-swap feature is independent of the presence of the optional TLV with the hot-swap switch status.

The result of this function can be used by the client to determine the number of slots supported by the specific target.

During power-up, the status of the card can be unknown for a short period of time. An optional TLV indicates whether the reported status is valid. If the TLV is missing, the client assumes that the status of the card is valid.

The extended card status is used on targets with three cards or more. A client must be able to handle the legacy card status as a response, even if it requested the extended card status.

If a session is associated with a card in slot 3 or higher, the legacy card status indicates the session as invalid.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.29 QMI_UIM_POWER_DOWN

Powers down the card.

UIM message ID

0x0030

Version introduced

Major - 1, Minor - 0

3.29.1 Request - QMI_UIM_POWER_DOWN_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.0	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

None

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3.29.2 Response - QMI_UIM_POWER_DOWN_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

3.29.3 Description of QMI_UIM_POWER_DOWN REQ/RESP

This function powers down the card. This is usually performed when the phone is switched off or when it is set to Airplane mode.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.30 QMI_UIM_POWER_UP

Powers up the card.

UIM message ID

0x0031

Version introduced

Major - 1, Minor - 0

3.30.1 Request - QMI_UIM_POWER_UP_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.0	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

Name	Version introduced	Version last modified
Ignore Hot-Swap Switch	1.15	1.15

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Ignore Hot-Swap Switch
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	boolean	ignore_hotswap_switch	1	Indicates if the hot-swap switch is to be ignored during the initialization of the card. Valid values: • 0 – Checks the hot-swap switch status • 1 – Ignores the hot-swap switch status

3.30.2 Response - QMI_UIM_POWER_UP_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.0	1.0

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.

3.30.3 Description of QMI_UIM_POWER_UP REQ/RESP

This function powers up the card after a previous powerdown command.

The client can request to ignore the status of the hot-swap switch during the initialization of the card. This functionality can be useful during testing. This feature is not available on all targets and the TLV is ignored if the request cannot be satisfied.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.31 QMI_UIM_STATUS_CHANGE_IND

Indicates that the status of the card has changed.

UIM message ID

0x0032

Version introduced

Major - 1, Minor - 0

3.31.1 Indication - QMI_UIM_STATUS_CHANGE_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Card Status	1.0	1.37
Valid Card Status	1.25	1.25
Extended Card Status	1.30	1.37

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Status
Length	Var			2	
Value	→	uint16	index_gw_pri	2	Index of the primary GW provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	index_1x_pri	2	Index of the primary 1X provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint16	index_gw_sec	2	Index of the secondary GW provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint16	index_1x_sec	2	Index of the secondary 1X provisioning application. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint8	num_slot	1	Indicates the number of slots available on the device. The following block is repeated for each slot.
		enum8	card_state	1	Indicates the state of the card. Valid values: • UIM_CARD_STATE_ABSENT (0x00) – Absent • UIM_CARD_STATE_PRESENT (0x01) – Present • UIM_CARD_STATE_ERROR (0x02) – Error
		enum8	upin_state	1	Indicates the state of UPIN. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	upin_retries	1	Indicates the number of retries remaining to verify UPIN.
		uint8	upuk_retries	1	Indicates the number of retries remaining to unblock UPIN.
		enum8	error_code	1	Indicates the reason for the card error, and is valid only when the card state is Error. Valid values: • UIM_CARD_ERROR_CODE_ UNKNOWN (0x00) – Unknown • UIM_CARD_ERROR_CODE_ POWER_DOWN (0x01) – Power down • UIM_CARD_ERROR_CODE_POLL_ ERROR (0x02) – Poll error • UIM_CARD_ERROR_CODE_NO_ ATR_RECEIVED (0x03) – No ATR received • UIM_CARD_ERROR_CODE_VOLT_ MISMATCH (0x04) – Volt mismatch • UIM_CARD_ERROR_CODE_ PARITY_ERROR (0x05) – Parity error • UIM_CARD_ERROR_CODE_ POSSIBLY_REMOVED (0x06) – Unknown, possibly removed • UIM_CARD_ERROR_CODE_SIM_ TECHNICAL_PROBLEMS (0x07) – Card returned technical problems • UIM_CARD_ERROR_CODE_NULL_ BYTES (0x08) – Card returned NULL bytes • UIM_CARD_ERROR_CODE_SAP_ CONNECTED (0x09) – Terminal in SAP mode Other values are possible and reserved for future use. When an unknown value is received, it is to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	num_app	1	Number of sets of the following elements: • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	app_type	1	Indicates the type of the application. Valid values: • UIM_APP_TYPE_UNKNOWN (0x00) – Unknown • UIM_APP_TYPE_SIM (0x01) – SIM card • UIM_APP_TYPE_USIM (0x02) – USIM application • UIM_APP_TYPE_RUIM (0x03) – RUIM card • UIM_APP_TYPE_CSIM (0x04) – CSIM application • UIM_APP_TYPE_ISIM (0x05) – ISIM application Other values are reserved for the future and are to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	app_state	1	Indicates the state of the application. Valid values: • UIM_APP_STATE_UNKNOWN (0x00) – Unknown • UIM_APP_STATE_DETECTED (0x01) – Detected • UIM_APP_STATE_PIN1_OR_ UPIN_REQ (0x02) – PIN1 or UPIN is required • UIM_APP_STATE_PUK1_OR_ PUK_REQ (0x03) – PUK1 or PUK for UPIN is required • UIM_APP_STATE_PERSON_ CHECK_REQ (0x04) – Personalization state must be checked • UIM_APP_STATE_PIN1_PERM_ BLOCKED (0x05) – PIN1 is blocked • UIM_APP_STATE_ILLEGAL (0x06) – Illegal • UIM_APP_STATE_READY (0x07) – Ready
		enum8	perso_state	1	Indicates the state of the personalization for the application. Valid values: • UIM_PERSO_STATE_UNKNOWN (0x00) – Unknown • UIM_PERSO_STATE_IN_PROGRESS (0x01) – Personalization operation is in progress • UIM_PERSO_STATE_READY (0x02) – Ready • UIM_PERSO_STATE_CODE_ REQ (0x03) – Personalization code is required • UIM_PERSO_STATE_PUK_REQ (0x04) – PUK for personalization code is required • UIM_PERSO_STATE_ PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	perso_feature	1	Indicates the personalization feature. This applies only when a personalization code is required to deactivate or unblock personalization. Valid values: • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_STATUS_ GW_SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_STATUS_ GW_CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_STATUS_ GW_UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_STATUS_ 1X_SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_STATUS_ 1X_CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_STATUS_ 1X_RUIM (0x0A) – 1X RUIM personalization • UIM_PERSO_FEATURE_STATUS_ UNKNOWN (0x0B) – Unknown
		uint8	perso_retries	1	Indicates the number of retries remaining to disable the personalization. This value is set to 0xFF if the modem configuration allows unlimited retries.
		uint8	perso_unblock_retries	1	Indicates the number of retries remaining to unblock the personalization.
		uint8	aid_len	1	Number of sets of the following elements: • aid_value

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	aid_value	Var	Application identifier value.
		enum8	univ_pin	1	Indicates whether UPIN replaces PIN1. Valid values: • UIM_UNIV_PIN_PIN1_USED (0x00) – PIN1 is used • UIM_UNIV_PIN_UPIN_REPLACES_PIN1 (0x01) – UPIN replaces PIN1
		enum8	pin1_state	1	Indicates the state of PIN1. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin1_retries	1	Indicates the number of retries remaining to verify PIN1.
		uint8	puk1_retries	1	Indicates the number of retries remaining to unblock PIN1.
		enum8	pin2_state	1	Indicates the state of PIN2. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin2_retries	1	Indicates the number of retries remaining to verify PIN2.
		uint8	puk2_retries	1	Indicates the number of retries remaining to unblock PIN2.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x11			1	Valid Card Status
Length	Var			2	
Value	→	uint8	card_status_validity_len	1	Number of sets of the following elements: • card_status_validity
		boolean	card_status_validity	Var	Indicates whether the reported status of the card on the corresponding slot is valid. Valid values: • 0 – Status of the card is unknown (even if reported as absent) • 1 – Status of the card is valid
Type	0x12			1	Extended Card Status
Length	Var			2	
Value	→	uint8	index_gw_len	1	Number of sets of the following elements: • index_gw
		uint16	index_gw	Var	Index of the GW provisioning applications. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.
		uint8	index_1x_len	1	Number of sets of the following elements: • index_1x
		uint16	index_1x	Var	Index of the 1X provisioning applications. The most significant byte indicates the slot (starting from 0), while the least significant byte indicates the application for that slot (starting from 0). The value 0xFFFF identifies that the session does not exist.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	num_slot	1	Number of sets of the following elements: • card_state • pin_state • pin_retries • puk_retries • error_code • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	card_state	1	Indicates the state of the card. Valid values: • UIM_EXTENDED_CARD_STATE_ABSENT (0x00) – Absent • UIM_EXTENDED_CARD_STATE_PRESENT (0x01) – Present • UIM_EXTENDED_CARD_STATE_ERROR (0x02) – Error • UIM_EXTENDED_CARD_STATE_UNKNOWN (0x03) – Unknown
		enum8	upin_state	1	Indicates the state of UPIN. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	upin_retries	1	Indicates the number of retries remaining to verify UPIN.
		uint8	upuk_retries	1	Indicates the number of retries remaining to unblock UPIN.
		enum8	error_code	1	Indicates the reason for the card error, and is valid only when the card state is Error. Valid values: • UIM_CARD_ERROR_CODE_ UNKNOWN (0x00) – Unknown • UIM_CARD_ERROR_CODE_ POWER_DOWN (0x01) – Power down • UIM_CARD_ERROR_CODE_POLL_ ERROR (0x02) – Poll error • UIM_CARD_ERROR_CODE_NO_ ATR_RECEIVED (0x03) – No ATR received • UIM_CARD_ERROR_CODE_VOLT_ MISMATCH (0x04) – Volt mismatch • UIM_CARD_ERROR_CODE_ PARITY_ERROR (0x05) – Parity error • UIM_CARD_ERROR_CODE_ POSSIBLY_REMOVED (0x06) – Unknown, possibly removed • UIM_CARD_ERROR_CODE_SIM_ TECHNICAL_PROBLEMS (0x07) – Card returned technical problems • UIM_CARD_ERROR_CODE_NULL_ BYTES (0x08) – Card returned NULL bytes • UIM_CARD_ERROR_CODE_SAP_ CONNECTED (0x09) – Terminal in SAP mode Other values are possible and reserved for future use. When an unknown value is received, it is to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	num_app	1	Number of sets of the following elements: • app_type • app_state • perso_state • perso_feature • perso_retries • perso_unblock_retries • aid_len • aid_value • univ_pin • pin_state • pin_retries • puk_retries • pin_state • pin_retries • puk_retries
		enum8	app_type	1	Indicates the type of the application. Valid values: • UIM_APP_TYPE_UNKNOWN (0x00) – Unknown • UIM_APP_TYPE_SIM (0x01) – SIM card • UIM_APP_TYPE_USIM (0x02) – USIM application • UIM_APP_TYPE_RUIM (0x03) – RUIM card • UIM_APP_TYPE_CSIM (0x04) – CSIM application • UIM_APP_TYPE_ISIM (0x05) – ISIM application Other values are reserved for the future and are to be handled as “Unknown”.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	app_state	1	Indicates the state of the application. Valid values: • UIM_APP_STATE_UNKNOWN (0x00) – Unknown • UIM_APP_STATE_DETECTED (0x01) – Detected • UIM_APP_STATE_PIN1_OR_ UPIN_REQ (0x02) – PIN1 or UPIN is required • UIM_APP_STATE_PUK1_OR_ PUK_REQ (0x03) – PUK1 or PUK for UPIN is required • UIM_APP_STATE_PERSON_ CHECK_REQ (0x04) – Personalization state must be checked • UIM_APP_STATE_PIN1_PERM_ BLOCKED (0x05) – PIN1 is blocked • UIM_APP_STATE_ILLEGAL (0x06) – Illegal • UIM_APP_STATE_READY (0x07) – Ready
		enum8	perso_state	1	Indicates the state of the personalization for the application. Valid values: • UIM_PERSO_STATE_UNKNOWN (0x00) – Unknown • UIM_PERSO_STATE_IN_PROGRESS (0x01) – Personalization operation is in progress • UIM_PERSO_STATE_READY (0x02) – Ready • UIM_PERSO_STATE_CODE_ REQ (0x03) – Personalization code is required • UIM_PERSO_STATE_PUK_REQ (0x04) – PUK for personalization code is required • UIM_PERSO_STATE_ PERMANENTLY_BLOCKED (0x05) – Permanently blocked

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	perso_feature	1	Indicates the personalization feature. This applies only when a personalization code is required to deactivate or unblock personalization. Valid values: • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_STATUS_ GW_NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_STATUS_ GW_SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_STATUS_ GW_CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_STATUS_ GW_UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_STATUS_ 1X_HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_STATUS_ 1X_SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_STATUS_ 1X_CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_STATUS_ 1X_RUIM (0x0A) – 1X RUIM personalization • UIM_PERSO_FEATURE_STATUS_ UNKNOWN (0x0B) – Unknown
		uint8	perso_retries	1	Indicates the number of retries remaining to disable the personalization. This value is set to 0xFF if the modem configuration allows unlimited retries.
		uint8	perso_unblock_retries	1	Indicates the number of retries remaining to unblock the personalization.
		uint8	aid_len	1	Number of sets of the following elements: • aid_value

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	aid_value	Var	Application identifier value.
		enum8	univ_pin	1	Indicates whether UPIN replaces PIN1. Valid values: • UIM_UNIV_PIN_PIN1_USED (0x00) – PIN1 is used • UIM_UNIV_PIN_UPIN_REPLACES_PIN1 (0x01) – UPIN replaces PIN1
		enum8	pin1_state	1	Indicates the state of PIN1. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin1_retries	1	Indicates the number of retries remaining to verify PIN1.
		uint8	puk1_retries	1	Indicates the number of retries remaining to unblock PIN1.
		enum8	pin2_state	1	Indicates the state of PIN2. Valid values: • UIM_PIN_STATE_UNKNOWN (0x00) – Unknown • UIM_PIN_STATE_ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_PIN_STATE_ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_PIN_STATE_DISABLED (0x03) – Disabled • UIM_PIN_STATE_BLOCKED (0x04) – Blocked • UIM_PIN_STATE_PERMANENTLY_BLOCKED (0x05) – Permanently blocked
		uint8	pin2_retries	1	Indicates the number of retries remaining to verify PIN2.
		uint8	puk2_retries	1	Indicates the number of retries remaining to unblock PIN2.

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3.31.2 Description of QMI_UIM_STATUS_CHANGE_IND

This event indicates that the status of the card has changed. The event does not indicate what changed from the previous event. The application is responsible for this.

The indication might not have the TLV with the card status. In this case, the application can retrieve the current card status using the QMI_UIM_GET_CARD_STATUS command.

During power-up, it is possible that the status of the card is unknown for a short period of time. The Valid Card Status optional TLV indicates whether the reported status is valid. If the TLV is missing, the client assumes that the status of the card is valid.

The event is sent to the application only when the application has registered for notifications using the QMI_UIM_EVENT_REG command.

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3.32 QMI_UIM_REFRESH_IND

Received when a refresh command comes from the card.

UIM message ID

0x0033

Version introduced

Major - 1, Minor - 0

3.32.1 Indication - QMI_UIM_REFRESH_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Refresh Event	1.0	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Refresh Event
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	stage	1	Indicates the stage of the Refresh procedure. Valid values: • UIM_REFRESH_STAGE_WAIT_FOR_OK (0x00) – Waiting for OK to refresh • UIM_REFRESH_STAGE_START (0x01) – Refresh started • UIM_REFRESH_STAGE_END_WITH_SUCCESS (0x02) – Refresh ended successfully • UIM_REFRESH_STAGE_END_WITH_FAILURE (0x03) – Refresh failed
		enum8	mode	1	Indicates the Refresh mode. Valid values: • UIM_REFRESH_MODE_RESET (0x00) – Reset • UIM_REFRESH_MODE_INIT (0x01) – Init • UIM_REFRESH_MODE_INIT_ FCN (0x02) – Init and FCN • UIM_REFRESH_MODE_FCN (0x03) – FCN • UIM_REFRESH_MODE_INIT_FULL_FCN (0x04) – Init and Full FCN • UIM_REFRESH_MODE_APP_RESET (0x05) – Application reset • UIM_REFRESH_MODE_3G_RESET (0x06) – 3G session reset
		enum8	session_type	1	Indicates the session type. Valid values: 0 through 7 and 10 through 21. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value. This value is required only for a nonprovisioning session type and is ignored in all other cases.
		uint16	num_of_files	2	Number of sets of the following elements: • file_id • path_len • path
		uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	Path value. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).

3.32.2 Description of QMI_UIM_REFRESH_IND

This indication is received when a refresh command comes from the card.

Details regarding the Refresh procedure and the stages and actions that an application must perform are described in 80-VM566-1.

The event is sent to the application only when the application has registered for notifications using the QMI_UIM_REFRESH_REGISTER command.

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3.33 QMI_UIM_AUTHENTICATE

Executes the authentication algorithm on the card.

UIM message ID

0x0034

Version introduced

Major - 1, Minor - 1

3.33.1 Request - QMI_UIM_AUTHENTICATE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.1	1.30
Authentication Data	1.1	1.1

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Authentication Data
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	context	1	Authenticate context. Valid values: • UIM_AUTH_CONTEXT_RUN_ GSM_ALG (0x00) – Runs the GSM alogrithm (valid only on a 2G SIM card, as specified in 3GPP TS 51.011) • UIM_AUTH_CONTEXT_RUN_ CAVE_ALG (0x01) – Runs the CAVE algorithm (valid only on a RUIM card, as specified in C.S0023-D) • UIM_AUTH_CONTEXT_GSM_ SEC (0x02) – GSM security context (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_3G_SEC (0x03) – 3G security context (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_VGCS_ VBS_SEC (0x04) – VGCS/VBS security context (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_GBA_ SEC_BOOTSTRAPPING (0x05) – GBA security context, Bootstrapping mode (valid only on a USIM or ISIM application, as specified in 3GPP TS 31.102 and 3GPP TS 31.103) • UIM_AUTH_CONTEXT_GBA_ SEC_NAF_DERIVATION (0x06) – GBA security context, NAF Derivation mode (valid only on a USIM or ISIM application, as specified in 3GPP TS 31.102 and 3GPP TS 31.103) • UIM_AUTH_CONTEXT_MBMS_ SEC_MSK_UPDATE (0x07) – MBMS security context, MSK Update mode (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_MBMS_ SEC_MTK_GENERATION (0x08) – MBMS security context, MTK Generation mode (valid only on a USIM application, as specified in 3GPP TS 31.102)

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	context (cont.)	1	• UIM_AUTH_CONTEXT_MBMS_SEC_MSK_DELETION (0x09) – MBMS security context, MSK Deletion mode (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_MBMS_SEC_MUK_DELETION (0x0A) – MBMS security context, MUK Deletion mode (valid only on a USIM application, as specified in 3GPP TS 31.102) • UIM_AUTH_CONTEXT_IMS_AKA_SEC (0x0B) – IMS AKA security context (valid only on a ISIM application, as specified in 3GPP TS 31.103) • UIM_AUTH_CONTEXT_HTTP_DIGEST_SEC (0x0C) – HTTP-digest security context (valid only on an ISIM application, as specified in 3GPP TS 31.103) • UIM_AUTH_CONTEXT_COMPUTE_IP_CHAP (0x0D) – Compute IP authentication, CHAP (valid only on RUIM or CSIM, as specified in C.S0023-D and C.S0065-0) • UIM_AUTH_CONTEXT_COMPUTE_IP_MN_HA (0x0E) – Compute IP authentication, MN-HA authenticator (valid only on RUIM or CSIM, as specified in C.S0023-D and C.S0065-0) • UIM_AUTH_CONTEXT_COMPUTE_IP_MIP_RRQ_HASH (0x0F) – Compute IP authentication, MIP-RRQ hash (valid only on RUIM or CSIM, as specified in C.S0023-D and C.S0065-0) • UIM_AUTH_CONTEXT_COMPUTE_IP_MN_AAA (0x10) – Compute IP authentication, MN-AAA authenticator (valid only on RUIM or CSIM, as specified in C.S0023-D and C.S0065-0) • UIM_AUTH_CONTEXT_COMPUTE_IP_HRPD_ACCESS (0x11) – Compute IP authentication, HRPD access authenticator (valid only on RUIM or CSIM, as specified in C.S0023-D and C.S0065-0) Other values are possible and reserved for future use.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint16	data_len	2	Number of sets of the following elements: • data
		uint8	data	Var	Authenticate data.

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.33.2 Response - QMI_UIM_AUTHENTICATE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.1	1.1

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.1	1.1
Authenticate Result	1.1	1.1
Response in Indication	1.12	1.12

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Authenticate Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Authenticate data. This value is a sequence of bytes returned from the card.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Context used for the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_AUTHENTICATION_ FAILED	Authentication error.

3.33.3 Description of QMI_UIM_AUTHENTICATE REQ/RESP

This function executes a security command on the card that depends on the card type.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the read request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_AUTHENTICATE_IND indication. In this case, the immediate response indicates only when the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate the case where the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceeding table are handled as QMI_ERR_INTERNAL.

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3.34 QMI_UIM_AUTHENTICATE_IND

Indication with an authenticate command result.

UIM message ID

0x0034

Version introduced

Major - 1, Minor - 12

3.34.1 Indication - QMI_UIM_AUTHENTICATE_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.12	1.12
Token of the Original Request	1.12	1.12

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.12	1.12
Authenticate Result	1.12	1.12

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Authenticate Result
Length	Var			2	
Value	→	uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Authenticate data. This value is a sequence of bytes returned from the card.

3.34.2 Description of QMI_UIM_AUTHENTICATE_IND

This indication is received when an authenticate request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_AUTHENTICATE_RESP.

Even when the client requests that the data be passed in a subsequent indication, it must be able to accommodate the case where the optional indication is not supported by the service and the complete result is returned in the response.

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3.35 QMI_UIM_CLOSE_SESSION

Forcibly closes a session to a nonprovisioning application.

UIM message ID

0x0035

Version introduced

Major - 1, Minor - 1

3.35.1 Request - QMI_UIM_CLOSE_SESSION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.1	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.

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Optional TLVs

Name	Version introduced	Version last modified
Termination of Application	1.50	1.50

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Termination of Application
Length	1			2	
Value	→	boolean	terminate_application	1	Indicates whether the application is terminated with a STATUS command before closing the associated logical channel. If the Termination of Application TLV is missing, the default behavior is TRUE, which terminates the application only for USIM, CSIM, ISIM, or 2G applications.

3.35.2 Response - QMI_UIM_CLOSE_SESSION_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.1	1.26

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Session type passed in the request is invalid.
QMI_ERR_ARG_TOO_LONG	Argument is too long.

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QMI_ERR_NO_EFFECT	Operation has no effect. This can occur when the client tries to close a session that is not open.

3.35.3 Description of QMI_UIM_CLOSE_SESSION REQ/RESP

This command forces the closure of a nonprovisioning session. A nonprovisioning session is automatically opened by the QMI when the client tries to use it for the first time and it remains open until it is forcibly closed with this command. This command is not intended to be used for channels opened with QMI_UIM_LOGICAL_CHANNEL or QMI_UIM_OPEN_LOGICAL_CHANNEL.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.36 QMI_UIM_GET_SERVICE_STATUS

Retrieves the status of a service on the card.

UIM message ID

0x0036

Version introduced

Major - 1, Minor - 2

3.36.1 Request - QMI_UIM_GET_SERVICE_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.2	1.30
Capabilities Mask	1.2	1.41

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Capabilities Mask
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	uint32	mask	4	Bitmask of the capabilities that the client retrieves from the card. Additional bits are reserved for future use and are to be set to zero. • Bit 0 – FDN status • Bit 1 – Hidden key PIN status (for USIM cards only) • Bit 2 – Index in the EF-DIR (for UICC cards only) • Bit 3 – ESN status (for 1X sessions only) • Bit 4 – ACL status (for USIM cards only) • Bit 5 – Service status as indicated in the service table

Optional TLVs

Name	Version introduced	Version last modified
Service Number	1.41	1.41

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Service Number
Length	1			2	
Value	→	uint8	service_number	1	This TLV must be present if the service status bit is enabled in the mask. When present, this TLV must contain a valid service number as indicated in the service table per the following specifications: • For CSIM, as specified in C.S0065-0 Section 5.2.80 • For USIM, as specified in 3GPP TS 31.102 Section 4.2.8 • For RUIM, as specified in C.S0023-D Section 3.4.18 • For GSM, as specified in 3GPP TS 51.011 Section 10.3.7

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3.36.2 Response - QMI_UIM_GET_SERVICE_STATUS_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.2	1.2

Optional TLVs

Name	Version introduced	Version last modified
FDN Status	1.2	1.2
Hidden Key Status	1.4	1.4
Index	1.4	1.4
ESN Status	1.21	1.21
ACL Status	1.26	1.26
Service Status	1.41	1.41

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	FDN Status
Length	1			2	
Value	→	enum8	fdn_status	1	Fixed dialing number status. Valid values: • UIM_FDN_STATUS_NOT_AVAILABLE (0x00) – FDN is not available • UIM_FDN_STATUS_AVAILABLE_DISABLED (0x01) – FDN is available and disabled • UIM_FDN_STATUS_AVAILABLE_ENABLED (0x02) – FDN is available and enabled
Type	0x11			1	Hidden Key Status
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	hidden_key_status	1	Status of the hidden key PIN. Valid values: • UIM_HIDDEN_KEY_STATUS_ NOT_SUPPORTED (0x00) – Not supported • UIM_HIDDEN_KEY_STATUS_ ENABLED_NOT_VERIFIED (0x01) – Enabled and not verified • UIM_HIDDEN_KEY_STATUS_ ENABLED_VERIFIED (0x02) – Enabled and verified • UIM_HIDDEN_KEY_STATUS_ DISABLED (0x03) – Disabled
Type	0x12			1	Index
Length	1			2	
Value	→	uint8	index	1	Index of the application in the EF_DIR file, starting from 1.
Type	0x13			1	ESN Status
Length	1			2	
Value	→	enum8	esn_status	1	ESN status. Valid values: • UIM_ESN_STATUS_NO_CHANGE (0x00) – ESN is not changed • UIM_ESN_STATUS_CHANGED (0x01) – ESN is changed
Type	0x14			1	ACL Status
Length	1			2	
Value	→	enum8	acl_status	1	Access point name control list service status. Valid values: • UIM_ACL_STATUS_NOT_ AVAILABLE (0x00) – ACL is not available • UIM_ACL_STATUS_AVAILABLE_ DISABLED (0x01) – ACL is available and disabled • UIM_ACL_STATUS_AVAILABLE_ ENABLED (0x02) – ACL is available and enabled
Type	0x15			1	Service Status
Length	1			2	
Value	→	boolean	service_status	1	Service status. Valid values: • 0 – Service is not available • 1 – Service is available

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Session type passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.36.3 Description of QMI_UIM_GET_SERVICE_STATUS REQ/RESP

This function retrieves the status of the services on the card. As a current limitation, it is possible to request only one service status in each request.

The function supports only provisioning sessions.

The status of the hidden key PIN can be set using the request used for all other PIN codes on the card: QMI_UIM_SET_PIN_PROTECTION.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.37 QMI_UIM_SET_SERVICE_STATUS

Changes the status of a service on the card.

UIM message ID

0x0037

Version introduced

Major - 1, Minor - 2

3.37.1 Request - QMI_UIM_SET_SERVICE_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.2	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.

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Optional TLVs

Name	Version introduced	Version last modified
FDN Service Enable/Disable	1.2	1.2
ACL Service Enable/Disable	1.26	1.26

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	FDN Service Enable/Disable
Length	1			2	
Value	→	boolean	fdn_status	1	Enables or disables the FDN service. Valid values: • 0 – Disables the FDN service • 1 – Enables the FDN service
Type	0x11			1	ACL Service Enable/Disable
Length	1			2	
Value	→	boolean	enable_acl	1	Enables or disables the ACL service. Valid values: • 0 – Disables the ACL service • 1 – Enables the ACL service

3.37.2 Response - QMI_UIM_SET_SERVICE_STATUS_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.2	1.2

Optional TLVs

None

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Session type passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.37.3 Description of QMI_UIM_SET_SERVICE_STATUS REQ/RESP

This function sets the status of a service on the card. If the optional TLVs are not sent, corresponding services do not change.

The status of FDN and ACL services cannot be set in the same request. If both TLVs are included, QMI_ERR_INTERNAL is returned.

The function supports only provisioning sessions.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.38 QMI_UIM_CHANGE_PROVISIONING_SESSION

Changes the provisioning session.

UIM message ID

0x0038

Version introduced

Major - 1, Minor - 3

3.38.1 Request - QMI_UIM_CHANGE_PROVISIONING_SESSION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Change	1.3	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Change
Length	2			2	
Value	→	enum8	session_type	1	Indicates the session type. Only provisioning sessions can be used. Valid values: 0 through 3 and 10 through 15. See Table A-1 for the list of available session types with their values and descriptions.
		boolean	activate	1	Indicates whether to deactivate or activate the provisioning session. Valid values: • 0 – Deactivate • 1 – Activate

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Optional TLVs

Name	Version introduced	Version last modified
Application Information	1.3	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Application Information
Length	Var			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	

3.38.2 Response - QMI_UIM_CHANGE_PROVISIONING_SESSION_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.3	1.26

Optional TLVs

None

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Session type passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NO_EFFECT	Operation has no effect. This can occur when the client tries to deactivate a session that is already inactive.

3.38.3 Description of QMI_UIM_CHANGE_PROVISIONING_SESSION REQ/RESP

This function activates, deactivates, or switches the provisioning sessions. The modem can be set to automatically select the provisioning sessions at powerup. When the provisioning sessions are not automatically selected, the application can select sessions using this function.

The function can also be used to deactivate or switch a provisioning session after this was successfully enabled.

If a session is already used, it must be deactivated before it can be activated on a different application. When the session is activated, the application must not have any other provisioning session linked to it, otherwise an error occurs.

The optional TLV with application information is required only when a session is activated.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.39 QMI_UIM_GET_LABEL

Gets the label of an application on a UICC card.

UIM message ID

0x0039

Version introduced

Major - 1, Minor - 3

3.39.1 Request - QMI_UIM_GET_LABEL_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Application Information	1.3	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Application Information
Length	Var			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	

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Optional TLVs

None

3.39.2 Response - QMI_UIM_GET_LABEL_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.3	1.3

Optional TLVs

Name	Version introduced	Version last modified
Application Label	1.3	1.3

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Application Label
Length	Var			2	
Value	→	uint8	label_len	1	Number of sets of the following elements: • label
		char	label	Var	Value of the label in ASCII characters.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Slot value passed in the request is invalid.

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3.39.3 Description of QMI_UIM_GET_LABEL REQ/RESP

This function retrieves the label of an application on a UICC card, as stored in the EF-DIR of the card. The function returns an error when used on an ICC card or when the specified application does not exist.

Other error codes not listed in the prceeding table are handled as QMI_ERR_INTERNAL.

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3.40 QMI_UIM_GET_CONFIGURATION

Gets the modem configuration for the UIM module.

UIM message ID

0x003A

Version introduced

Major - 1, Minor - 5

3.40.1 Request - QMI_UIM_GET_CONFIGURATION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Configuration Mask	1.17	1.54

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Configuration Mask
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	mask32	configuration_mask	4	Requested configurations: • UIM_GET_CONFIGURATION_ AUTOMATIC_SELECTION (0x01) – Automatic selection • UIM_GET_CONFIGURATION_ PERSONALIZATION_STATUS (0x02) – Personalization status • UIM_GET_CONFIGURATION_ HALT_SUBSCRIPTION (0x04) – Halt subscription • UIM_GET_CONFIGURATION_ USB_UICC_SUPPORTED (0x08) – USB UICC is supported • UIM_GET_CONFIGURATION_ SAP_CLIENT_SUPPORTED (0x10) – SAP in client mode is supported • UIM_GET_CONFIGURATION_ PERSO_REACTIVATION_STATUS (0x20) – Personalization reactivation status All other bits are reserved for future use; if the TLV is missing, the service returns all configuration items in the response.

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3.40.2 Response - QMI_UIM_GET_CONFIGURATION_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.5	1.5

Optional TLVs

Name	Version introduced	Version last modified
Automatic Selection	1.5	1.5
Personalization Status	1.5	1.5
Halt Subscription Publication	1.9	1.9
Personalization Status of Other Slots	1.40	1.40
USB UICC supported flag	1.46	1.46
Support for SAP in client mode	1.46	1.46
Temporary Depersonalization Status	1.52	1.52
Personalization Reactivation Status	1.54	1.54

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Automatic Selection
Length	1			2	
Value	→	boolean	automatic_selection	1	Indicates whether the modem is configured to automatically select the provisioning sessions at powerup. Valid values: • 0 – Automatic provisioning is off • 1 – Automatic provisioning is on
Type	0x11			1	Personalization Status
Length	Var			2	
Value	→	uint8	num_features	1	Number of active personalization features. The following block is repeated for each feature.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	feature	1	Indicates the personalization feature to deactivate or unblock. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		uint8	verify_left	1	Number of the remaining attempts to verify the personalization feature.
		uint8	unblock_left	1	Number of the remaining attempts to unblock the personalization feature.
Type	0x12			1	Halt Subscription Publication
Length	1			2	
Value	→	boolean	halt_subscription	1	Indicates if the modem is configured to publish the subscription after successful initialization. Valid values: • 0 – Modem proceeds with publishing the subscription • 1 – Modem does not publish the subscription
Type	0x13			1	Personalization Status of Other Slots
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	uint8	extended_personalization_ status_ len	1	Number of sets of the following elements: • feature • verify_left • unblock_left
		uint8	personalization_status_len	1	Number of sets of the following elements: • feature • verify_left • unblock_left
		enum8	feature	1	Indicates the personalization feature to deactivate or unblock. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		uint8	verify_left	1	Number of the remaining attempts to verify the personalization feature.
		uint8	unblock_left	1	Number of the remaining attempts to unblock the personalization feature.
Type	0x14			1	USB UICC supported flag

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Field	Field value	Field type	Parameter	Size (byte)	Description
Length	1			2	
Value	→	boolean	usb_uicc_supported	1	Indicates if the modem is configured to support the USB UICC feature. Valid values: • 0 – Modem does not support the USB UICC feature • 1 – Modem supports the USB UICC feature
Type	0x15			1	Support for SAP in client mode
Length	1			2	
Value	→	boolean	sap_client_supported	1	Indicates if the modem is configured to support for SAP in client mode. Valid values: • 0 – Modem does not support SAP in client mode • 1 – Modem supports SAP in client mode
Type	0x16			1	Temporary Depersonalization Status
Length	Var			2	
Value	→	uint8	temporary_ depersonalization_ status_len	1	Number of sets of the following elements: • feature • temporary_unlock_time_ left
		uint8	temporary_unlock_status_ len	1	Number of sets of the following elements: • feature • temporary_unlock_time_ left

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	feature	1	Indicates the personalization feature that is temporarily deactivated. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		uint32	temporary_unlock_time_ left	4	
Type	0x17			1	Personalization Reactivation Status
Length	Var			2	
Value	→	uint8	perso_reactivation_status_ len	1	Number of sets of the following elements: • slot • feature • verify_left • unblock_left

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
		enum8	feature	1	Indicates the personalization feature to deactivate or unblock. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		uint8	verify_left	1	Number of the remaining attempts to verify the personalization feature.
		uint8	unblock_left	1	Number of the remaining attempts to unblock the personalization feature.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.40.3 Description of QMI_UIM_GET_CONFIGURATION REQ/RESP

This command retrieves the parameters of the modem UIM configuration that are commonly required by a control point.

This command can be executed even when the SIM card is not present on the device. This command returns a personalization status that is more complete than the personalization status a control point can retrieve using only the QMI_UIM_GET_CARD_STATUS command. This is because a feature might be enabled when no card is currently blocked by that feature. Personalization features not listed in the response are considered inactive.

The command response lists additional items as follows:

- Personalization features that are temporarily deactivated, and the amount of time, in seconds, before they are to be reactivated
- Personalization features that were previously activated but are currently deactivated

On targets that support multiple slots with a different personalization configuration on each slot, the uim_personalization_status_other_slots_type optional TLV provides the personalization status starting with the second slot.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.41 QMI_UIM_SEND_APDU

Sends an APDU to the card.

UIM message ID

0x003B

Version introduced

Major - 1, Minor - 6

3.41.1 Request - QMI_UIM_SEND_APDU_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.6	1.30
APDU	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	APDU
Length	Var			2	
Value	→	uint16	apdu_len	2	Number of sets of the following elements: • apdu
		opaque	apdu	Var	Content of the APDU.

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Optional TLVs

Name	Version introduced	Version last modified
Logical Channel	1.7	1.7
Intermediate Procedure Bytes	1.21	1.21

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Logical Channel
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel to be used.
Type	0x11			1	Intermediate Procedure Bytes
Length	1			2	
Value	→	enum8	procedure_bytes	1	Indicates whether the client is expecting the intermediate procedure bytes or the final result from the card. Valid values: • UIM_APDU_RETURN_PROCEDURE_BYTES (0x00) – Return intermediate procedure bytes (Default) • UIM_APDU_SKIP_PROCEDURE_BYTES (0x01) – Return final result and status words

3.41.2 Response - QMI_UIM_SEND_APDU_RESP

Message type

Response

Sender

Control point

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.6	1.6

Optional TLVs

Name	Version introduced	Version last modified
APDU Response	1.6	1.6
Long APDU Response	1.22	1.22

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	APDU Response
Length	Var			2	
Value	→	uint16	apdu_len	2	Number of sets of the following elements: • apdu
		opaque	apdu	Var	Content of the application protocol data unit response from the card.
Type	0x11			1	Long APDU Response Token and total length to reconstruct the APDU response from the card.
Length	6			2	
Value	→	uint16	total_length	2	Indicates the total length of the response APDU from the card
		uint32	token	4	Token used in the subsequent indications with APDU response

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Argument value passed in the request is invalid.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_ACCESS_DENIED	Requested operation is not allowed for security reasons.
QMI_ERR_INSUFFICIENT_ RESOURCES	Response is longer than the maximum supported size.

3.41.3 Description of QMI_UIM_SEND_APDU REQ/RESP

This function sends an APDU directly to the card. Control points using this function can interfere with the normal behavior of the modem. When a UICC card is in use, control points are to open a new logical channel and then use the channel to stream the APDUs.

The control point is responsible for handling cases where the card returns a procedure byte in the response indicating that a GET RESPONSE command is needed to retrieve the data. In such cases, the client can specify that it wants only the final result by using the procedure_bytes TLV. If the TLV is absent, the client must handle it.

If the APDU response from the SIM card exceeds the maximum size supported by QMI_UIM_SEND_APDU_RESP, the service returns the QMI_ERR_INSUFFICIENT_ RESOURCES error with the additional indication of a token and total length. The APDU response is sent in subsequent QMI_UIM_SEND_APDU_IND indications to the client, and the client is responsible for recreating the original response from the card.

The optional logical_channel parameter passes the channel ID when the logical channel is opened using the QMI_UIM_LOGICAL_CHANNEL request to stream the APDU on the appropriate logical channel.

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For security reasons, the modem can be configured to restrict the usage of specific APDUs. In this case, the service returns the error code QMI_ERR_ACCESS_DENIED to the control point. More specifically, the commands MANAGE CHANNEL and SELECT by DF name can be restricted, and the control point is to use the QMI_UIM_LOGICAL_CHANNEL or QMI_UIM_OPEN_LOGICAL_CHANNEL requests to access an application on the UICC card.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.42 QMI_UIM_SEND_APDU_IND

Indication with a Send APDU command result.

UIM message ID

0x003B

Version introduced

Major - 1, Minor - 22

3.42.1 Indication - QMI_UIM_SEND_APDU_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Chunk of APDU Response	1.22	1.22

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Chunk of APDU Response
Length	Var			2	
Value	→	uint32	token	4	Unique token used in the previous QMI_UIM_SEND_APDU response.
		uint16	total_length	2	Total length of the response APDU from the card.
		uint16	offset	2	Offset of the current chunk, starting from 0.
		uint16	apdu_len	2	Number of sets of the following elements: • apdu
		opaque	apdu	Var	Content of the data chunk of the APDU response from the card.

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Optional TLVs

None

3.42.2 Description of QMI_UIM_SEND_APDU_IND

This indication is received when a send APDU request is completed and the overall response from the card is larger than the maximum buffer of the QMI_UIM_SEND_APDU_RESP message.

In this case, the overall response is divided into multiple chunks and each chunk is sent to the client in a separate indication. The client is responsible for reconstructing the APDU response from the card, combining all the indications.

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3.43 QMI_UIM_SAP_CONNECTION

Establishes and releases a connection to the UIM module as a SAP client.

UIM message ID

0x003C

Version introduced

Major - 1, Minor - 6

3.43.1 Request - QMI_UIM_SAP_CONNECTION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
SAP Connect	1.6	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	SAP Connect
Length	2			2	
Value	→	enum8	connect	1	Indicates the operation to perform. Valid values: • UIM_SAP_OP_DISCONNECT (0x00) – Disconnect • UIM_SAP_OP_CONNECT (0x01) – Connect • UIM_SAP_OP_CHECK_STATUS (0x02) – Check status
		enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

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Optional TLVs

Name	Version introduced	Version last modified
Disconnect Mode	1.6	1.6
Intermediate GET RESPONSE	1.16	1.16
Condition to Enable a SAP Connection	1.26	1.26

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Disconnect Mode (This TLV is used only for disconnection and is ignored in other cases.)
Length	1			2	
Value	→	enum8	disconnect_mode	1	Indicates the type of disconnection to perform. Valid values: • UIM_SAP_DISCONNECT_ IMMEDIATE (0x00) – Immediate disconnect • UIM_SAP_DISCONNECT_ GRACEFULL (0x01) – Graceful shutdown of the connection
Type	0x11			1	Intermediate GET RESPONSE
Length	1			2	
Value	→	boolean	intermediate_get_response	1	Indicates whether the client is expecting the intermediate GET RESPONSE procedure bytes or the final result from the card. Valid values: • 0 – Return final result and status words • 1 – Return intermediate GET RESPONSE procedure bytes
Type	0x12			1	Condition to Enable a SAP Connection
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum	connection_condition	4	Indicates the type of condition to apply during a connection request. When this TLV is missing, the default behavior is to block a SAP connection when a voice or data call is active. Valid values: • UIM_SAP_CONNECTION_COND_BLOCK_VOICE_OR_DATA (0x00) – Block a SAP connection when a voice or data call is active (Default) • UIM_SAP_CONNECTION_COND_BLOCK_DATA (0x01) – Block a SAP connection when a data call is active • UIM_SAP_CONNECTION_COND_BLOCK_VOICE (0x02) – Block a SAP connection when a voice call is active • UIM_SAP_CONNECTION_COND_BLOCK_NONE (0x03) – Allow a SAP connection in all cases

3.43.2 Response - QMI_UIM_SAP_CONNECTION_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.6	1.6

Optional TLVs

Name	Version introduced	Version last modified
SAP State	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	SAP State (This TLV is returned only when the client requested the SAP state.)
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	sap_state	1	Indicates the SAP connection status. Valid values: • UIM_SAP_STATE_NOT_ENABLED (0x00) – Not enabled • UIM_SAP_STATE_CONNECTING (0x01) – Connecting • UIM_SAP_STATE_CONNECTED_ SUCCESSFULLY (0x02) – Connected successfully • UIM_SAP_STATE_CONNECTION_ ERROR (0x03) – Connection error • UIM_SAP_STATE_DISCONNECTING (0x04) – Disconnecting • UIM_SAP_STATE_DISCONNECTED_ SUCCESSFULLY (0x05) – Disconnected successfully

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Argument value passed in the request is invalid.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device. This can occur when the device does not support SAP functionality.
QMI_ERR_NO_EFFECT	Operation has no effect. This can occur when the client tries to disconnect a SAP connection that is already disconnected.
QMI_ERR_DEVICE_IN_USE	Device already has a SAP connection and cannot accept a new one.
QMI_ERR_DEVICE_NOT_READY	Device is busy or is not ready to accept an SAP connection.

3.43.3 Description of QMI_UIM_SAP_CONNECTION REQ/RESP

This command establishes and releases a connection to the UIM module as a SAP client while the modem acts as a SAP server. A control point can use this request to check the initial SAP status and register for SAP indications to monitor the connection status using the QMI_UIM_EVENT_REG request.

During the disconnection, the client can specify whether to terminate the connection gracefully or immediately. If the TLV with this information is missing, the connection is terminated immediately.

During connection establishment, the client can specify whether it wants the intermediate GET RESPONSE procedure bytes to be relayed to it. If the TLV is absent, the modem provides the final result only. If the modem does not support this TLV, the client must handle the situation where the modem only provides the final result.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.44 QMI_UIM_SAP_REQUEST

Executes a SAP request.

UIM message ID

0x003D

Version introduced

Major - 1, Minor - 6

3.44.1 Request - QMI_UIM_SAP_REQUEST_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
SAP Request	1.6	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	SAP Request
Length	2			2	
Value	→	enum8	sap_request	1	Indicates the SAP request. Valid values: • UIM_SAP_REQUEST_GET_ATR (0x00) – Gets the ATR • UIM_SAP_REQUEST_SEND_ APDU (0x01) – Sends the APDU • UIM_SAP_REQUEST_POWER_ SIM_OFF (0x02) – Powers off the SIM • UIM_SAP_REQUEST_POWER_ SIM_ON (0x03) – Powers on the SIM • UIM_SAP_REQUEST_RESET_ SIM (0x04) – Resets the SIM • UIM_SAP_REQUEST_CARD_ READER_STATUS (0x05) – Indicates the card reader status

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

Name	Version introduced	Version last modified
APDU Data	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	APDU Data (This TLV must be present for the Send APDU request. For all other cases, if present, the TLV is ignored.)
Length	Var			2	
Value	→	uint16	apdu_len	2	Number of sets of the following elements: • apdu
		uint8	apdu	Var	APDU value.

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3.44.2 Response - QMI_UIM_SAP_REQUEST_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.6	1.6

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Optional TLVs

Name	Version introduced	Version last modified
ATR Value	1.6	1.6
APDU Result	1.6	1.6
Card Reader Status	1.6	1.6

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	ATR Value (This TLV is present only for the Get ATR request.)
Length	Var			2	
Value	→	uint8	ATR_len	1	Number of sets of the following elements: • ATR_value
		uint8	ATR_value	Var	Value of the ATR from the card.
Type	0x11			1	APDU Result (This TLV is present only for the Send APDU request.)
Length	Var			2	
Value	→	uint16	apdu_len	2	Number of sets of the following elements: • apdu
		uint8	apdu	Var	APDU response.
Type	0x12			1	Card Reader Status (This TLV is present only for a card reader status request.)
Length	Var			2	
Value	→	uint8	status_len	1	Number of sets of the following elements: • status
		uint8	status	Var	Value of the card reader status.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Argument value passed in the request is invalid.
QMI_ERR_MISSING_ARG	TLV was missing in the request.

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3.44.3 Description of QMI_UIM_SAP_REQUEST REQ/RESP

This function executes a SAP request after the connection has been established using the QMI_UIM_SAP_CONNECTION request.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.45 QMI_UIM_SAP_CONNECTION_IND

Indicates a change in the status of the SAP connection.

UIM message ID

0x003E

Version introduced

Major - 1, Minor - 6

3.45.1 Indication - QMI_UIM_SAP_CONNECTION_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Card Status	1.6	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Status
Length	2			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	sap_state	1	Indicates the SAP connection status. Valid values: • UIM_SAP_STATE_NOT_ENABLED (0x00) – Not enabled • UIM_SAP_STATE_CONNECTING (0x01) – Connecting • UIM_SAP_STATE_CONNECTED_SUCCESSFULLY (0x02) – Connected successfully • UIM_SAP_STATE_CONNECTION_ERROR (0x03) – Connection error • UIM_SAP_STATE_DISCONNECTING (0x04) – Disconnecting • UIM_SAP_STATE_DISCONNECTED_SUCCESSFULLY (0x05) – Disconnected successfully
		enum8	slot	1	Indicates the slot in use. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

3.45.2 Description of QMI_UIM_SAP_CONNECTION_IND

This indication is received when there is a change in the SAP connection.

When a client starts a SAP connection using the QMI_UIM_SAP_CONNECTION request, it implictly registers for SAP indications.

After receiving a disconnection indication, the client does not need to explicitly send the QMI_UIM_SAP_CONNECTION request to terminate the SAP connection. The QMI UIM cleans the status internally.

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3.46 QMI_UIM_LOGICAL_CHANNEL

Opens and closes a logical channel to an application on a UICC card.

UIM message ID

0x003F

Version introduced

Major - 1, Minor - 7

3.46.1 Request - QMI_UIM_LOGICAL_CHANNEL_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.7	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

Name	Version introduced	Version last modified
Open the Logical Channel	1.7	1.7
Close the Logical Channel	1.7	1.7
Template Requested from the Card in SELECT	1.10	1.10
Termination of Application	1.25	1.25

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Open the Logical Channel
Length	Var			2	
Value	→	uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value.
Type	0x11			1	Close the Logical Channel
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel to be closed.
Type	0x12			1	Template Requested from the Card in SELECT
Length	1			2	
Value	→	enum8	file_control_information	1	Template requested from the card in the SELECT command when the application is selected. When this TLV is missing, FCP is requested by default. Valid values: • UIM_FILE_CONTROL_ INFORMATION_NO_DATA (0x00) – No data • UIM_FILE_CONTROL_ INFORMATION_FCP (0x01) – FCP • UIM_FILE_CONTROL_ INFORMATION_FCI (0x02) – FCI • UIM_FILE_CONTROL_ INFORMATION_FCI_WITH_ INTERFACES (0x03) – FCI with interfaces • UIM_FILE_CONTROL_ INFORMATION_FMD (0x04) – FMD
Type	0x13			1	Termination of Application
Length	1			2	
Value	→	boolean	terminate_application	1	Indicates whether the application is terminated with a SELECT request before closing the associated logical channel. When the Termination of Application TLV is missing, the default behavior is TRUE to terminate the application.

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3.46.2 Response - QMI_UIM_LOGICAL_CHANNEL_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.7	1.24

Optional TLVs

Name	Version introduced	Version last modified
Channel ID	1.7	1.7
Card Result	1.10	1.10
Response to SELECT	1.10	1.10

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Channel ID
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel that was opened.
Type	0x11			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x12			1	Response to SELECT
Length	Var			2	
Value	→	uint8	select_response_len	1	Number of sets of the following elements: • select_response
		uint8	select_response	Var	Raw value of the response sent by the card to the initial SELECT command.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	One of the parameters passed in the request is invalid.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INSUFFICIENT_ RESOURCES	SIM card does not have sufficient resources to allocate a new logical channel.
QMI_ERR_SIM_FILE_NOT_FOUND	AID specified in the request is not present on the card.
QMI_ERR_ACCESS_DENIED	Requested operation is not allowed for security reasons.
QMI_ERR_INCOMPATIBLE_STATE	Application indicated in the AID is already in use and is not-shareable.

3.46.3 Description of QMI_UIM_LOGICAL_CHANNEL REQ/RESP

This function opens and closes a logical channel to an application on a UICC card, with the specified AID value. The logical channel is exclusive for the control point and is not shared with any other client. The logical channel can be used to stream APDUs without interfering with the logical channels used by the modem or by other clients. The channel ID can be used in the QMI_UIM_SEND_APDU request to send the APDU on the logical channel.

For security reasons, the modem can be configured to restrict the access to some applications on the UICC card. In this case, the service returns the error code QMI_ERR_ACCESS_DENIED to the control point.

The function returns an error when the specified application does not exist. This function can also be used on an ICC card, if it supports logical channels.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.47 QMI_UIM_SUBSCRIPTION_OK

Enables the client to indicate whether it is OK to proceed with publishing the subscription.

UIM message ID

0x0040

Version introduced

Major - 1, Minor - 9

3.47.1 Request - QMI_UIM_SUBSCRIPTION_OK_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.9	1.30
OK for Subscription	1.9	1.9

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	OK for Subscription
Length	1			2	
Value	→	boolean	ok_for_subscription	1	Indicates whether it is OK to publish the subscription. Valid values: • 0 – Not OK to publish the subscription • 1 – OK to publish the subscription

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Optional TLVs

Name	Version introduced	Version last modified
Encrypted IMSI	1.18	1.18

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted IMSI
Length	Var			2	
Value	→	uint8	encrypted_imsi_len	1	Number of sets of the following elements: • encrypted_imsi
		opaque	encrypted_imsi	Var	Encrypted value of the IMSI of the session. The decrypted value is checked against the IMSI of the card, and the command is executed only if it matches.

3.47.2 Response - QMI_UIM_SUBSCRIPTION_OK_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.9	1.9

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_INVALID_ARG	One of the parameters passed in the request is invalid.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device. This can occur when the device is configured to automatically publish the subscription status.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.

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QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_AUTHENTICATION_ FAILED	Originator of the request could not be authenticated.

3.47.3 Description of QMI_UIM_SUBSCRIPTION_OK REQ/RESP

This function enables the client to indicate whether it is OK to publish the subscription allowing the modem to move from limited service to full service on that subscription. This message is used when the configuration indicates that the modem does not publish the subscription (see QMI_UIM_GET_CONFIGURATION) and the perso state of the subscription is ready indicating the modem has completed its part of the initialization successfully.

If a provisioning session is not supplied, QMI_ERR_INVALID_ARG is returned.

If a provisioning session is supplied but the subscription is inactive or already published, QMI_ERR_INTERNAL is returned.

If a provisioning session is supplied but the modem is not configured to accept this command, QMI_ERR_OP_DEVICE_UNSUPPORTED is returned.

For security reasons, the modem can be configured to accept this request only if the optional encrypted IMSI value is passed as a parameter. This guarantees the authenticity of the requesting client and enables the subscription process to proceed. In this case, if the encrypted IMSI value is missing, QMI_ERR_MISSING_ARG is returned.

The security algorithm and keys used for the encryption are not part of this document.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.48 QMI_UIM_GET_ATR

Retrieves the Answer to Reset of a specific card.

UIM message ID

0x0041

Version introduced

Major - 1, Minor - 10

3.48.1 Request - QMI_UIM_GET_ATR_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.10	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

None

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3.48.2 Response - QMI_UIM_GET_ATR_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.10	1.10

Optional TLVs

Name	Version introduced	Version last modified
ATR Value	1.10	1.10

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	ATR Value
Length	Var			2	
Value	→	uint8	atr_value_len	1	Number of sets of the following elements: • atr_value
		opaque	atr_value	Var	Raw value of the ATR sent by the card during the initialization.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.

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3.48.3 Description of QMI_UIM_GET_ATR REQ/RESP

This function retrieves the ATR value sent by the card when it was initialized and cached in the modem. The value is returned as raw data, as received from the card and coded as indicated in ISO 7816-3. The client is responsible to interpret it.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.49 QMI_UIM_OPEN_LOGICAL_CHANNEL

Opens a logical channel on a UICC card.

UIM message ID

0x0042

Version introduced

Major - 1, Minor - 20

3.49.1 Request - QMI_UIM_OPEN_LOGICAL_CHANNEL_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.20	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

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Optional TLVs

Name	Version introduced	Version last modified
AID	1.20	1.20
Template Requested from the Card in SELECT	1.20	1.20

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	AID
Length	Var			2	
Value	→	uint8	aid_len	1	Number of sets of the following elements: • aid
		opaque	aid	Var	Application identifier value.
Type	0x11			1	Template Requested from the Card in SELECT
Length	1			2	
Value	→	enum8	file_control_information	1	Template requested from the card in the SELECT command when the application is selected. When this TLV is missing, FCP is requested by default. Valid values: • UIM_FILE_CONTROL_ INFORMATION_NO_DATA (0x00) – No data • UIM_FILE_CONTROL_ INFORMATION_FCP (0x01) – FCP • UIM_FILE_CONTROL_ INFORMATION_FCI (0x02) – FCI • UIM_FILE_CONTROL_ INFORMATION_FCI_WITH_ INTERFACES (0x03) – FCI with interfaces • UIM_FILE_CONTROL_ INFORMATION_FMD (0x04) – FMD

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3.49.2 Response - QMI_UIM_OPEN_LOGICAL_CHANNEL_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.20	1.24

Optional TLVs

Name	Version introduced	Version last modified
Channel ID	1.20	1.20
Card Result	1.20	1.20
Response to SELECT	1.20	1.20

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Channel ID
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel that was opened.
Type	0x11			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x12			1	Response to SELECT
Length	Var			2	
Value	→	uint8	select_response_len	1	Number of sets of the following elements: • select_response
		opaque	select_response	Var	Raw value of the response sent by the card to the initial SELECT command.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	One of the parameters passed in the request is invalid.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INSUFFICIENT_ RESOURCES	SIM card does not have sufficient resources to allocate a new logical channel.
QMI_ERR_SIM_FILE_NOT_FOUND	AID specified in the request is not present on the card.
QMI_ERR_ACCESS_DENIED	Requested operation is not allowed for security reasons.
QMI_ERR_INCOMPATIBLE_STATE	Application indicated in the AID is already in use and is not-shareable.

3.49.3 Description of QMI_UIM_OPEN_LOGICAL_CHANNEL REQ/RESP

This function is equivalent to QMI_UIM_LOGICAL_CHANNEL, with additional capabilities.

This function opens a logical channel to an application on a UICC card, with the specified AID value. The logical channel is exclusive for the control point and is not shared with any other client. The logical channel can be used to stream APDUs without interfering with the logical channels used by the modem or by other clients. The channel ID can be used in the QMI_UIM_SEND_APDU request to send the APDU on the logical channel.

If the optional TLV with the AID is missing, the logical channel is created without selecting any application.

For security reasons, the modem can be configured to restrict the access to some applications on the UICC card. In this case, the service returns the error code QMI_ERR_ACCESS_DENIED to the control point.

The function returns an error when the specified application does not exist. This function can also be used on an ICC card, if it supports logical channels.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.50 QMI_UIM_SESSION_CLOSED_IND

Indicates either a provisioning or a nonprovisioning session is closed by the modem.

UIM message ID

0x0043

Version introduced

Major - 1, Minor - 24

3.50.1 Indication - QMI_UIM_SESSION_CLOSED_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.24	1.30

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

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Optional TLVs

Name	Version introduced	Version last modified
Application ID	1.24	1.24
Logical Channel	1.24	1.24
Provisioning Session	1.33	1.33
Session Closure Cause	1.33	1.33
File ID	1.33	1.33

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Application ID
Length	Var			2	
Value	→	uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value.
Type	0x11			1	Logical Channel
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel being closed.
Type	0x12			1	Provisioning Session
Length	1			2	
Value	→	enum8	provisioning_session	1	Indicates the provisioning session type that is closed. Valid values: 0 to 3 and 10 to 15. See Table A-1 for the list of available session types with their values and descriptions.
Type	0x13			1	Session Closure Cause
Length	4			2	
Value	→	enum	cause	4	Indicates the cause of a session closure. See Table A-2 for the causes of session closures with their values and descriptions. Other values are reserved for the future use and are to be handled as an unknown cause by the client.
Type	0x14			1	File ID
Length	2			2	
Value	→	uint16	file_id	2	Indicates the file ID of the mandatory file that is missing, or is present but contains invalid content.

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3.50.2 Description of QMI_UIM_SESSION_CLOSED_IND

This indication is received when either a provisioning or a nonprovisioning session is closed by the modem. In the case of a nonprovisioning session, this can be a result of the internal recovery mechanism when the modem is not able to recover the session.

A control point must register using the QMI_UIM_EVENT_REG request to receive the indication that a provisioning session is closed. In this case, the type of the session closed is also provided in the indication.

In the case of a nonprovisioning session, the event is sent only to the control point that opened the session. If the session is opened through QMI_UIM_LOGICAL_CHANNEL or QMI_UIM_OPEN_LOGICAL_CHANNEL, only the channel ID is provided in the indication. In all other cases, only the application ID is provided.

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3.51 QMI_UIM_REFRESH_REGISTER_ALL

Registers for file change notifications triggered by the card for all files.

UIM message ID

0x0044

Version introduced

Major - 1, Minor - 25

3.51.1 Request - QMI_UIM_REFRESH_REGISTER_ALL_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.25	1.30
Register for Refresh	1.25	1.25

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	Register for Refresh
Length	1			2	
Value	→	boolean	register_for_refresh	1	Indicates whether to register or deregister for refresh indications. Valid values: • 0 – Deregister • 1 – Register

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Optional TLVs

Name	Version introduced	Version last modified
Vote for Initialization	1.25	1.25

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Vote for Initialization
Length	1			2	
Value	→	boolean	vote_for_init	1	Indicates whether to vote for initialization when there is a refresh. When the Vote for Initialization TLV is missing, the default behavior is to not wait for the client vote. Valid values: • 0 – Client does not vote for initialization (Default) • 1 – Client votes for initialization

3.51.2 Response - QMI_UIM_REFRESH_REGISTER_ALL_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.25	1.25

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

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3.51.3 Description of QMI_UIM_REFRESH_REGISTER_ALL REQ/RESP

This function registers for file change notifications for any file triggered by the card.

The client is notified only when any file that belongs to the requested session type is modified by the Refresh procedure. The client does not need to specify a list of files.

This function can be invoked multiple times for each session type. If the function is invoked twice with the same session type, the new values overwrite the previous values.

The client can specify whether it wants to vote for the refresh. If the client indicates that it wants to vote, it receives a QMI_UIM_REFRESH_IND indication before the FCN Refresh procedure is started and is requested to indicate whether it is OK to proceed with the refresh. For more information on the Refresh procedure, refer to 80-VM566-1.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.52 QMI_UIM_SET_FILE_STATUS

Changes the activation status of the file on the card.

UIM message ID

0x0045

Version introduced

Major - 1, Minor - 32

3.52.1 Request - QMI_UIM_SET_FILE_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.32	1.32
File ID	1.32	1.32
File Status	1.32	1.32

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	File Status
Length	4			2	
Value	→	enum	file_status	4	Activates or deactivates the file. Valid values: • UIM_FILE_STATUS_DEACTIVATED (0x00) – File is deactivated. • UIM_FILE_STATUS_ACTIVATED (0x01) – File is activated.

Optional TLVs

None

3.52.2 Response - QMI_UIM_SET_FILE_STATUS_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.32	1.32

Optional TLVs

None

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when there is an attempt to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path or invalid session type).
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.52.3 Description of QMI_UIM_SET_FILE_STATUS REQ/RESP

This command sets the status of any file on the card based on the access path.

This command is applicable only to UICC cards and returns QMI_ERR_NO_EFFECT on ICC cards.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.53 QMI_UIM_SWITCH_SLOT

Switches the binding between a logical slot and a physical slot.

UIM message ID

0x0046

Version introduced

Major - 1, Minor - 34

3.53.1 Request - QMI_UIM_SWITCH_SLOT_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Logical Slot	1.34	1.34
Physical Slot	1.34	1.34

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Logical Slot
Length	1			2	
Value	→	enum8	logical_slot	1	Logical slot that is switched to the physical slot. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	Physical Slot
Length	4			2	
Value	→	enum	physical_slot	4	Physical slot that the logical slot is to be switched to. Valid values: • UIM_PHY_SLOT_1 (0x01) – Slot 1 • UIM_PHY_SLOT_2 (0x02) – Slot 2 • UIM_PHY_SLOT_3 (0x03) – Slot 3 • UIM_PHY_SLOT_4 (0x04) – Slot 4 • UIM_PHY_SLOT_5 (0x05) – Slot 5

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Optional TLVs

None

3.53.2 Response - QMI_UIM_SWITCH_SLOT_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.34	1.34

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NOT_SUPPORTED	Operation is not supported.
QMI_ERR_NO_EFFECT	Operation had no effect.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device. This can occur when the device does not support the slots specified in the request.

3.53.3 Description of QMI_UIM_SWITCH_SLOT REQ/RESP

This command switches (activates) a physical slot for the given logical slot. It deactivates the current active physical slot and activates physical slot in the request (if allowed). The physical slot may or may not have a card in it.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.54 QMI_UIM_GET_SLOTS_STATUS

Retrieves the current status of the physical and logical slots.

UIM message ID

0x0047

Version introduced

Major - 1, Minor - 34

3.54.1 Request - QMI_UIM_GET_SLOTS_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

None

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3.54.2 Response - QMI_UIM_GET_SLOTS_STATUS_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.34	1.34

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Optional TLVs

Name	Version introduced	Version last modified
Physical Slot Status	1.34	1.34

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Physical Slot Status Information regarding the various physical slots on the device. The number of elements correspond with the number of physical slots on the device.
Length	Var			2	
Value	→	uint8	physical_slot_status_len	1	Number of sets of the following elements: • physical_card_status • physical_slot_state • logical_slot • iccid_len • iccid
		enum	physical_card_status	4	Status of the card in the physical slot. Valid values: • UIM_PHYSICAL_CARD_STATE_ UNKNOWN (0x00) – Unknown • UIM_PHYSICAL_CARD_STATE_ ABSENT (0x01) – Absent • UIM_PHYSICAL_CARD_STATE_ PRESENT (0x02) – Present
		enum	physical_slot_state	4	State of the physical slot. Valid values: • UIM_PHYSICAL_SLOT_STATE_ INACTIVE (0x00) – Inactive • UIM_PHYSICAL_SLOT_STATE_ ACTIVE (0x01) – Active
		enum8	logical_slot	1	Logical slot associated with this physical slot. This is valid only if the physical slot is active. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
		uint8	iccid_len	1	Number of sets of the following elements: • iccid
		opaque	iccid	Var	Contains the ICCID of the card in the physical slot.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NOT_SUPPORTED	Operation is not supported.

3.54.3 Description of QMI_UIM_GET_SLOTS_STATUS REQ/RESP

This command gets the mapping of logical slots to physical slots, the current activity status of each physical slot, and additional information, such as the ICCID of the physical slots.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.55 QMI_UIM_SLOT_STATUS_CHANGE_IND

Indicates that the status of the physical slot has changed.

UIM message ID

0x0048

Version introduced

Major - 1, Minor - 34

3.55.1 Indication - QMI_UIM_SLOT_STATUS_CHANGE_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Physical Slot Status	1.34	1.34

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Physical Slot Status
Length	Var			2	
Value	→	uint8	num_physical_slots	1	Number of sets of the following elements: • physical_card_status • physical_slot_state • logical_slot • iccid_len • iccid

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum	physical_card_status	4	Status of the card in the physical slot. Valid values: • UIM_PHYSICAL_CARD_STATE_ UNKNOWN (0x00) – Unknown • UIM_PHYSICAL_CARD_STATE_ ABSENT (0x01) – Absent • UIM_PHYSICAL_CARD_STATE_ PRESENT (0x02) – Present
		enum	physical_slot_state	4	State of the physical slot. Valid values: • UIM_PHYSICAL_SLOT_STATE_ INACTIVE (0x00) – Inactive • UIM_PHYSICAL_SLOT_STATE_ ACTIVE (0x01) – Active
		enum8	logical_slot	1	Logical slot associated with this physical slot. This is valid only if the physical slot is active. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
		uint8	iccid_len	1	Number of sets of the following elements: • iccid
		opaque	iccid	Var	Contains the ICCID of the card in the physical slot.

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3.55.2 Description of QMI_UIM_SLOT_STATUS_CHANGE_IND

This indication communicates that the status of one or more physical slots on the device has changed.

The indication is sent in one of the following cases:

- The physical slot to logical slot mapping has changed as a result of switching the slot
- A card has been removed from, or inserted to, the physical slot
- A physical slot is powered up or down.

The event does not indicate what changed from the previous event. The application is responsible for this.

The event is sent to the application only if it has registered for notifications using the QMI_UIM_EVENT_REG command.

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3.56 QMI_UIM_READ_TRANSPARENT_LONG_IND

Indication with a read transparent command result.

UIM message ID

0x0049

Version introduced

Major - 1, Minor - 35

3.56.1 Indication - QMI_UIM_READ_TRANSPARENT_LONG_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Chunk of Read Transparent Data	1.35	1.35

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Chunk of Read Transparent Data
Length	Var			2	
Value	→	uint32	long_data_token	4	Unique long data token used in the previous QMI_UIM_READ_TRANSPARENT response or indication.
		uint16	total_length	2	Total length of the read data.
		uint16	offset	2	Offset of the current chunk (starting from 0).
		uint16	content_len	2	Number of sets of the following elements: • content
		uint8	content	Var	Read content; sequence of bytes as read from the card.

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Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.35	1.35
Encrypted Data	1.35	1.35

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Encrypted Data
Length	1			2	
Value	→	boolean	encryption	1	Indicates whether the data from the card passed in read_result is encrypted. Valid values: • 0 – Data in read_result is unencrypted • 1 – Data in read_result is encrypted If the Encrypted Data TLV is missing, the read_result data is in unencrypted format.

3.56.2 Description of QMI_UIM_READ_TRANSPARENT_LONG_IND

This indication is sent when a read transparent request is completed and the service determines that the requested data is too long to be sent in the response. Therefore, QMI_ERR_INSUFFICIENT_RESOURCES is returned in the response along with the File Length TLV and Long Data Token TLV.

Subsequently, the entire read result data is sent in one or more chunks, with each chunk sent in a separate indication. The client is responsible for reconstructing the entire response data, combining all of the indications received.

If the client requested encrypted response data, the data in the read result is in encrypted format. This procedure guarantees protection against a man-in-the-middle attack between the service and the client. The security algorithm and keys used for the encryption are not a part of this document. If the service does not support the encryption, it might provide an unencrypted response.

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3.57 QMI_UIM_SIM_BUSY_STATUS_IND

Indicates the busy status of a SIM card.

UIM message ID

0x004A

Version introduced

Major - 1, Minor - 35

3.57.1 Indication - QMI_UIM_SIM_BUSY_STATUS_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
SIM Busy Status	1.35	1.35

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	SIM Busy Status
Length	Var			2	
Value	→	uint8	sim_busy_status_len	1	Number of sets of the following elements: • sim_busy_status
		boolean	sim_busy_status	Var	Indicates whether the card on the corresponding slot is busy. Valid values: • 0 – SIM card is not busy • 1 – SIM card is busy

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3.57.2 Description of QMI_UIM_SIM_BUSY_STATUS_IND

This indication communicates the busy status of all the available SIM cards.

The indication is sent when the modem detects that the SIM card has entered or exited the busy state. The busy state occurs when the card cannot accept any command from the modem for a variable time duration. This indication informs the client about the busy state so the client can take the necessary action.

The event is sent to the client only if it has registered for SIM busy notifications using the QMI_UIM_EVENT_REG command.

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3.58 QMI_UIM_GET_PLMN_NAME_TABLE_INFO

Retrieves information about the SE.13 table used by the modem.

UIM message ID

0x004B

Version introduced

Major - 1, Minor - 35

3.58.1 Request - QMI_UIM_GET_PLMN_NAME_TABLE_INFO_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

None

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3.58.2 Response - QMI_UIM_GET_PLMN_NAME_TABLE_INFO_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.35	1.35

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Optional TLVs

Name	Version introduced	Version last modified
Release Date	1.35	1.35
Table Source	1.35	1.35

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Release Date Information regarding the release date of the SE.13 table database used by the modem.
Length	4			2	
Value	→	uint8	month	1	Month.
		uint8	day	1	Day of the month.
		uint16	year	2	Year.
Type	0x11			1	Table Source
Length	4			2	
Value	→	enum	table_source	4	Information regarding the source of the used SE.13 table database. Valid values: • UIM_PLMN_NAME_TABLE_ SOURCE_HARDCODED (0x01) – Table is hardcoded in the modem source code • UIM_PLMN_NAME_TABLE_ SOURCE_EFS (0x02) – Table is fetched from the modem file system Other values are reserved for future use and are to be ignored by the client.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NOT_SUPPORTED	PLMN table was not found in the file system or in the source code.

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3.58.3 Description of QMI_UIM_GET_PLMN_NAME_TABLE_INFO REQ/RESP

This command gets the information about the SE.13 table database that is currently supported by the EONS algorithm in the modem. Information includes the GSMA SE.13 table release date and the SE.13 table source used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.59 QMI_UIM_PERSONALIZATION

Activates and sets the personalization data on the phone.

UIM message ID

0x004C

Version introduced

Major - 1, Minor - 36

3.59.1 Request - QMI_UIM_PERSONALIZATION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Control Key	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Control Key
Length	Var			2	
Value	→	uint8	ck_value_len	1	Number of sets of the following elements: • ck_value
		char	ck_value	Var	Control key value. This value is a sequence of ASCII characters.

Optional TLVs

Name	Version introduced	Version last modified
GW Network Personalization	1.36	1.36
GW Network Subset Personalization	1.36	1.36
GW Service Provider Personalization	1.36	1.36
GW Corporate Personalization	1.36	1.36
GW SIM Personalization	1.36	1.36
1X Network Type 1 Personalization	1.36	1.36
1X Network Type 2 Personalization	1.36	1.36
1X RUIM Personalization	1.36	1.36

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Name	Version introduced	Version last modified
Slot	1.40	1.40
Number of Iterations	1.40	1.40
Number of Retries	1.40	1.40
Reactivate Personalization Feature	1.49	1.49

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	GW Network Personalization
Length	Var			2	
Value	→	uint8	feature_gw_network_ perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.
Type	0x11			1	GW Network Subset Personalization
Length	Var			2	
Value	→	uint8	feature_gw_network_ subset_perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc • digit6 • digit7
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.
		char	digit6	1	Digit 6 of the IMSI in ASCII characters.
		char	digit7	1	Digit 7 of the IMSI in ASCII characters.
Type	0x12			1	GW Service Provider Personalization
Length	Var			2	
Value	→	uint8	feature_gw_sp_perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc • gid1
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	gid1	1	Service provider code found in GID1.
Type	0x13			1	GW Corporate Personalization
Length	Var			2	
Value	→	uint8	feature_gw_corporate_ perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc • gid1 • gid2
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.
		uint8	gid1	1	Service provider code found in GID1.
		uint8	gid2	1	Corporate customer code found in GID2.
Type	0x14			1	GW SIM Personalization
Length	Var			2	
Value	→	uint8	feature_gw_sim_perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc • msin_len • msin
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.
		uint8	msin_len	1	Number of sets of the following elements: • msin
		char	msin	Var	MSIN value stored on the card in ASCII characters.
Type	0x15			1	1X Network Type 1 Personalization
Length	Var			2	
Value	→	uint8	feature_1x_network1_ perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc

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Field	Field value	Field type	Parameter	Size (byte)	Description
		char	mnc	Var	MNC value in ASCII characters.
Type	0x16			1	1X Network Type 2 Personalization
Length	Var			2	
Value	→	uint8	feature_1x_network2_ perso_len	1	Number of sets of the following elements: • irm_code
		char	irm_code	4	First four digits of the IRM-based MIN of IMSI_M in ASCII characters.
Type	0x17			1	1X RUIM Personalization
Length	Var			2	
Value	→	uint8	feature_1x_ruim_ perso_len	1	Number of sets of the following elements: • mcc • mnc_len • mnc • msin_len • msin
		char	mcc	3	MCC value in ASCII characters.
		uint8	mnc_len	1	Number of sets of the following elements: • mnc
		char	mnc	Var	MNC value in ASCII characters.
		uint8	msin_len	1	Number of sets of the following elements: • msin
		char	msin	Var	MSIN value stored on the card in ASCII characters.
Type	0x18			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x19			1	Number of Iterations
Length	4			2	
Value	→	uint32	num_iterations	4	The number of iterations to be used in the personalization algorithm.
Type	0x1A			1	Number of Retries
Length	4			2	
Value	→	uint32	num_retries	4	The number of retries allowed for depersonalization. Setting the value to 0 allows unlimited retries.
Type	0x1B			1	Reactivate Personalization Feature
Length	1			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum8	reactivate_feature	1	Indicates the personalization feature to reactivate. Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization Other values are reserved for future use and are to be ignored by the client.

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3.59.2 Response - QMI_UIM_PERSONALIZATION_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.36	1.45

Optional TLVs

Name	Version introduced	Version last modified
Retries Remaining	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Retries Remaining This value is returned only when re-activating the personalization feature.
Length	2			2	
Value	→	uint8	verify_left	1	Number of the remaining attempts to verify the personalization.
		uint8	unblock_left	1	Number of the remaining attempts to unblock the personalization.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_INCORRECT_PIN	Control key in the request is incorrect.
QMI_ERR_PIN_BLOCKED	Simlock feature is blocked. An unblock operation must be issued.
QMI_ERR_PIN_PERM_BLOCKED	Simlock feature is permanently blocked. The UE is unusable.
QMI_ERR_NO_SIM	Action is being performed on a SIM that is not initialized.

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QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_INCOMPATIBLE_STATE	Personalization is already activated and must be deactivated before reprogramming personalization data.
QMI_ERR_OP_PARTIAL_FAILURE	Some personalization codes were set but an error prevented additional processing.

3.59.3 Description of QMI_UIM_PERSONALIZATION REQ/RESP

This command sets the data and then activates personalization on the device, as described in 3GPP TS 22.022. Each feature can be activated independently of one another. However, network data configurations must be consistent across activated personalization modes to prevent contradicting featurization, and only one feature can be activated per message (one optional TLV per message).

If personalization is already activated, personalization must be deactivated before being reactivated with new data.

If the number of iterations is missing, the modem internally chooses an appropriate number.

If the number of retries is present, the modem might use it to determine the maximum number of retries allowed for depersonalization. If the number of retries is missing, the modem chooses the number of retries based on the internal support.

The Reactivate Personalization feature TLV can be used to reactivate the feature that was previously activated by passing the same control key. In this case, the client must not pass the corresponding data TLV. The data passed in the previous activation request is used for reactivation.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.60 QMI_UIM_INCREASE

Performs an increase operation for any file on the card and provides access by the path.

UIM message ID

0x004D

Version introduced

Major - 1, Minor - 36

3.60.1 Request - QMI_UIM_INCREASE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.36	1.36
File ID	1.36	1.36
Value to be Added	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Value to be Added
Length	Var			2	
Value	→	uint8	increase_data_len	1	Number of sets of the following elements: • increase_data
		opaque	increase_data	Var	Increase data.

Optional TLVs

Name	Version introduced	Version last modified
Response in Indication	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.60.2 Response - QMI_UIM_INCREASE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.36	1.36

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Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.36	1.36
Value of Increase Record	1.36	1.36
Response in Indication	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Value of Increase Record
Length	Var			2	
Value	→	uint8	increase_result_len	1	Number of sets of the following elements: • increase_result
		opaque	increase_result	Var	Value of the increased record result and the value that has been added.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

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3.60.3 Description of QMI_UIM_INCREASE REQ/RESP

This command provides increase access to a specific file in the card and provides access by the path.

The response contains the status code received from the card (SW1 and SW2) when the card responded to the increase request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_INCREASE_IND indication. In this case, the immediate response indicates only whether the command was received correctly for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

A client must be able to accommodate a case in which the optional indication is not supported by the service and the return of the complete result in the response, even when the request indicates that the indication is to be used.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.61 QMI_UIM_INCREASE_IND

Indicates the clients with an increase confirmation.

UIM message ID

0x004D

Version introduced

Major - 1, Minor - 36

3.61.1 Indication - QMI_UIM_INCREASE_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.36	1.36
Token of the Original Request	1.36	1.36

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.36	1.36
Value of Increase Record	1.36	1.36

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Value of Increase Record
Length	Var			2	
Value	→	uint8	increase_result_len	1	Number of sets of the following elements: • increase_result
		opaque	increase_result	Var	Value of the increased record result and the value that has been added.

3.61.2 Description of QMI_UIM_INCREASE_IND

This indication is received when an increase request is completed and a subsequent indication is requested for the result.

The standard response in the indication is the same as the response a client receives when it requests the result. The complete list of error codes is the same as QMI_UIM_INCREASE_RESP.

When the client requests the data to be passed in a subsequent indication, it must be able to accommodate a case in which the optional indication is not supported by the service and the complete result is returned in the response.

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3.62 QMI_UIM_RECOVERY

Performs recovery on an identified slot.

UIM message ID

0x004E

Version introduced

Major - 1, Minor - 38

3.62.1 Request - QMI_UIM_RECOVERY_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.38	1.38

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

None

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3.62.2 Response - QMI_UIM_RECOVERY_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.38	1.38

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_NO_EFFECT	Operation had no effect.

3.62.3 Description of QMI_UIM_RECOVERY REQ/RESP

This command triggers the recovery process within UIM, restoring an identified slot after a reset.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.63 QMI_UIM_RESELECT

Performs a SELECT of an application on the specified logical channel.

UIM message ID

0x004F

Version introduced

Major - 1, Minor - 39

3.63.1 Request - QMI_UIM_RESELECT_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.39	1.39
Channel ID	1.39	1.39
Selection Mode	1.39	1.39

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot to be used. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	Channel ID
Length	1			2	
Value	→	uint8	channel_id	1	ID of the logical channel on which to perform the SELECT.
Type	0x03			1	Selection Mode
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum	select_mode	4	SELECT command mode. Valid values: • UIM_SELECT_MODE_FIRST_ OR_ONLY (0x00) – First or only occurrence of the AID • UIM_SELECT_MODE_LAST (0x01) – Last occurrence of the AID • UIM_SELECT_MODE_NEXT (0x02) – Next occurrence of the AID • UIM_SELECT_MODE_PREVIOUS (0x03) – Previous occurrence of the AID

Optional TLVs

None

3.63.2 Response - QMI_UIM_RESELECT_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.39	1.39

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.39	1.39
Response to SELECT	1.39	1.39

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Response to SELECT
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	uint8	select_response_len	1	Number of sets of the following elements: • select_response
		opaque	select_response	Var	Raw value of the response sent by the card to the SELECT command.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an invalid SELECT mode).
QMI_ERR_NOT_SUPPORTED	Operation is not supported.
QMI_ERR_ACCESS_DENIED	Functionality is not supported by the card.
QMI_ERR_SIM_FILE_NOT_FOUND	Application is not found on the card.

3.63.3 Description of QMI_UIM_RESELECT REQ/RESP

This command selects an application on a logical channel that was opened by the control point. The AID selected is the one that matches the partial AID provided in a previous QMI_UIM_LOGICAL_CHANNEL or QMI_UIM_OPEN_LOGICAL_CHANNEL command.

The matching AID is found using the selection mode specified in the command (i.e, first, last, next, or previous AID) that matches the partial AID provided when opening the logical channel.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.64 QMI_UIM_RECOVERY_IND

Indicates that a SIM recovery completed successfully.

UIM message ID

0x0050

Version introduced

Major - 1, Minor - 40

3.64.1 Indication - QMI_UIM_RECOVERY_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.40	1.40

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the recovered slot. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

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Optional TLVs

None

3.64.2 Description of QMI_UIM_RECOVERY_IND

This indication is sent after the modem successfully recovers the SIM card. Recovery occurs in the following conditions:

- Physical errors, e.g., Rx breaks or parity errors
- Other unexpected behavior, e.g., command response timeouts

The event is only sent to the client if it has been registered for Recovery Complete notifications using the QMI_UIM_EVENT_REG command.

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3.65 QMI_UIM_SEND_STATUS

Sends a STATUS command.

UIM message ID

0x0051

Version introduced

Major - 1, Minor - 40

3.65.1 Request - QMI_UIM_SEND_STATUS_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.40	1.40

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.

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Optional TLVs

Name	Version introduced	Version last modified
Status Command Mode	1.40	1.40
Response Requested	1.40	1.40

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Status Command Mode
Length	4			2	
Value	→	enum	status_cmd_mode	4	Mode for the STATUS command. Valid values: • UIM_STATUS_CMD_NO_ INDICATION (0x00) – No indication • UIM_STATUS_CMD_APP_ INITIALIZED (0x01) – Current application is initialized in the terminal • UIM_STATUS_CMD_WILL_ TERMINATE_APP (0x02) – Terminal initiates the termination of the current application This value defaults to UIM_STATUS_CMD_NO_ INDICATION if the TLV is missing.
Type	0x11			1	Response Requested
Length	4			2	
Value	→	enum	status_cmd_resp	4	Response required for the STATUS command. Valid values: • UIM_STATUS_CMD_FCP_RESP (0x00) – Response is identical to the SELECT command, i.e., FCP • UIM_STATUS_CMD_AID_RESP (0x01) – DF name TLV-object (AID) of the current application is returned • UIM_STATUS_CMD_NO_DATA_ RESP (0x02) – No data is returned This value defaults to UIM_STATUS_CMD_NO_DATA_RESP if the TLV is missing.

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3.65.2 Response - QMI_UIM_SEND_STATUS_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.40	1.40

Optional TLVs

Name	Version introduced	Version last modified
STATUS Response	1.40	1.40
Card Result	1.41	1.41

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	STATUS Response
Length	Var			2	
Value	→	uint8	status_response_len	1	Number of sets of the following elements: • status_response
		opaque	status_response	Var	Depending on the parameters in the request, this value is either identical to the SELECT command response or it is the AID.
Type	0x11			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Session type passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

3.65.3 Description of QMI_UIM_SEND_STATUS REQ/RESP

This command sends a STATUS command to the card. Per the specification, the command can be sent as follows:

- With no indication
- The terminal is to indicate that the current application is initialized in the terminal
- The terminal is to initiate the termination of the current application

The caller can dictate the following:

- No data is to be returned
- The response parameters are to be identical to the SELECT command (i.e., FCP)
- The AID of the current application can be returned

Error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.66 QMI_UIM_GET_SIM_PROFILE

Queries the profile information from the card.

UIM message ID

0x0052

Version introduced

Major - 1, Minor - 42

3.66.1 Request - QMI_UIM_GET_SIM_PROFILE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.42	1.42

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the recovered slot. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

None

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3.66.2 Response - QMI_UIM_GET_SIM_PROFILE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.42	1.42

Optional TLVs

Name	Version introduced	Version last modified
Profile information	1.42	1.42
Present active profile information	1.42	1.42

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Profile information
Length	Var			2	
Value	→	uint8	profile_info_len	1	Number of sets of the following elements: • profile_id • profile_type
		enum	profile_id	4	Indicates the profile ID of the card. Valid values: • UIM_PROFILE_ID_1 (0x01) – Profile 1 • UIM_PROFILE_ID_2 (0x02) – Profile 2 • UIM_PROFILE_ID_3 (0x03) – Profile 3 • UIM_PROFILE_ID_4 (0x04) – Profile 4 • UIM_PROFILE_ID_5 (0x05) – Profile 5 • UIM_PROFILE_ID_6 (0x06) – Profile 6 • UIM_PROFILE_ID_7 (0x07) – Profile 7 • UIM_PROFILE_ID_8 (0x08) – Profile 8

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Field	Field value	Field type	Parameter	Size (byte)	Description
		enum	profile_type	4	Indicates the profile type of the card. Valid values: • UIM_PROFILE_TYPE_REGULAR (0x00) – Regular • UIM_PROFILE_TYPE_EMERGENCY (0x01) – Emergency
Type	0x11			1	Present active profile information
Length	8			2	
Value	→	enum	profile_id	4	Indicates the profile ID of the card. Valid values: • UIM_PROFILE_ID_1 (0x01) – Profile 1 • UIM_PROFILE_ID_2 (0x02) – Profile 2 • UIM_PROFILE_ID_3 (0x03) – Profile 3 • UIM_PROFILE_ID_4 (0x04) – Profile 4 • UIM_PROFILE_ID_5 (0x05) – Profile 5 • UIM_PROFILE_ID_6 (0x06) – Profile 6 • UIM_PROFILE_ID_7 (0x07) – Profile 7 • UIM_PROFILE_ID_8 (0x08) – Profile 8
		enum	profile_type	4	Indicates the profile type of the card. Valid values: • UIM_PROFILE_TYPE_REGULAR (0x00) – Regular • UIM_PROFILE_TYPE_EMERGENCY (0x01) – Emergency

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Slot ID passed in the request is invalid.
QMI_ERR_NOT_SUPPORTED	Functionality is not supported by the card or by the terminal.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.

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3.66.3 Description of QMI_UIM_GET_SIM_PROFILE REQ/RESP

This command queries the profiles supported by the card. The response contains a list of profiles supported by the card and the profile information of the currently active profile.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.67 QMI_UIM_SET_SIM_PROFILE

Switches the profile on the SIM card.

UIM message ID

0x0053

Version introduced

Major - 1, Minor - 42

3.67.1 Request - QMI_UIM_SET_SIM_PROFILE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.42	1.42
Profile ID	1.42	1.42

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the recovered slot. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	Profile ID
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum	profile_id	4	Indicates the profile ID to switch. Valid values: • UIM_PROFILE_ID_1 (0x01) – Profile 1 • UIM_PROFILE_ID_2 (0x02) – Profile 2 • UIM_PROFILE_ID_3 (0x03) – Profile 3 • UIM_PROFILE_ID_4 (0x04) – Profile 4 • UIM_PROFILE_ID_5 (0x05) – Profile 5 • UIM_PROFILE_ID_6 (0x06) – Profile 6 • UIM_PROFILE_ID_7 (0x07) – Profile 7 • UIM_PROFILE_ID_8 (0x08) – Profile 8

Optional TLVs

None

3.67.2 Response - QMI_UIM_SET_SIM_PROFILE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.42	1.42

Optional TLVs

None

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Slot ID passed in the request is invalid.
QMI_ERR_NOT_SUPPORTED	Functionality is not supported by the card or by the terminal.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NO_EFFECT	Operation had no effect. This can occur when the client tries to switch to a profile that is already active.

3.67.3 Description of QMI_UIM_SET_SIM_PROFILE REQ/RESP

This command allows clients to switch to other profiles present in the card.

If the profile switch is successful, a successful response returns.

If the profile ID provided in the TLV is same as the profile that is activated on the card, the error QMI_ERR_NO_EFFECT is returned.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.68 QMI_UIM_SUPPLY_VOLTAGE

Informs the modem to proceed with the supply voltage Vcc deactivation.

UIM message ID

0x0054

Version introduced

Major - 1, Minor - 43

3.68.1 Request - QMI_UIM_SUPPLY_VOLTAGE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.43	1.43

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot where the supply voltage Vcc is to be deactivated. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

Optional TLVs

None

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3.68.2 Response - QMI_UIM_SUPPLY_VOLTAGE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.43	1.43

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready. This can occur if the modem is not deactivating the Vcc line.

3.68.3 Description of QMI_UIM_SUPPLY_VOLTAGE REQ/RESP

This command indicates that it is alright to deactivate the supply voltage Vcc line of the UICC. This command can only be used after receiving a QMI_UIM_SUPPLY_VOLTAGE_IND message for deactivation from the modem.

The command allows an external client to perform additional operations that might be required during the deactivation of the UICC. For example, an external CLF can deactivate the SWP line.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.69 QMI_UIM_SUPPLY_VOLTAGE_IND

Indicates that the modem must deactivate or activate the supply voltage Vcc line of the UICC.

UIM message ID

0x0054

Version introduced

Major - 1, Minor - 43

3.69.1 Indication - QMI_UIM_SUPPLY_VOLTAGE_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.43	1.43
Vcc Command Type	1.43	1.43

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot where the supply voltage Vcc line must be activated or deactivated. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	Vcc Command Type
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum	vcc_command	4	Indicates whether supply voltage Vcc is to be deactivated or activated. Valid values: • UIM_SUPPLY_VOLTAGE_ACTIVATE (0x00) – Supply voltage Vcc is activated • UIM_SUPPLY_VOLTAGE_ DEACTIVATE (0x01) – Supply voltage Vcc must be deactivated

Optional TLVs

None

3.69.2 Description of QMI_UIM_SUPPLY_VOLTAGE_IND

This indication is sent by the modem when it must deactivate the supply voltage Vcc line of a UICC, regardless of the reason. It might occur as part of a normal power-down of the UICC or during a recovery procedure. After receiving this indication, the client can perform any required action, such as deactivating the SWP line of the UICC. Then the client invokes QMI_UIM_SUPPLY_VOLTAGE_REQ to indicate to the modem to proceed with the supply voltage Vcc deactivation.

This indication is also sent by the modem when it activates the supply voltage Vcc line of the UICC. Unlike deactivation, after the Vcc line is activated, this indication is sent and subsequently QMI UIM proceeds with the initialization without waiting for acknowledgement of the indication.

This indication is only received if a client has registered for the event via the QMI_UIM_EVENT_REG command.

The deactivation or activation of other UICC lines controlled by the modem can occur before or after this indication is sent.

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3.70 QMI_UIM_CARD_ACTIVATION_STATUS_IND

Indicates the card activation status.

UIM message ID

0x0055

Version introduced

Major - 1, Minor - 44

3.70.1 Indication - QMI_UIM_CARD_ACTIVATION_STATUS_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Slot	1.44	1.44
Card Activation Status	1.44	1.44

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Slot
Length	1			2	
Value	→	enum8	slot	1	Indicates the slot where card activation has started or ended. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5
Type	0x02			1	Card Activation Status
Length	4			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	enum	card_activation_status	4	Indicates the card activation status on the corresponding slot. Valid values: • CARD_ACTIVATION_START (0x00) – Card activation started • CARD_ACTIVATION_END_WITH_SUCCESS (0x01) – Card activation ended successfully • CARD_ACTIVATION_END_WITH_FAILURE (0x02) – Card activation failed

Optional TLVs

None

3.70.2 Description of QMI_UIM_CARD_ACTIVATION_STATUS_IND

This indication communicates the card activation status on a specific slot.

The indication is sent when the modem detects that card activation has started or ended. For example, card activation can be triggered using over-the-air service provisioning. This indication informs the client about the activation status so the client can take the necessary action.

The event is only sent to the client if it has registered for card activation notifications using the QMI_UIM_EVENT_REG command.

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3.71 QMI_UIM_DEPERSONALIZATION_SECURE

Performs a temporary unlock or unlock without the control key.

UIM message ID

0x0056

Version introduced

Major - 1, Minor - 45

3.71.1 Request - QMI_UIM_DEPERSONALIZATION_SECURE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Encrypted Depersonalization Request Data	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Encrypted Depersonalization Request Data
Length	Var			2	
Value	→	uint16	encrypted_ depersonalization_data_ len	2	Number of sets of the following elements: • encrypted_depersonalization_data
		opaque	encrypted_ depersonalization_data	Var	See Section D.1 for the request format.

Optional TLVs

None

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3.71.2 Response - QMI_UIM_DEPERSONALIZATION_SECURE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.45	1.45

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Result Code	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Result Code
Length	Var			2	
Value	→	uint16	encrypted_resp_len	2	Number of sets of the following elements: • encrypted_resp
		opaque	encrypted_resp	Var	See Section D.6 for the response format.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device.
QMI_ERR_NO_EFFECT	Operation had no effect. This can occur if depersonalization is attempted on a feature which is not personalized.
QMI_ERR_AUTHENTICATION_ FAILED	Originator of the request could not be authenticated.

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3.71.3 Description of QMI_UIM_DEPERSONALIZATION_SECURE REQ/RESP

This command deactivates a personalization feature on the terminal without a control key or enables a temporary deactivation of a personalization feature on the terminal for a specified duration. This command can also disable temporary deactivation of a personalization feature.

For security reasons, the modem is configured to accept this request in encrypted format only. This guarantees the authenticity of the requesting client and enables the deactivation of personalization or temporarily setting or unsetting personalization on the terminal.

The result in the standard response indicates whether the encrypted response is constructed correctly.

The security algorithm and keys used for the encryption are not in the scope of this document.

Error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.72 QMI_UIM_PERSONALIZATION_SECURE

Activates and locks the device to simlock codes.

UIM message ID

0x0057

Version introduced

Major - 1, Minor - 45

3.72.1 Request - QMI_UIM_PERSONALIZATION_SECURE_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Encrypted Personalization Data	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Encrypted Personalization Data
Length	Var			2	
Value	→	uint16	encrypted_ personalization_data_len	2	Number of sets of the following elements: • encrypted_personalization_data
		opaque	encrypted_ personalization_data	Var	See Section D.2 for the request format.

Optional TLVs

None

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3.72.2 Response - QMI_UIM_PERSONALIZATION_SECURE_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.45	1.45

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Result Code	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Result Code
Length	Var			2	
Value	→	uint16	encrypted_resp_len	2	Number of sets of the following elements: • encrypted_resp
		opaque	encrypted_resp	Var	See Section D.6 for the response format.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Device cannot handle the PIN length in the request.
QMI_ERR_OP_DEVICE_ UNSUPPORTED	Operation is not supported by the device.
QMI_ERR_MISSING_ARG	TLV was missing in the request.
QMI_ERR_INCOMPATIBLE_STATE	Personalization is already activated and must be deactivated before reprogramming personalization data.
QMI_ERR_AUTHENTICATION_ FAILED	Originator of the request could not be authenticated.
QMI_ERR_OP_PARTIAL_FAILURE	Some personalization codes were set but an error prevented additional processing.

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3.72.3 Description of QMI_UIM_PERSONALIZATION_SECURE REQ/RESP

This command activates personalization on the device without a control key. All data in the request message must be encrypted before being populated. Once the encrypted message has been sent, the data is decrypted and processed according to the formats specified in Section D.6.

If personalization is already activated, personalization must be deactivated to reactivate with the new data.

For security reasons, the modem is configured to accept this request in the encrypted format only. This guarantees the authenticity of the requesting client and enables the activation of personalization.

The result in the standard response indicates whether the encrypted response is constructed correctly.

The security algorithm and keys used for the encryption are not in the scope of this document.

Error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.73 QMI_UIM_EMERGENCY_ONLY

Sets the phone in emergency-only mode.

UIM message ID

0x0058

Version introduced

Major - 1, Minor - 45

3.73.1 Request - QMI_UIM_EMERGENCY_ONLY_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Encrypted Emergency Only Request Data	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Encrypted Emergency Only Request Data
Length	Var			2	
Value	→	uint16	encrypted_emergency_data_len	2	Number of sets of the following elements: • encrypted_emergency_data
		opaque	encrypted_emergency_data	Var	See Section D.3 for the request format.

Optional TLVs

None

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3.73.2 Response - QMI_UIM_EMERGENCY_ONLY_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.45	1.45

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Result Code	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Result Code
Length	Var			2	
Value	→	uint16	encrypted_resp_len	2	Number of sets of the following elements: • encrypted_resp
		opaque	encrypted_resp	Var	See Section D.6 for the response format.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.
QMI_ERR_NO_EFFECT	Operation had no effect. This can occur if emergency-only mode is requested when the terminal is already in that mode.
QMI_ERR_AUTHENTICATION_ FAILED	Originator of the request could not be authenticated.

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3.73.3 Description of QMI_UIM_EMERGENCY_ONLY REQ/RESP

This command sets the emergency-only mode or the normal operation mode.

For security reasons, the modem is configured to accept this request in the encrypted format only. This guarantees the authenticity of the requesting client and enables setting or unsetting the emergency-only mode.

The result in the standard response indicates whether the encrypted response is constructed correctly.

The security algorithm and keys used for the encryption are not in the scope of this document.

Error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.74 QMI_UIM_SIMLOCK_CONFIGURATION

Provides access to the simlock configuration data.

UIM message ID

0x0059

Version introduced

Major - 1, Minor - 45

3.74.1 Request - QMI_UIM_SIMLOCK_CONFIGURATION_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Encrypted Simlock Configuration Request Data	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Encrypted Simlock Configuration Request Data
Length	Var			2	
Value	→	uint16	encrypted_simlock_ config_data_len	2	Number of sets of the following elements: • encrypted_simlock_config_data
		opaque	encrypted_simlock_ config_data	Var	See Section D.4.1 for the request format.

Optional TLVs

None

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3.74.2 Response - QMI_UIM_SIMLOCK_CONFIGURATION_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.45	1.45

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Result Code	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Result Code
Length	Var			2	
Value	→	uint16	encrypted_resp_len	2	Number of sets of the following elements: • encrypted_resp
		opaque	encrypted_resp	Var	See Section D.6 for the response format.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_INVALID_OPERATION	Device is not expecting the request.
QMI_ERR_AUTHENTICATION_FAILED	Originator of the request could not be authenticated.

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3.74.3 Description of QMI_UIM_SIMLOCK_CONFIGURATION REQ/RESP

This command enables or disables the storage of simlock configuration data in external secure storage. This command also retrieves or stores the simlock configuration data required by the modem in external secure storage. Additionally, this command provides the write request result issued by the modem to store simlock configuration data in external secure storage.

At least one client must be registered for the remote simlock configuration event via QMI_UIM_EVENT_REG to enable or disable simlock configuration data in external secure storage.

For security reasons, the modem is configured to accept this request in encrypted format only. This guarantees the authenticity of the requesting client.

The result in the standard response indicates whether the encrypted response is constructed correctly.

The security algorithm and keys used for the encryption are not in the scope of this document.

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3.75 QMI_UIM_SIMLOCK_CONFIGURATION_IND

Indication with the simlock configuration data access request.

UIM message ID

0x0059

Version introduced

Major - 1, Minor - 45

3.75.1 Indication - QMI_UIM_SIMLOCK_CONFIGURATION_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Encrypted Value: Simlock Configuration Request	1.45	1.45

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Encrypted Value: Simlock Configuration Request
Length	Var			2	
Value	→	uint16	encrypted_simlock_ config_ind_len	2	Number of sets of the following elements: • encrypted_simlock_config_ind
		opaque	encrypted_simlock_ config_ind	Var	See Section D.4.2 for the indication format.

Optional TLVs

None

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3.75.2 Description of QMI_UIM_SIMLOCK_CONFIGURATION_IND

This indication is received by an application when the modem must access the simlock configuration data from secure storage.

For read simlock configuration data requests, the modem sends the indication with the simlock configuration file to read the data (see Section D.4.1). Upon receiving the indication, the application is expected to provide the data requested, using the simlock configuration request message.

For write simlock configuration data requests, the modem sends the indication with the data to be written to the simlock configuration file and the simlock configuration file to write (see Section D.4.2). Upon receiving the indication, the application is expected to store the simlock configuration data in secure storage and send the result of the write request to the modem using the simlock configuration request message.

See Section D.4 for details regarding the indication, simlock configuration request, and response actions that the modem and application must perform.

For security reasons, the indication payload is in encrypted format only. This guarantees the authenticity of the client sending the indication.

The security algorithm and keys used for the encryption are not in the scope of this document.

This indication is received only if a client has registered for the event using QMI_UIM_EVENT_REG request.

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3.76 QMI_UIM_GBA

Initializes the GBA and generates a key that is shared with a NAF.

UIM message ID

0x005A

Version introduced

Major - 1, Minor - 47

3.76.1 Request - QMI_UIM_GBA_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.47	1.47
NAF ID	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	NAF ID
Length	Var			2	
Value	→	uint8	fqdn_len	1	Number of sets of the following elements: • fqdn

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Field	Field value	Field type	Parameter	Size (byte)	Description
		char	fqdn	Var	FQDN of the NAF.
		opaque	security_protocol_id	5	Identifies the security protocol of the bootstrapping application per 3GPP TS 33.220 Annex H.

Optional TLVs

Name	Version introduced	Version last modified
Force Bootstrapping	1.47	1.47
Indication Token	1.47	1.47
APN Name	1.51	1.51

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Force Bootstrapping
Length	1			2	
Value	→	boolean	force_bootstrapping	1	Indicates whether bootstrapping is required even if the key is still valid. Valid values: • 0 – Do not force bootstrap and reuse Ks if it is still valid • 1 – Force bootstrapping even if the Ks is still valid
Type	0x11			1	Indication Token
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.
Type	0x12			1	APN Name
Length	Var			2	
Value	→	uint8	apn_name_len	1	Number of sets of the following elements: • apn_name
		char	apn_name	Var	Identifies the APN name required for bootstrapping.

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3.76.2 Response - QMI_UIM_GBA_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.47	1.47

Optional TLVs

Name	Version introduced	Version last modified
Ks_NAF	1.47	1.47
Encrypted Ks_NAF	1.47	1.47
Bootstrapping Transaction ID	1.47	1.47
Key material lifetime	1.47	1.47
IMPI	1.47	1.47
Response in Indication	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Ks(_ext)_NAF
Length	32			2	
Value	→	opaque	ks_naf	32	Key shared between the NAF and the ME. Derivation of the Key is specified in 3GPP TS 33.220.
Type	0x11			1	Encrypted Ks(_ext)_NAF
Length	Var			2	
Value	→	uint16	encrypted_ks_naf_len	2	Number of sets of the following elements: • encrypted_ks_naf
		opaque	encrypted_ks_naf	Var	Encrypted value of Ks(_ext)_NAF, as specified in 3GPP TS 33.220.
Type	0x12			1	Bootstrapping Transaction ID
Length	Var			2	
Value	→	uint8	btid_len	1	Number of sets of the following elements: • btid
		char	btid	Var	ID that binds the subscriber identity to the keying material. For B-TID formatting, refer to 3GPP TS 33.220.

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x13			1	Key material lifetime
Length	Var			2	
Value	→	uint8	lifetime_len	1	Number of sets of the following elements: • lifetime
		char	lifetime	Var	Lifetime of the BSF-generated key material. The lifetime is expressed in UTC format with the trailing character indicating the timezone. For time formatting, refer to ISO 8601.
Type	0x14			1	IMPI
Length	Var			2	
Value	→	uint8	impi_len	1	Number of sets of the following elements: • impi
		char	impi	Var	Private user identity used for the GBA procedure specified in 3GPP TS 33.220.
Type	0x15			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.
QMI_ERR_DEVICE_NOT_READY	Device or SIM is not ready. (PIN/PERSO required)
QMI_ERR_NOT_SUPPORTED	Operation is not supported.
QMI_ERR_NETWORK_NOT_READY	Network is not ready.
QMI_ERR_ABORTED	Operation was aborted.
QMI_ERR_AUTHENTICATION_ FAILED	Originator of the request could not be authenticated.

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3.76.3 Description of QMI_UIM_GBA REQ/RESP

The request sends the application-requested GBA information to the GBA module to perform the bootstrapping procedure and produce a shared NAF key.

If the indication token is not included as part of the request, a response with the derived NAF key and corresponding information (B-TID, lifetime, etc.) and whether the command was processed successfully is sent through the response message.

If the indication token is included as part of the request, the message indicates whether the request was processed successfully while the payload (derived NAF key, B-TID, lifetime, etc.) is sent back with a subsequent indication. The response message is sent back immediately.

The client can optionally pass the APN required for the bootstrapping procedure. If the APN is not included, the modem internally selects the APN based on the NV configuration or other policies.

The response can contain either ks_naf or encrypted_ks_naf (not both), depending on internal policies and the modem configuration.

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3.77 QMI_UIM_GBA_IND

Delayed response indication containing status and shared key related information.

UIM message ID

0x005A

Version introduced

Major - 1, Minor - 47

3.77.1 Indication - QMI_UIM_GBA_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.47	1.47
Token of the Original Request	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Ks_NAF	1.47	1.47
Encrypted Ks_NAF	1.47	1.47
Bootstrapping Transaction ID	1.47	1.47
Key material lifetime	1.47	1.47
IMPI	1.47	1.47

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Ks(_ext)_NAF
Length	32			2	
Value	→	opaque	ks_naf	32	Key shared between the NAF and the ME. Key derivation is specified in 3GPP TS 33.220.
Type	0x11			1	Encrypted Ks(_ext)_NAF
Length	Var			2	
Value	→	uint16	encrypted_ks_naf_len	2	Number of sets of the following elements: • encrypted_ks_naf
		opaque	encrypted_ks_naf	Var	Encrypted value of Ks(_ext)_NAF, as specified in 3GPP TS 33.220.
Type	0x12			1	Bootstrapping Transaction ID
Length	Var			2	
Value	→	uint8	btid_len	1	Number of sets of the following elements: • btid
		char	btid	Var	ID that binds the subscriber identity to the keying material. For B-TID formatting, refer to 3GPP TS 33.220.
Type	0x13			1	Key material lifetime
Length	Var			2	
Value	→	uint8	lifetime_len	1	Number of sets of the following elements: • lifetime
		char	lifetime	Var	The lifetime of the BSF-generated key material. Lifetime is expressed in UTC form with the trailing character indicating the timezone. For time formatting, refer to ISO 8601.
Type	0x14			1	IMPI
Length	Var			2	
Value	→	uint8	impi_len	1	Number of sets of the following elements: • impi
		char	impi	Var	Private user identity used for the GBA procedure specified in 3GPP TS 33.220.

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3.77.2 Description of QMI_UIM_GBA_IND

This indication is sent if the GBA request includes an indication token, indicating an asynchronous delayed response payload.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_GBA_RESP.

The response can contain either ks_naf or encrypted_ks_naf (not both), depending on internal policies and the modem configuration.

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3.78 QMI_UIM_GET_GBA_IMPI

Retrieves the IMPI used for the GBA corresponding to the session.

UIM message ID

0x005B

Version introduced

Major - 1, Minor - 47

3.78.1 Request - QMI_UIM_GET_GBA_IMPI_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.

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Optional TLVs

Name	Version introduced	Version last modified
Encrypt Data	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypt Data
Length	1			2	
Value	→	boolean	encrypt_data	1	Indicates whether the data read from the card is to be encrypted.

3.78.2 Response - QMI_UIM_GET_GBA_IMPI_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.47	1.47

Optional TLVs

Name	Version introduced	Version last modified
IMPI	1.47	1.47
Encrypted IMPI	1.47	1.47

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	IMPI
Length	Var			2	
Value	→	uint8	impi_len	1	Number of sets of the following elements: • impi
		char	impi	Var	Private user identity used for the GBA procedure specified in 3GPP TS 33.220.
Type	0x11			1	Encrypted IMPI
Length	Var			2	

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Field	Field value	Field type	Parameter	Size (byte)	Description
Value	→	uint16	encrypted_impi_len	2	Number of sets of the following elements: • encrypted_impi
		opaque	encrypted_impi	Var	Encrypted version of the IMPI.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when there is an attempt to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.

3.78.3 Description of QMI_UIM_GET_GBA_IMPI REQ/RESP

This command retrieves the IMPI value of a specified session for the GBA.

A client can request to encrypt the IMPI data, which guarantees protection against a man-in-the-middle attack between the service and the client. The security algorithm and keys used for the encryption are not in the scope this document. The service might not support the encryption and provide a non-encrypted response.

Response can contain either IMPI or encrypted IMPI (not both).

Error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.79 QMI_UIM_SEARCH_RECORD

Provides access to search for records in the card using a given pattern.

UIM message ID

0x005C

Version introduced

Major - 1, Minor - 48

3.79.1 Request - QMI_UIM_SEARCH_RECORD_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

Name	Version introduced	Version last modified
Session Information	1.48	1.48
File ID	1.48	1.48
Search Pattern	1.48	1.48

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Session Information
Length	Var			2	
Value	→	enum8	session_type	1	Indicates the session type. Valid values: 0 through 24. See Table A-1 for the list of available session types with their values and descriptions.
		uint8	aid_len	1	Number of sets of the following elements: • aid
		uint8	aid	Var	Application identifier value or channel ID. This value is required for nonprovisioning and for logical channel session types. It is ignored in all other cases.
Type	0x02			1	File ID
Length	Var			2	
Value	→	uint16	file_id	2	File ID.

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	path_len	1	Number of sets of the following elements: • path
		uint8	path	Var	File path. This value must be the complete path of the file, which is a sequence block of 2 bytes (e.g., 0x3F00 0x7FFF).
Type	0x03			1	Search Pattern
Length	Var			2	
Value	→	uint8	pattern_len	1	Number of sets of the following elements: • pattern
		opaque	pattern	Var	Pattern used to search a file for all possible records.

Optional TLVs

Name	Version introduced	Version last modified
Search Data	1.46	1.48
Response in Indication	1.48	1.48

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Search Data When this TLV is absent, use default values for search_type and first_record.
Length	5			2	
Value	→	enum	search_type	4	Indicates the direction in which the pattern search starts from record. Valid values: • UIM_SEARCH_FORWARD (0x00) – Forward Search (default if the TLV is not present) • UIM_SEARCH_BACKWARD (0x01) – Backward Search Other values are reserved for future use.
		uint8	first_record	1	First record number in the search. The default value is 1 if the TLV is not present.
Type	0x11			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result must be provided in a subsequent indication.

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3.79.2 Response - QMI_UIM_SEARCH_RECORD_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.48	1.48

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.48	1.48
Search Result	1.48	1.48
Response in Indication	1.48	1.48

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Search Result
Length	Var			2	
Value	→	uint8	records_len	1	Number of sets of the following elements: • records
		uint8	records	Var	List of the number of records matching the pattern provided in the request.
Type	0x12			1	Response in Indication
Length	4			2	
Value	→	uint32	indication_token	4	When this TLV is present, it indicates that the result is provided in a subsequent indication.

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Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_ACCESS_DENIED	Access to the requested file is denied. This can occur when an attempt is made to access a PIN-protected file.
QMI_ERR_SIM_FILE_NOT_FOUND	File is not present on the card.
QMI_ERR_ARG_TOO_LONG	Path in the request was too long.
QMI_ERR_INVALID_ARG	Parameters passed as input were invalid (e.g., an odd length of the path).
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_NOT_PROVISIONED	File is not provisioned on the card.

3.79.3 Description of QMI_UIM_SEARCH_RECORD REQ/RESP

This command provides search access for a specific pattern in records of a linear fixed or cyclic file in the card and provides access by path.

The response contains the status code received from the card (SW1 and SW2) and record numbers matching the pattern in the request.

The client can pass a token in the request to receive the result in a subsequent QMI_UIM_SEARCH_RECORD_IND indication. In this case, the immediate response indicates only whether the command was correctly received for processing. The service passes the token back in the immediate response to indicate that it supports the indication. The service then passes the same token in the indication, so that the client can match it with the original request.

When the client requests a search with a given pattern in the records of a file, the search starts from the record number specified in the request and continues until the last record of the file has been searched. The number of each record containing the given pattern is returned.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.80 QMI_UIM_SEARCH_RECORD_IND

Indication with a search record command result.

UIM message ID

0x005C

Version introduced

Major - 1, Minor - 48

3.80.1 Indication - QMI_UIM_SEARCH_RECORD_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.48	1.48
Token of the Original Request	1.48	1.48

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Token of the Original Request
Length	4			2	
Value	→	uint32	indication_token	4	Indicates the token of the original request.

Optional TLVs

Name	Version introduced	Version last modified
Card Result	1.48	1.48
Search Result	1.48	1.48

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Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Card Result
Length	2			2	
Value	→	uint8	sw1	1	SW1 status code received from the card.
		uint8	sw2	1	SW2 status code received from the card.
Type	0x11			1	Search Result
Length	Var			2	
Value	→	uint8	records_len	1	Number of sets of the following elements: • records
		uint8	records	Var	List of the number of records matching the pattern given in the request.

3.80.2 Description of QMI_UIM_SEARCH_RECORD_IND

This indication is received when a search record request is completed and a subsequent indication is requested for the result.

The standard response present in the indication is the same that a client gets when it asks for the result in the response. The complete list of error codes is the same as in QMI_UIM_SEARCH_RECORD_RESP.

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3.81 QMI_UIM_REMOTE_UNLOCK

Performs operations related to remote unlock.

UIM message ID

0x005D

Version introduced

Major - 1, Minor - 52

3.81.1 Request - QMI_UIM_REMOTE_UNLOCK_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
Integrity Protected Simlock Configuration Data	1.52	1.52

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Integrity Protected Simlock Configuration Data
Length	Var			2	
Value	→	uint16	simlock_data_len	2	Number of sets of the following elements: • simlock_data
		opaque	simlock_data	Var	Simlock configuration data.

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3.81.2 Response - QMI_UIM_REMOTE_UNLOCK_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.52	1.52

Optional TLVs

Name	Version introduced	Version last modified
Encrypted Key	1.52	1.52

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	Encrypted Key
Length	Var			2	
Value	→	uint16	encrypted_key_len	2	Number of sets of the following elements: • encrypted_key
		opaque	encrypted_key	Var	Encrypted value of the key.

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NOT_SUPPORTED	Operation is not supported.
QMI_ERR_DEVICE_NOT_READY	Device is not ready.
QMI_ERR_ARG_TOO_LONG	Argument is too long.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.
QMI_ERR_AUTHENTICATION_ FAILED	Data in the request could not be authenticated.

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3.81.3 Description of QMI_UIM_REMOTE_UNLOCK REQ/RESP

This command performs the operations related to remote unlock of the SIM on the device. The response returns a random key, which can be used for remote simlock provisioning by an external entity. The random key in the response is encrypted with a public key provisioned in the device.

The request can also update the simlock configuration data in the device, upon successfully verifying the integrity of the data. The updated simlock configuration is securely stored.

The integrity verification mechanism of the simlock data is not part of this document.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.82 QMI_UIM_VERIFY_IMSI

Verifies whether a subscription related to IMSI is validated by simlock.

UIM message ID

0x005E

Version introduced

Major - 1, Minor - 53

3.82.1 Request - QMI_UIM_VERIFY_IMSI_REQ

Message type

Request

Sender

Control point

Mandatory TLVs

None

Optional TLVs

Name	Version introduced	Version last modified
IMSI	1.53	1.53

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x10			1	IMSI
Length	Var			2	
Value	→	uint8	imsi_len	1	Number of sets of the following elements: • imsi
		opaque	imsi	Var	Content of the IMSI, in the same format as it is stored in the SIM card.

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3.82.2 Response - QMI_UIM_VERIFY_IMSI_RESP

Message type

Response

Sender

Service

Mandatory TLVs

The Result Code TLV (defined in Section 2.3.1) is always present in the response.

Name	Version introduced	Version last modified
Result Code	1.53	1.53

Optional TLVs

None

Error codes

QMI_ERR_NONE	No error in the request.
QMI_ERR_INVALID_ID	IMSI verification fails.
QMI_ERR_INTERNAL	Unexpected error occurred during processing.
QMI_ERR_MALFORMED_MSG	Message was not formulated correctly by the control point or the message was corrupted during transmission.
QMI_ERR_NO_MEMORY	Device could not allocate memory to formulate the response.
QMI_ERR_NOT_SUPPORTED	Operation is not supported.
QMI_ERR_INVALID_ARG	Argument passed in the request is invalid.

3.82.3 Description of QMI_UIM_VERIFY_IMSI REQ/RESP

This command verifies whether a subscription related to IMSI is validated by simlock. If the optional IMSI TLV is not present, it verifies the IMSI for all active provisioning sessions, and returns an error if any session is not properly verified.

The response returns QMI_ERR_INVALID_ID if IMSI verification fails. During normal modem operation, the response should always return success. The action to take if a failure occurs is not specified in this document and it is up to the client to decide how to proceed.

Other error codes not listed in the preceding table are handled as QMI_ERR_INTERNAL.

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3.83 QMI_UIM_TEMPORARY_UNLOCK_STATUS_IND

Indicates the temporary unlock status of the modem.

UIM message ID

0x005F

Version introduced

Major - 1, Minor - 54

3.83.1 Indication - QMI_UIM_TEMPORARY_UNLOCK_STATUS_IND

Message type

Indication

Sender

Service

Scope

Unicast (per control point)

Mandatory TLVs

Name	Version introduced	Version last modified
Temporary Unlock Status	1.54	1.54

Field	Field value	Field type	Parameter	Size (byte)	Description
Type	0x01			1	Temporary Unlock Status
Length	Var			2	
Value	→	uint8	temporary_unlock_status_len	1	Number of sets of the following elements: • slot_id • feature • duration
		enum8	slot_id	1	Indicates the slot on which the temporary unlock indication is received. Valid values: • UIM_SLOT_1 (0x01) – Slot 1 • UIM_SLOT_2 (0x02) – Slot 2 • UIM_SLOT_3 (0x03) – Slot 3 • UIM_SLOT_4 (0x04) – Slot 4 • UIM_SLOT_5 (0x05) – Slot 5

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Field	Field value	Field type	Parameter	Size (byte)	Description
		uint8	feature_info_len	1	Number of sets of the following elements: • feature • duration
		enum8	feature	1	Indicates the personalization feature Valid values: • UIM_PERSO_FEATURE_GW_ NETWORK (0x00) – GW network personalization • UIM_PERSO_FEATURE_GW_ NETWORK_SUBSET (0x01) – GW network subset personalization • UIM_PERSO_FEATURE_GW_ SERVICE_PROVIDER (0x02) – GW service provider personalization • UIM_PERSO_FEATURE_GW_ CORPORATE (0x03) – GW corporate personalization • UIM_PERSO_FEATURE_GW_ UIM (0x04) – GW UIM personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_1 (0x05) – 1X network type 1 personalization • UIM_PERSO_FEATURE_1X_ NETWORK_TYPE_2 (0x06) – 1X network type 2 personalization • UIM_PERSO_FEATURE_1X_ HRPD (0x07) – 1X HRPD personalization • UIM_PERSO_FEATURE_1X_ SERVICE_PROVIDER (0x08) – 1X service provider personalization • UIM_PERSO_FEATURE_1X_ CORPORATE (0x09) – 1X corporate personalization • UIM_PERSO_FEATURE_1X_ RUIM (0x0A) – 1X RUIM personalization
		uint32	duration	4	Time before the temporary unlock expires, in seconds.

Optional TLVs

None

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3.83.2 Description of QMI_UIM_TEMPORARY_UNLOCK_STATUS_IND

This indication is sent to inform the recipient which personalization features are temporarily unlocked. When the temporary unlock expires for all slots and all features, an indication with zero members in the Temporary Unlock Status TLV is sent.

This indication is only received if a client has registered for the event via the QMI_UIM_EVENT_REG command.

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A Sessions

Most QMI UIM commands require the passing of a session as a parameter. This is used internally to route the command to the correct card and to the correct application within the card.

Table A-1 summarizes the available sessions, the parameter values, and when the sessions are used.

Table A-1 Session types

Session type	Value	Description
Primary GW provisioning	0	Accesses the GSM DF (for ICC) or USIM application (for UICC) used to acquire a GSM, a WCDMA or an LTE network.
Primary 1X provisioning	1	Accesses the CDMA DF (for ICC) or CSIM application (for UICC) used to acquire a 1X/DO network.
Secondary GW provisioning	2	Accesses the GSM DF (for ICC) or USIM application (for UICC) used to acquire a GSM, a WCDMA or an LTE network (Dual Standby).
Secondary 1X provisioning	3	Accesses the CDMA DF (for ICC) or CSIM application (for UICC) used to acquire 1X/DO network (Dual Standby).
Nonprovisioning on slot 1	4	Accesses a nonprovisioning application available on the UICC card in slot 1. The nonprovisioning application can be an ISIM or a USIM currently not used to acquire the network. The application is specified using the AID, as reported in the card status. For UICC, the modem uses shared logical channels.
Nonprovisioning on slot 2	5	Accesses a nonprovisioning application available on the UICC card in slot 2. The nonprovisioning application can be an ISIM or a USIM currently not used to acquire the network. The application is specified using the AID, as reported in the card status. For UICC, the modem uses shared logical channels.
Card on slot 1	6	Accesses files that are not in any application of the card in slot 1 (i.e., to access the global phonebook or the EF-DIR).
Card on slot 2	7	Accesses files that are not in any application of the card in slot 2 (i.e., to access the global phonebook or the EF-DIR).
Logical channel on slot 1	8	Accesses an application available on the UICC card in slot 1, after opening an exclusive logical channel using a QMI_UIM_LOGICAL_CHANNEL_REQ request. The channel ID is passed in the request.

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Table A-1 Session types (cont.)

Session type	Value	Description
Logical channel on slot 2	9	Accesses an application available on the UICC card in slot 2, after opening an exclusive logical channel using a QMI_UIM_LOGICAL_CHANNEL_REQ request. The channel ID is passed in the request.
Tertiary GW provisioning	10	Accesses the GSM DF (for ICC) or USIM application (for UICC) used to acquire a GSM, a WCDMA or an LTE network (Triple Standby)
Tertiary 1X provisioning	11	Accesses the CDMA DF (for ICC) or CSIM application (for UICC) used to acquire 1X/DO network (Triple Standby).
Quaternary GW provisioning	12	Accesses the GSM DF (for ICC) or USIM application (for UICC) used to acquire a GSM, a WCDMA or an LTE network (Quadruple Standby).
Quaternary 1X provisioning	13	Accesses the CDMA DF (for ICC) or CSIM application (for UICC) used to acquire 1X/DO network (Quadruple Standby).
Quinary GW provisioning	14	Accesses the GSM DF (for ICC) or USIM application (for UICC) used to acquire a GSM, a WCDMA or an LTE network (Quintuple Standby).
Quinary 1X provisioning	15	Accesses the CDMA DF (for ICC) or CSIM application (for UICC) used to acquire 1X/DO network (Quintuple Standby).
Nonprovisioning on slot 3	16	Accesses a nonprovisioning application available on the UICC card in slot 3. The nonprovisioning application can be an ISIM or a USIM currently not used to acquire the network. The application is specified using the AID, as reported in the card status. For UICC, the modem uses shared logical channels.
Nonprovisioning on slot 4	17	Accesses a nonprovisioning application available on the UICC card in slot 4. The nonprovisioning application can be an ISIM or a USIM currently not used to acquire the network. The application is specified using the AID, as reported in the card status. For UICC, the modem uses shared logical channels.
Nonprovisioning on slot 5	18	Accesses a nonprovisioning application available on the UICC card in slot 5. The nonprovisioning application can be an ISIM or a USIM currently not used to acquire the network. The application is specified using the AID, as reported in the card status. For UICC, the modem uses shared logical channels.
Card on slot 3	19	Accesses files that are not in any application of the card in slot 3 (i.e., to access the global phonebook or the EF-DIR).
Card on slot 4	20	Accesses files that are not in any application of the card in slot 4 (i.e., to access the global phonebook or the EF-DIR).

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Table A-1 Session types (cont.)

Session type	Value	Description
Card on slot 5	21	Accesses files that are not in any application of the card in slot 5 (i.e., to access the global phonebook or the EF-DIR).
Logical channel on slot 3	22	Accesses an application available on the UICC card in slot 3, after opening an exclusive logical channel using a QMI_UIM_LOGICAL_CHANNEL_REQ request. The channel ID is passed in the request.
Logical channel on slot 4	23	Accesses an application available on the UICC card in slot 4, after opening an exclusive logical channel using a QMI_UIM_LOGICAL_CHANNEL_REQ request. The channel ID is passed in the request.
Logical channel on slot 5	24	Accesses an application available on the UICC card in slot 5, after opening an exclusive logical channel using a QMI_UIM_LOGICAL_CHANNEL_REQ request. The channel ID is passed in the request.

Provisioning sessions allow the client to determine which card or application is used to acquire the network, when many are available. The client can directly use the session to access the correct application and card.

The status of each session can be retrieved using the QMI_UIM_GET_CARD_STATUS_REQ request. For each session, the status indicates whether a session is active as well as which card and application is being used.

Provisioning sessions are usually activated automatically by the modem during the power-up sequence. However, the modem can be configured to allow the client to activate the provisioning sessions using the QMI_UIM_CHANGE_PROVISIONING_SESSION_REQ request.

Nonprovisioning sessions are opened on the fly by the QMI UIM service when the first request is received. The logical channel associated with these sessions is shared and can be used by multiple clients, including other modules running in the modem. When the client closes the session, the associated logical channel can remain open, if other clients are still using it.

Logical channel sessions are used to establish an exclusive logical channel to a specific application on the card. These sessions are commonly used to stream APDUs to the application on the card and to have better control of non 3GPP/3GPP2 applications on the card. A new logical channel is opened when the session is created and then closed when the session is closed.

Table A-2 summarizes the causes of session closures.

Table A-2 Session closure causes

Session closure cause	Value	Description
UIM_SESSION_CLOSED_CAUSE_UNKNOWN	0x00	Unknown cause.
UIM_SESSION_CLOSED_CAUSE_CLIENT_REQUEST	0x01	Client requested to deactivate the session.
UIM_SESSION_CLOSED_CAUSE_CARD_ERROR	0x02	Card error.
UIM_SESSION_CLOSED_CAUSE_CARD_POWER_DOWN	0x03	SIM card is powered down.
UIM_SESSION_CLOSED_CAUSE_CARD_REMOVED	0x04	SIM card is removed.
UIM_SESSION_CLOSED_CAUSE_REFRESH	0x05	Session is deactivated as a part of a Refresh procedure.

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Table A-2 Session closure causes (cont.)

Session closure cause	Value	Description
UIM_SESSION_CLOSED_CAUSE_PIN_STATUS_FAIL	0x06	Status of the PIN cannot be retrieved from the SIM.
UIM_SESSION_CLOSED_CAUSE_RECOVERY	0x07	Internal card recovery.
UIM_SESSION_CLOSED_CAUSE_FDN_SRV_ENABLED_ WITH_NO_ FDN_SUPPORT	0x08	FDN service is enabled on the SIM, but the terminal does not support it.
UIM_SESSION_CLOSED_CAUSE_PERSO_OPERATION_ FAIL	0x09	Personalzation failure.
UIM_SESSION_CLOSED_CAUSE_FILE_CONTENT_ INVALID	0x0A	Content of the file is invalid.
UIM_SESSION_CLOSED_CAUSE_MANDATORY_FILE_ MISSING	0x0B	Mandatory file is missing.

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B QMI UIM Call Flows

B.1 Registration for Events

Figure B-1 illustrates the events registration process.

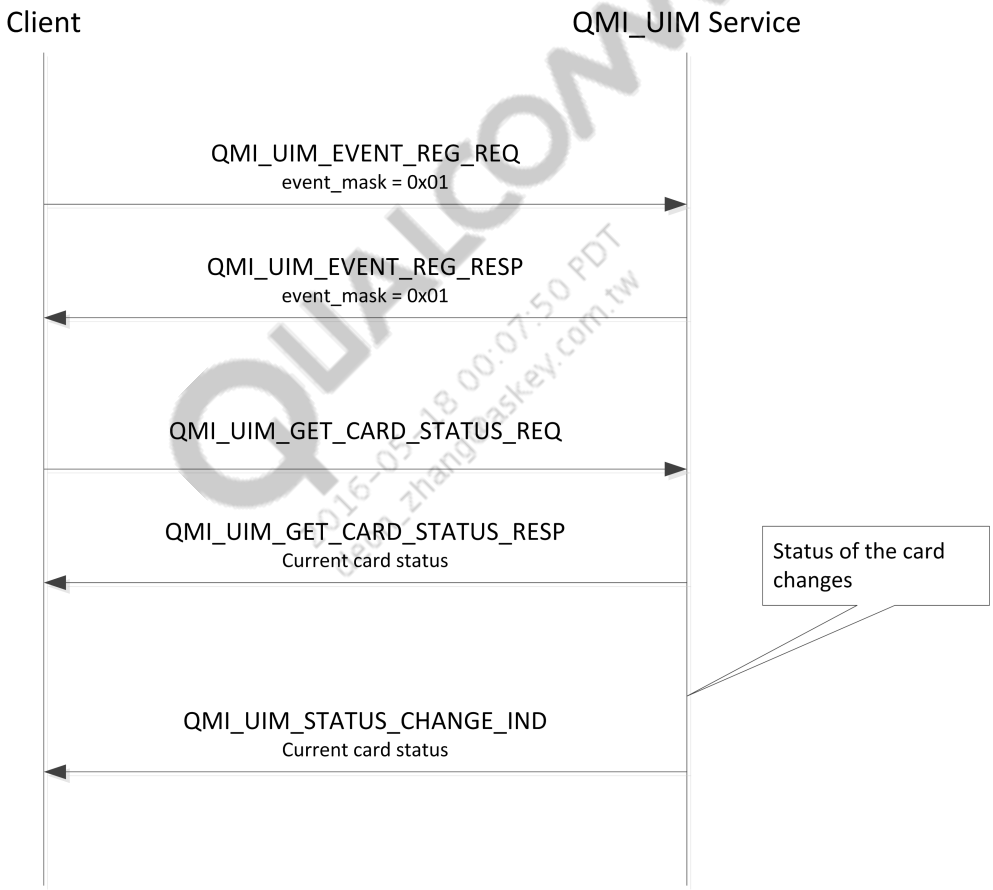


Figure B-1 Registration for events

The client registers with the QMI UIM service for notification of card status as the first step in the initialization process. This enables the client to receive indications from the service when there is a change in the card status. Because the service sends the indication only when a change occurs, the client must use the QMI_UIM_GET_CARD_STATUS_REQ command to retrieve the initial state of the card.

The QMI_UIM_GET_CARD_STATUS_REQ command is invoked after registration for the events to avoid missing changes in the card status that occur between the first and the second request.

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B.2 Shared Logical Channels

Figure B-2 illustrates the process for opening and closing shared logical channels.

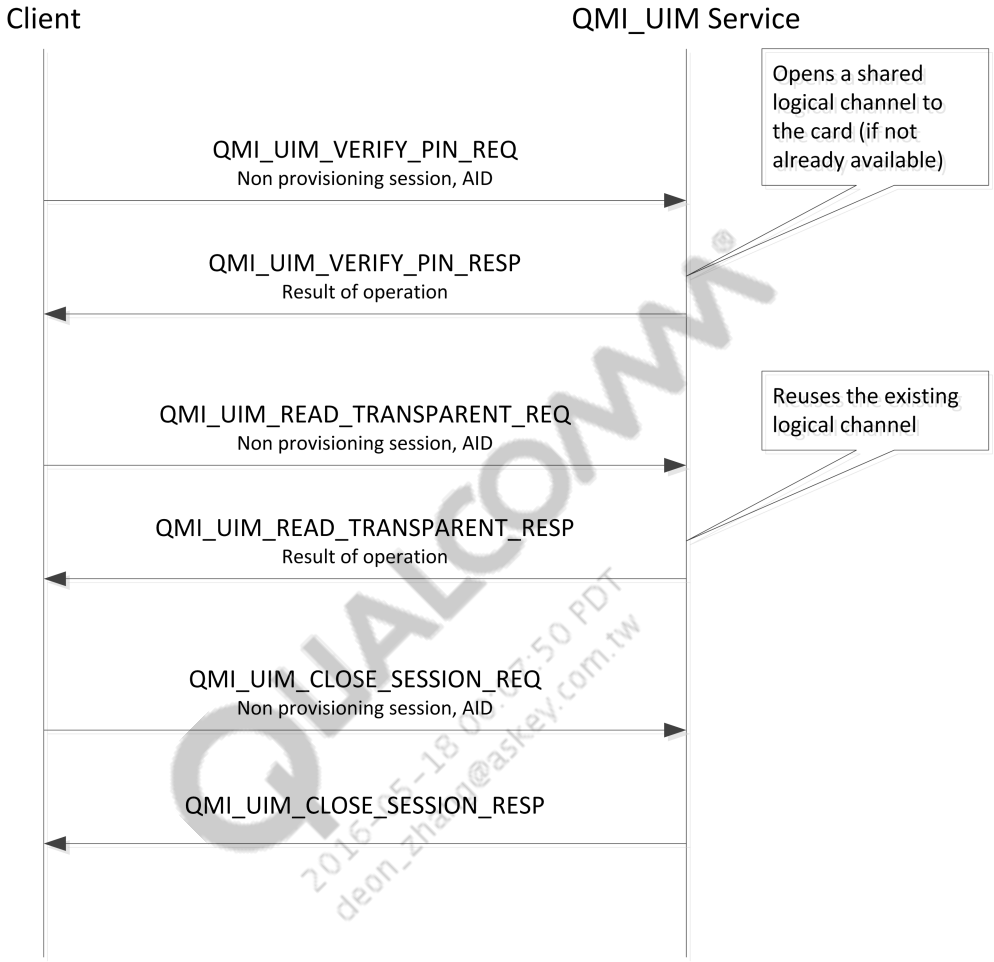


Figure B-2 Shared logical channels

Access to a nonprovisioning application, such as an ISIM, is normally handled transparently by the QMI UIM service. The service opens a shared logical channel to the specific application on the UICC card when the first request that requires access to it is issued and maintains it open for subsequent messages.

The client can specify to close the logical channel using the QMI_UIM_CLOSE_SESSION_REQ command at any point in time to free the logical channel.

Notes:

- The actual logical channel remains open after the request to close the session when other clients are using the channel, including modules running in the modem.
- This operation is valid for UICC cards only. ICC cards do not support logical channels.

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B.3 Exclusive Logical Channels

Figure B-3 illustrates the process for opening and closing exclusive logical channels.

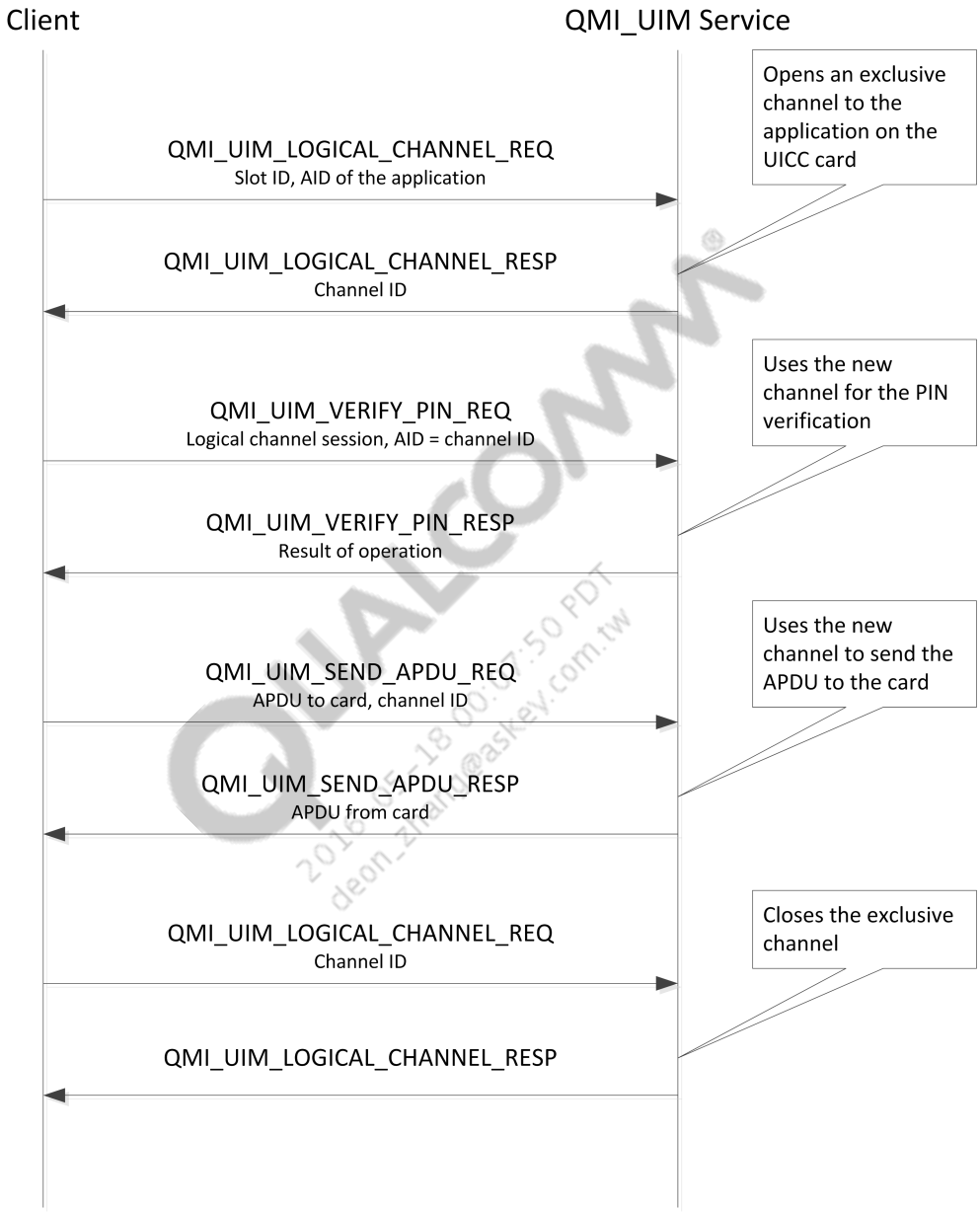


Figure B-3 Exclusive logical channels

Channels to nonprovisioning applications that reside on a UICC card are shared among all the clients available on the device. However, an exclusive channel may be required to avoid interference with other applications accessing the card (i.e., the client wants to compose and send APDUs directly to the UICC card).

The QMI_UIM_LOGICAL_CHANNEL_REQ command is used to request an exclusive channel to the UICC card. It returns a channel ID that can be used in subsequent requests. All operations, except Refresh, are allowed on the exclusive channel.

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The client can request closure of the logical channel by issuing the QMI_UIM_LOGICAL_CHANNEL_REQ command again. However, when the card is reset (i.e., power-down of the interface, REFRESH command with RESET mode), the client does not need to explicitly close the logical channel.

Exclusive channels must be explicitly re-opened after a reset of the card.

Note: This operation is valid for UICC cards only. ICC cards do not support logical channels.

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B.4 Subscription OK – Personalization on the Client

Figure B-4 illustrates the sequence for executing additional personalization algorithms prior to subscription publication.

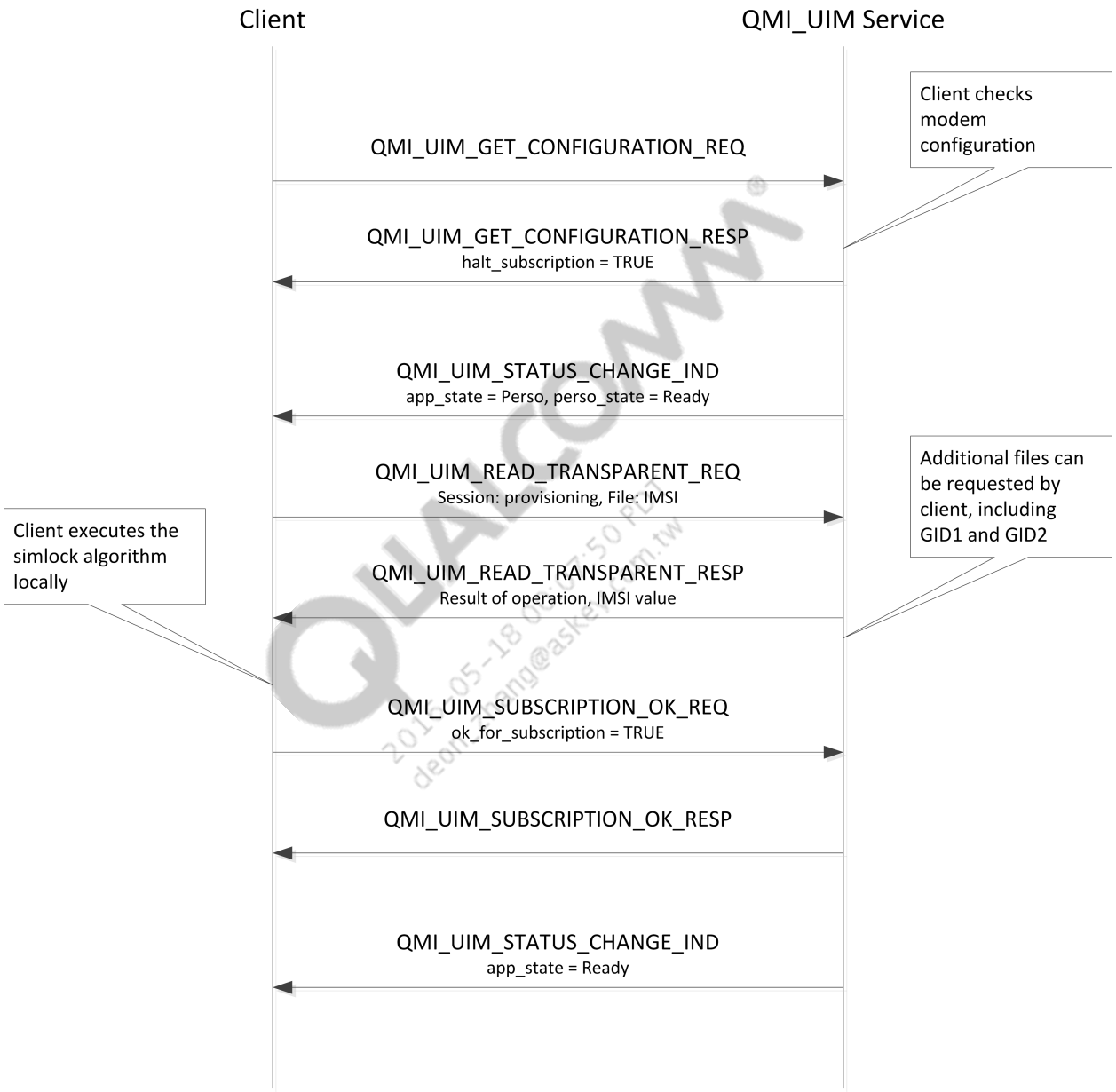


Figure B-4 Subscription OK

During card initialization, the modem performs personalization checks to verify that the SIM card inserted in the device is valid before the subscription is published and the modem starts the procedure to acquire the network.

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The modem can be configured so that a client can execute additional personalization checks to augment or replace the current checks. The client can check whether this feature is active during initialization using the QMI_UIM_GET_CONFIGURATION_REQ command. When the TLV with indication of "halt_subscription" is missing, the feature is not supported.

When the feature is supported and enabled, the modem stops the initialization of the provisioning applications at the step immediately before "Ready". The application state indicates that the personalization status must be checked, and the personalization state indicates Ready.

The client can use QMI UIM requests to read the IMSI and other required files from the card and to execute the personalization algorithm locally. If the SIM is valid, the client indicates the validity to the modem using the QMI_UIM_SUBSCRIPTION_OK_REQ command. The modem then proceeds with internal initialization and eventually publishes the subscription.

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B.5 File Access

Figure B-5 illustrates the process for accessing a file on the card with an immediate response and a subsequent indication containing the result.

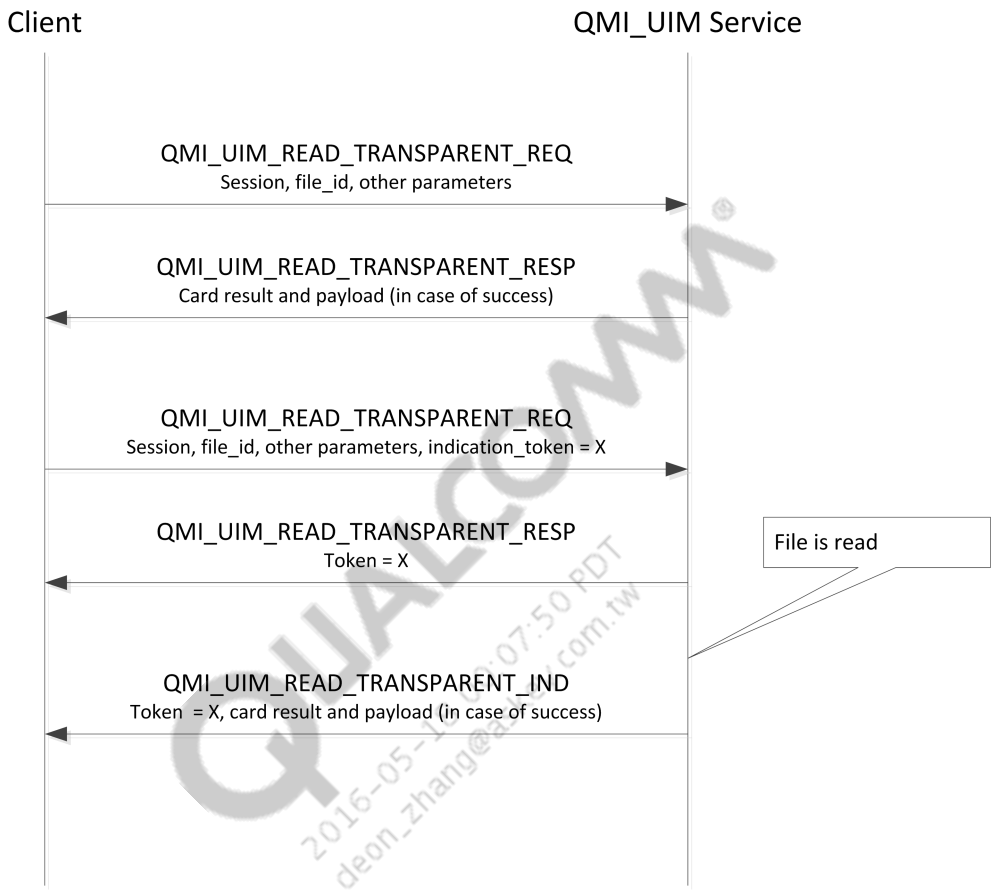


Figure B-5 File access

Operations that require access to the SIM card can experience a delay due to the interface between the device and the card. These delays become even longer in some circumstances, such as during powerup, when many card access requests are requested by many different clients. For this reason, an optional token can be specified in the command to request the modem to immediately send a response and provide the actual result later using a separate indication. In this case, the same token value is passed in both the response and in the subsequent indication so the client knows for which request the result is received.

The token can be any value and can be used by the client to carry information to process the indication when the result is received.

Note: This feature was introduced in revision 1.12 of the QMI UIM interface, so the client must be able to accommodate the unavailability of this feature. The client can check the response to see whether the token is available.

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B.6 Refresh

Figure B-6 illustrates the process for handling the REFRESH proactive command.

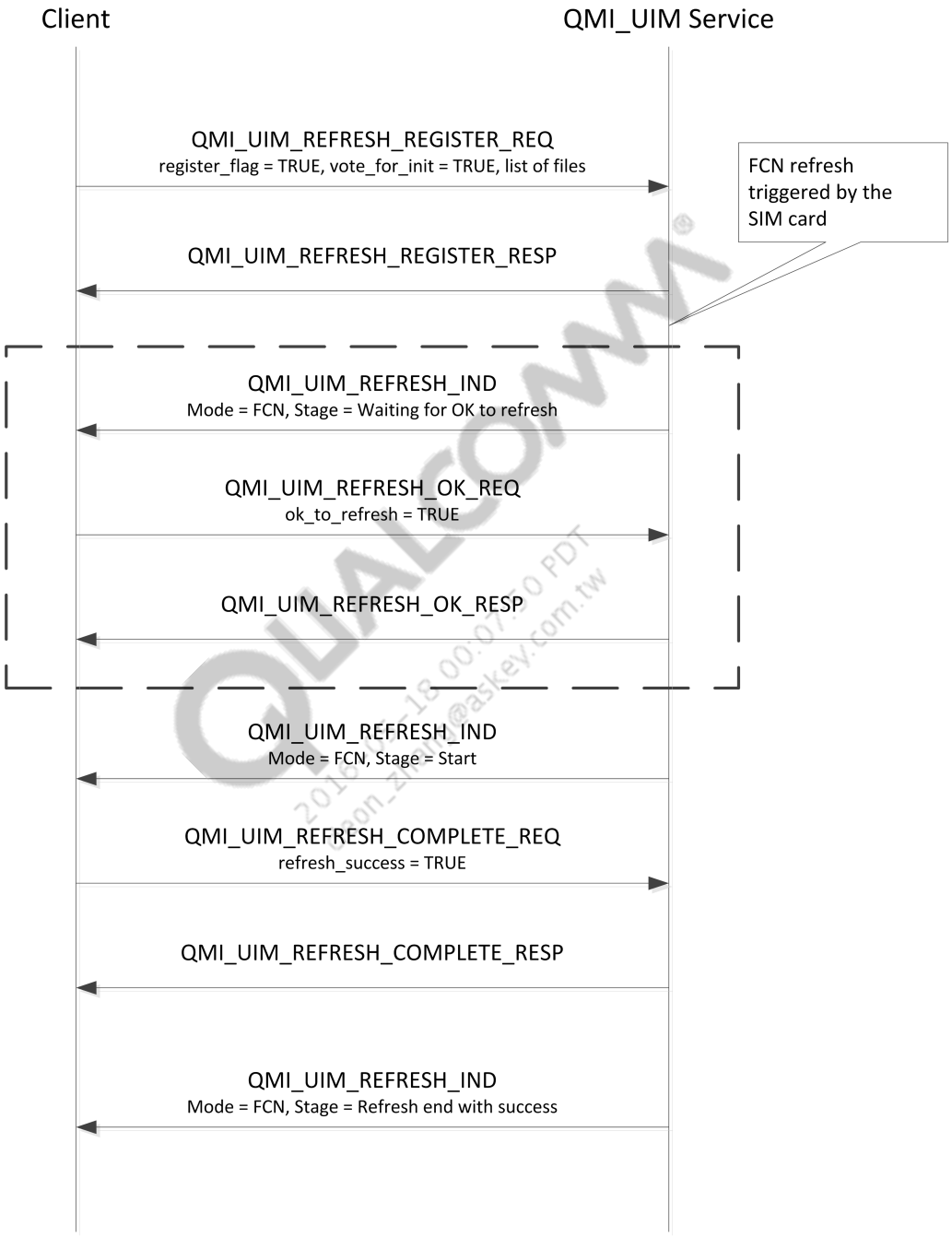


Figure B-6 Refresh

The client can request notification by the QMI UIM service of REFRESH proactive commands coming from the SIM card so it can take necessary actions, such as re-reading files.

When the client registers for the Refresh feature, it can specify whether it wants to vote. This is an optional intermediate step that is required by some clients. It ensures that a refresh is not performed under specific conditions (i.e., during data streaming) that can cause a negative user experience.

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Unless there is a requirement to vote for the Refresh feature, clients should set this flag to FALSE.

Each refresh indication contains two values:

- Mode – Indicates the refresh type, as indicated in the command details field of the REFRESH proactive command received from the SIM card.
- Stage – Identifies the phase of the refresh procedure:
  - Voting – This phase is optional and is used to check whether it is OK for the client to proceed with the refresh.
  - Start – This phase indicates that the refresh procedure is started. The exact sequence of operations executed by the modem depends on the refresh mode. In case of FCN refresh, the client is to reread the files from the card.
  - End – This event is sent after the Terminal Response is sent to the card to indicate that the refresh has completed.

For more information on the expected behavior of a client for each refresh event, see Appendix C.

The REFRESH command is provided to clients by the QMI UIM service and not by the QMI CAT service, even if it is a proactive command originated by the card. The reason for this behavior is that the REFRESH command has a deep impact on the modem, affecting multiple modules. For this reason, the single proactive command is split into various stages and requires logic to assemble the responses from many modules.

Notes:

- The dotted outline in the call flow indicates optional commands. These occur only when the client indicates that it wants to vote for the Refresh feature.
- A client must use the QMI_UIM_REFRESH_OK_REQ request after receiving any refresh indication that indicates the stage as “Waiting for OK to refresh”.
- A client must use the QMI_UIM_REFRESH_COMPLETE_REQ request after receiving a refresh indication that indicates the stage as “Refresh start” if applicable to the Refresh mode. For details, see Table C-1.
- After a RESET refresh, the indication with the end event can be missing and the client should not wait for it.
- The client can receive QMI_UIM_STATUS_CHANGE_IND indications during some of the refresh procedures. These events are not indicated in the call flow because the focus is on the Refresh procedure.

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B.7 Encrypted PIN1

Figure B-7 illustrates the process for PIN1 encryption.

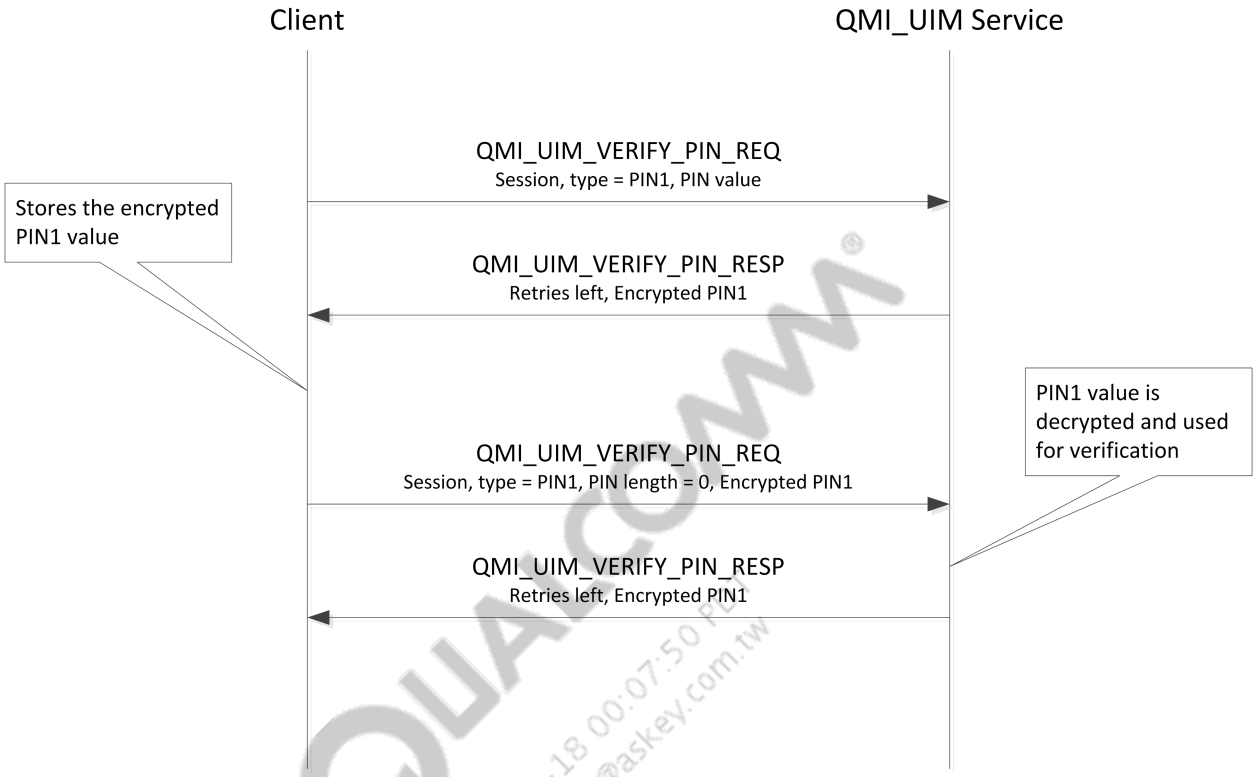


Figure B-7 Encrypted PIN1

The QMI UIM service provides support for PIN1 encryption to implement features such as silent PIN verification after a restart of the modem. After any successful PIN operation (i.e., verification, enablement, disablement, change, unblock) executed on PIN1, the QMI UIM service returns an encrypted value of PIN1 to the client.

The encrypted value can be stored by the client and re-used later to verify the PIN1 again. The encrypted PIN1 value cannot be used for any other PIN operation.

Although the client has the encrypted value of PIN1, it does not have the key to decrypt it. While the modem has the key, it does not store the encrypted value. This solution guarantees security against attacks performed on one side only.

Note: This feature is introduced in revision 1.7 of the QMI UIM interface and may not always be enabled on the modem. The client must not assume availability and must check the response for any PIN operation to determine whether the encrypted PIN1 value is present.

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B.8 Application Selection

Figure B-8 illustrates the process for the client to manually activate the provisioning sessions.

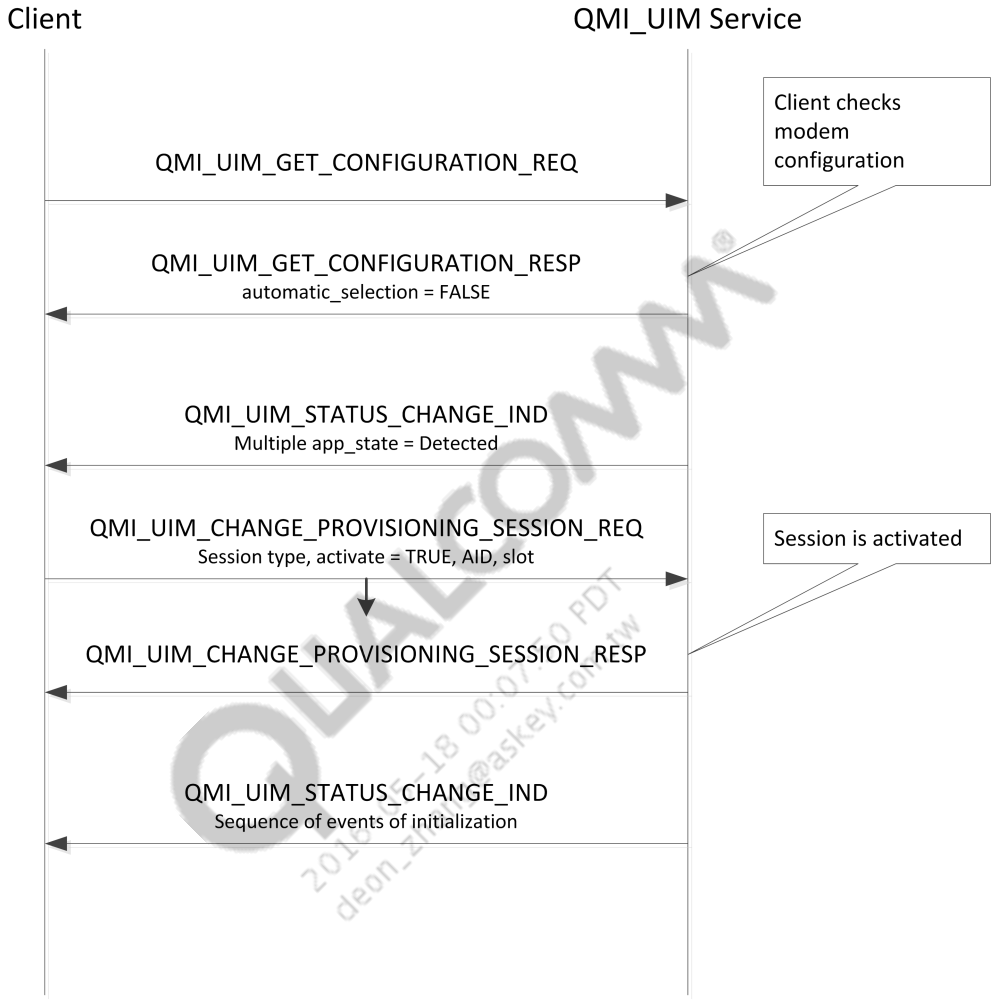


Figure B-8 Application selection

The modem selects the applications available on the SIM card and activates the corresponding provisioning sessions automatically. However, the modem can be configured to not select the applications automatically (i.e., when there are multiple SIM cards available on the device). The client can manually activate the provisioning sessions, either using an internal algorithm to define the appropriate sessions or presenting a list of possibilities to the user on the screen.

After the subscription is activated, the initialization of that subscription proceeds normally. The same QMI_UIM_CHANGE_PROVISIONING_SESSION_REQ command can also be used to deactivate an active session. This operation can be useful when the user wants to use a different subscription available on the card.

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B.9 Session Closed Indication

Figure B-9 illustrates the QMI UIM service sending the Session Closed indication to registered clients during card initialization when a mandatory file (e.g., EF-IMSI) is missing on the card.

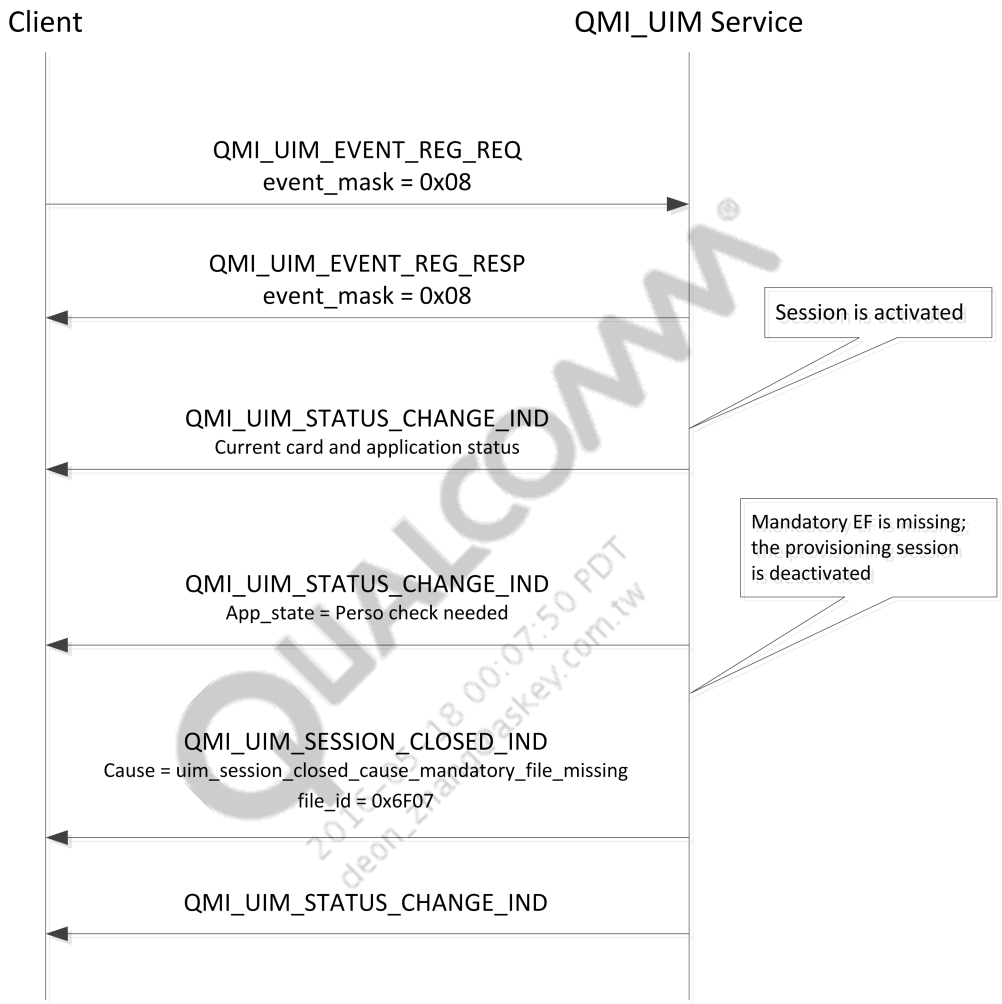


Figure B-9 Session Closed indication

During card initialization, the modem reads various mandatory files before the subscription is published. If a mandatory file is missing on the card, the modem stops the initialization procedure and closes the provisioning session. For the clients who register with the QMI UIM service for notification of the provisioning Session Closed indication, the QMI_UIM_SESSION_CLOSED_IND indication is sent with a cause for the session closure. This indication is sent in addition to the QMI_UIM_STATUS_CHANGE_IND indication.

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B.10 Reduced Card Indications

A client can register for a reduced number of card indications during card initialization by using the *reduced card status* bit in QMI_UIM_EVENT_REG_REQ in addition to the *card status* or *extended card status* bits. Setting the *reduced card status* bit does not impact events received by other clients.

Figure B-10 illustrates the events received during a typical UICC initialization with USIM and CSIM. The dotted lines represent indications that are not received if clients request a reduced number of indications.

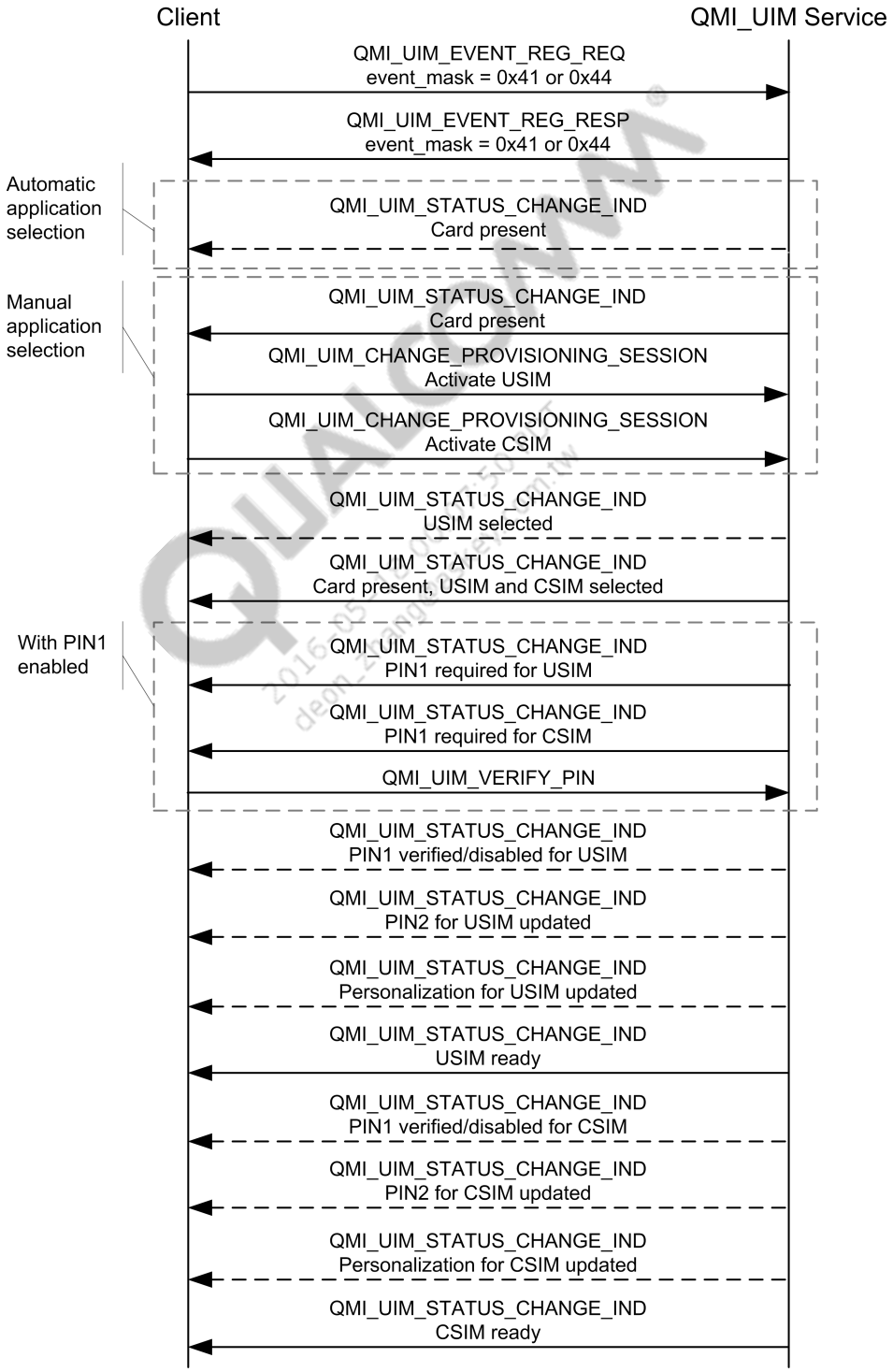


Figure B-10 Reduced card indications

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B.11 Vcc Power-down and Recovery

QMI_UIM_SUPPLY_VOLTAGE_IND notifies the client of the UICC Vcc line activation or deactivation status and informs the modem to proceed with a deactivation.

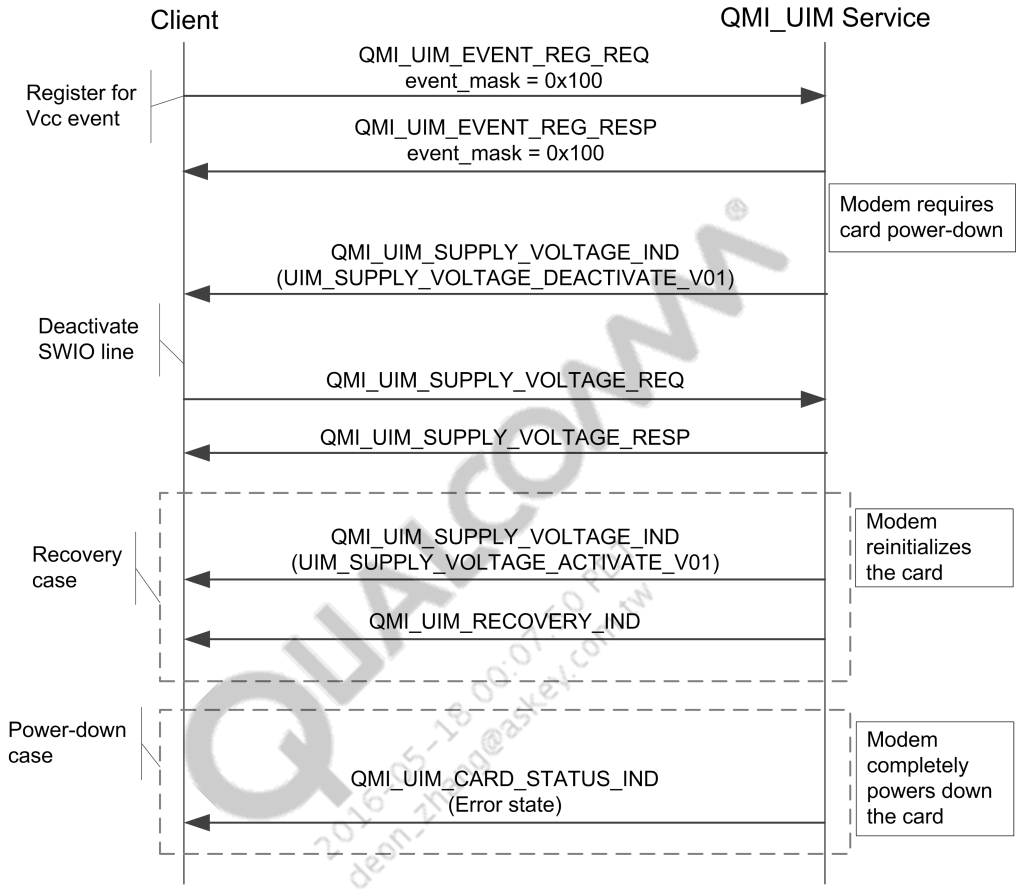


Figure B-11 Vcc power-down and recovery

The client must register to receive an indication via QMI_UIM_EVENT_REG. Before the modem powers down the UICC, QMI_UIM_SUPPLY_VOLTAGE_IND is sent to the client to reset its mode and deactivate the SWIO line. The client calls QMI_UIM_SUPPLY_VOLTAGE_REQ to inform it to proceed with the deactivation.

QMI_UIM_SUPPLY_VOLTAGE_IND also notifies the client in the following conditions:

- When the UICC is powered up
- During a normal power-up procedure
- During a recovery procedure

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B.12 Multiple Profiles

Only one profile is active at a time for a subscription service on a multiple SIM profile UICC. The inactive profile is not visible to the modem or any interface.

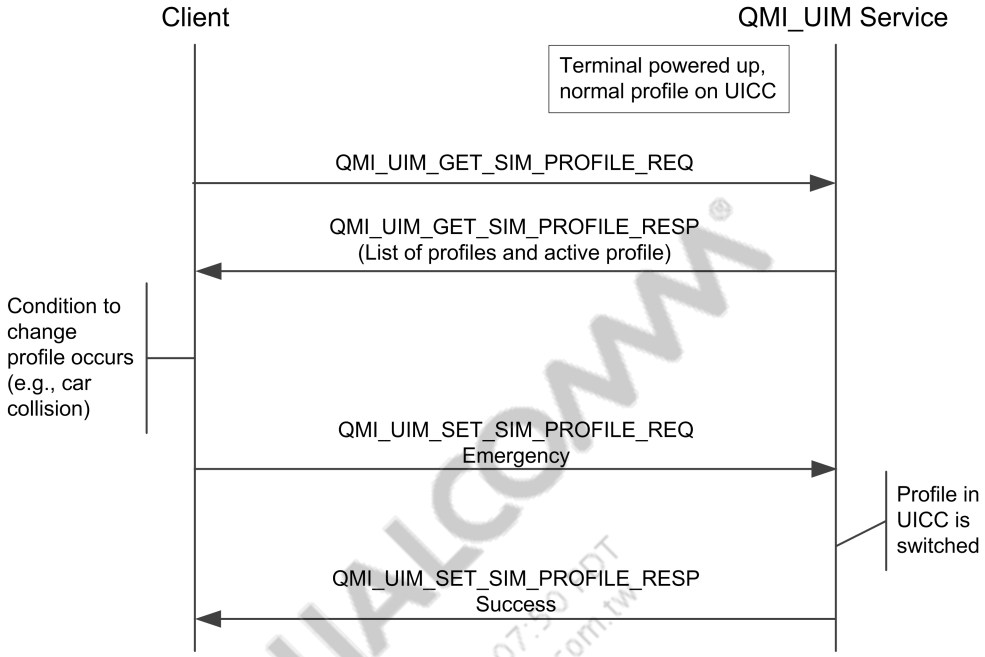


Figure B-12 Multiple profiles

QMI_UIM_GET_SIM_PROFILE retrieves the SIM profile list and active SIM profile data.

When a condition occurs to change the profile (e.g., a car collision), the client sends QMI_UIM_SET_SIM_PROFILE_REQ to switch profiles. The UICC resets and the active subscription belongs to the emergency profile.

QMI_UIM_SET_SIM_PROFILE provides a SIM profile ID to enable the client to switch back to the normal SIM profile.

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B.13 Enable Remote Simlock Configuration

Figure B-13 illustrates the process for enabling the remote simlock configuration.

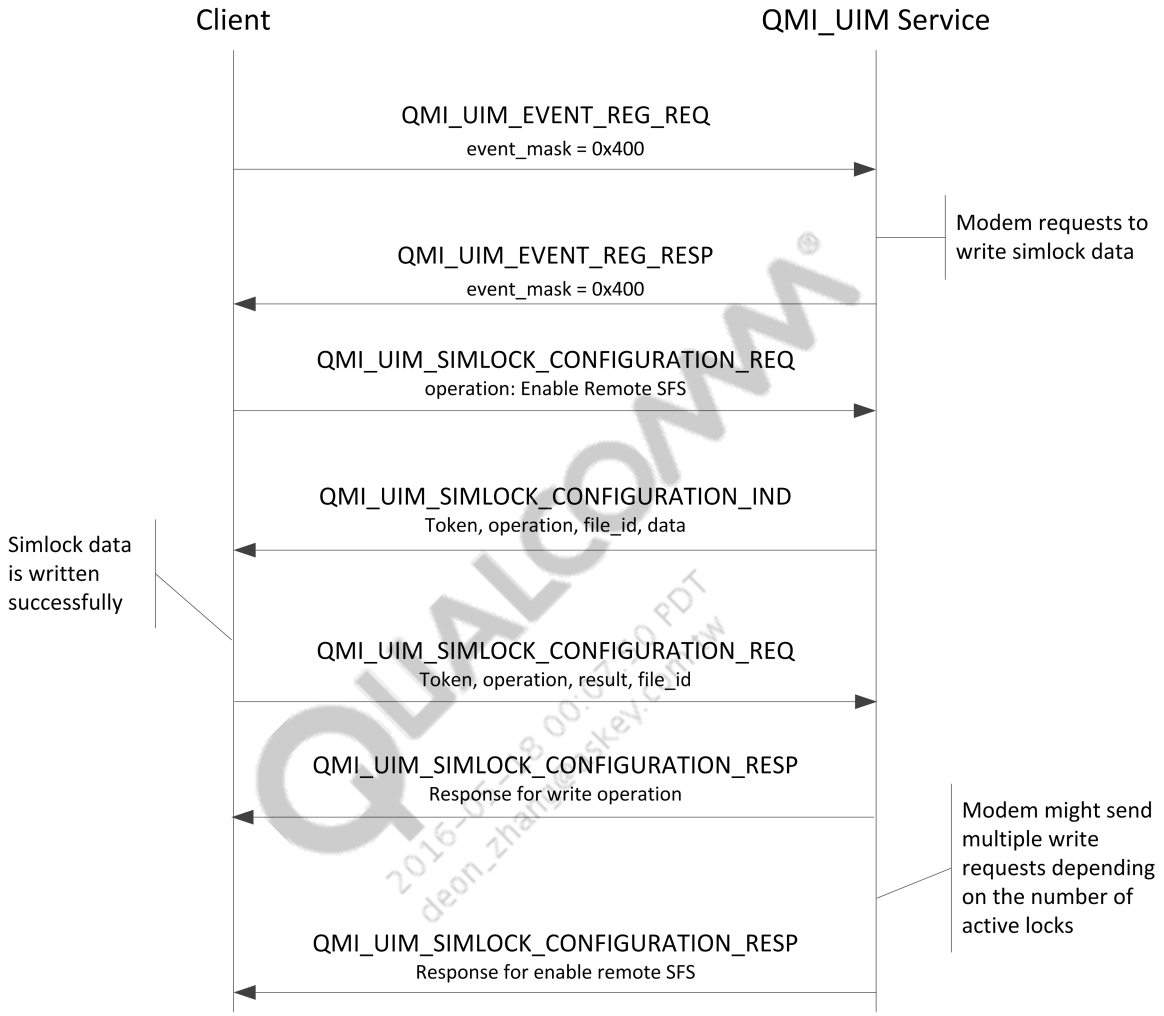


Figure B-13 Remote simlock configuration

When the client requests to enable the remote simlock configuration, the modem stores the simlock data in the remote storage location. When the modem requests to write the simlock data, the QMI_UIM service sends the simlock configuration indication to the registered clients. The client must store the simlock data and acknowledge the result of the operation to the QMI_UIM service via the simlock configuration request message.

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B.14 Reading or Writing Simlock Data

Figure B-14 illustrates the QMI_UIM service sending the simlock configuration indication to the registered clients when there is a request from the modem to read or write the simlock data.

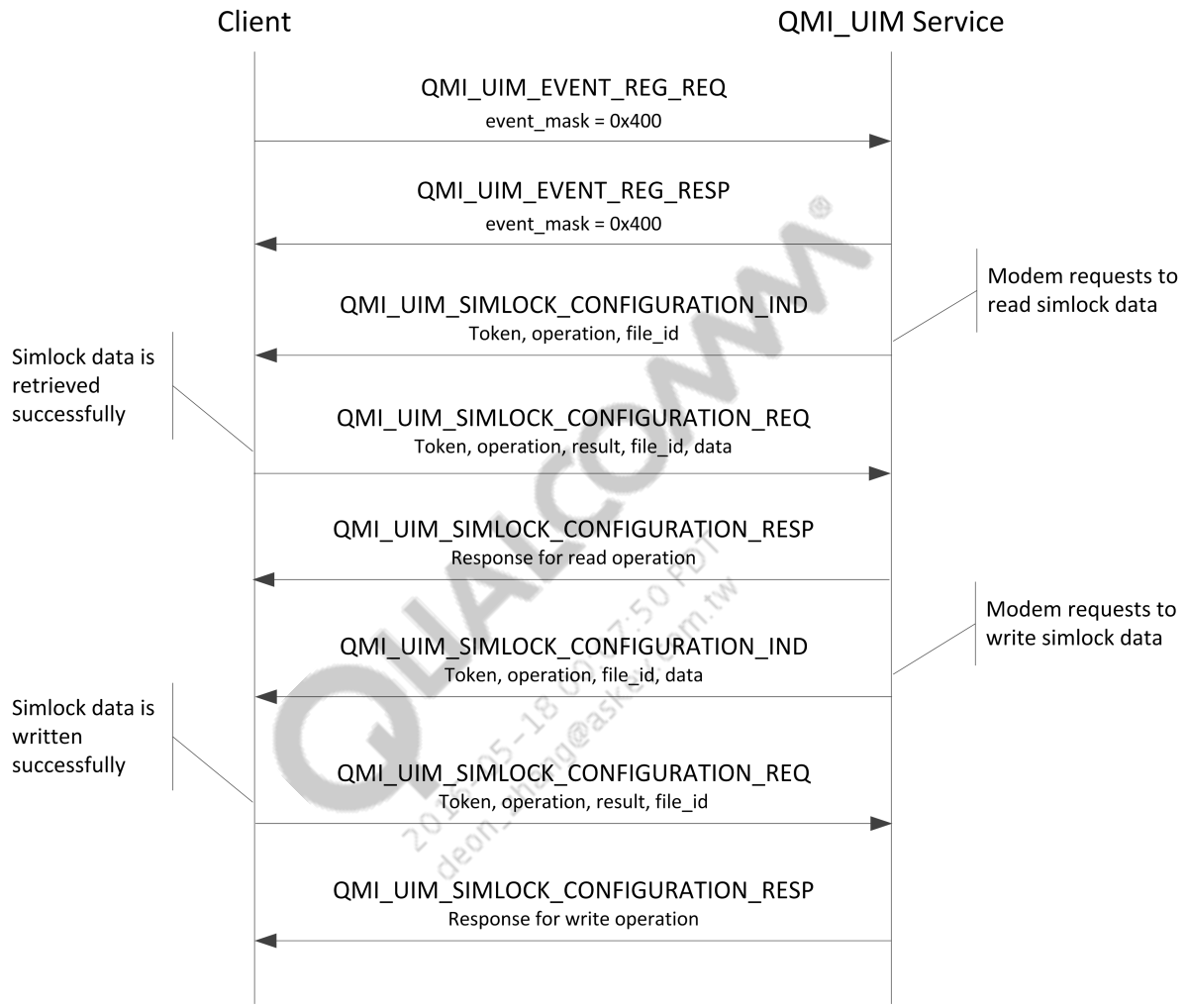


Figure B-14 Simlock configuration indication

During the simlock verification process, the modem reads the simlock data. If the modem determines that the simlock data is stored in a remote secure file system (SFS) location, it sends the simlock configuration indication to clients registered with the QMI_UIM service for remote simlock configuration notification. Upon receiving the indication, the client provides the simlock data using the simlock configuration request message.

During a simlock operation (such as depersonalization of a feature), the modem updates the simlock data. If the modem determines that the simlock data is stored in a remote SFS location, it sends the simlock configuration indication to clients registered with the QMI_UIM service for remote simlock configuration notification. Upon receiving the indication, the client stores the simlock data in the remote SFS location and acknowledges the result of the operation via the simlock configuration request message.

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C Refresh Indication Behaviors

Table C-1 describes the expected behavior of a client when it receives a refresh indication after registering for it. The behavior depends on the mode and the stage, as indicated in the refresh indication.

Table C-1 Refresh indication behaviors

Mode	Stage		
	WAIT_FOR_OK (Init or FCN)	**START**	**END SUCCESS**
Reset	Vote if it is OK to continue with the refresh.	Delete all cached values. The card is reinitialized and its status is updated.	This event might be missing. The client should look at the card status and application status.
FCN		Reread the files being refreshed, and then invoke the complete() function.	No action is required.
Init		Invalidate all cached values. Nonprovisioning session: Reread the files, and then invoke the complete() function.	Provisioning session: Reread all files (if not done when the application state is back to Ready).
Init + FCN		Invalidate cached values of files in the FCN list. Nonprovisioning session: Reread the files in the FCN list, and then invoke the complete() function.	Provisioning session: Reread files in the FCN list (if not done when the application state is back to Ready).
Init + FullFCN		Invalidate all cached values. Nonprovisioning session: Reread the files and then invoke the complete() function.	Provisioning session: Reread all files (if not done when the application state is back to Ready).
App reset		Invalidate all cached values. Nonprovisioning session: Invoke the complete() function and wait for End Stage. Provisioning session: Wait for the application state to be Ready or End Stage.	Provisioning session: Reread all files (if not done when the application state is back to Ready). Nonprovisioning session: Reread all files.
3G session reset		Qualcomm recommends deleting all cached values.* Nonprovisioning session: Invoke the complete() function.	Provisioning session: Reread all of the files discarded when the refresh was started (if not done when the application state returned to Ready).

* If a 3G session reset contains a FCN list, it is propagated to the client as part of the FCN refresh indication. To avoid interoperability issues, discard the entire cache, not just the EFs in the FCN list.

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D Encrypted Requests and Responses

The following rules apply to all of the payloads described in this appendix:

- All the numbers are represented in little-endian format
- All the messages can be extended in the future by adding more bytes at the end, so clients and services can handle it

D.1 QMI_UIM_DEPERSONALIZATION_SECURE_REQ

Table D-1 describes QMI_UIM_PERSONALIZATION_SECURE_REQ payload content before encryption.

Table D-1 Depersonalization secure payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0056) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – Reserved for future use (RFU) (set to 0)
slot	1	Slot to be used. Valid values: • 1 – Slot 1 • 2 – Slot 2 • 3 – Slot 3 • 4 – Slot 4 • 5 – Slot 5
feature	1	Personalization feature to deactivate: • 0 – GW network personalization • 1 – GW network subset personalization • 2 – GW service provider personalization • 3 – GW corporate personalization • 4 – GW UIM personalization • 5 – 1X network type 1 personalization • 6 – 1X network type 2 personalization • 7 – 1X High Rate Packet Data (HRPD) personalization • 8 – 1X service provider personalization • 9 – 1X corporate personalization • 10 – 1X RUIM personalization
operation	1	Depersonalization request type: • 0 – Unlock without the control key • 1 – Enable temporary depersonalization • 2 – Disable temporary depersonalization

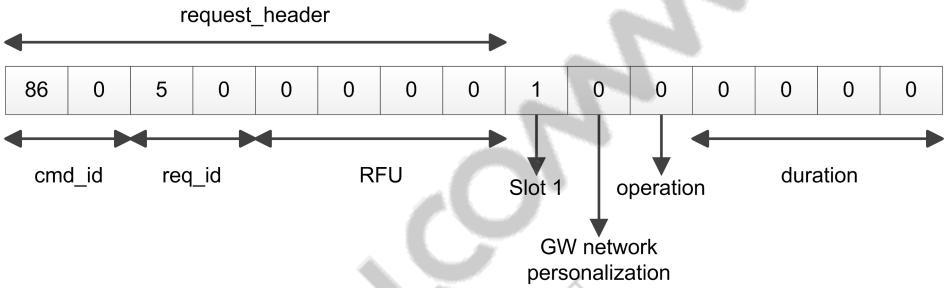
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Table D-1 Secure depersonalization payload (cont.)

Parameter	Size (byte)	Description
duration	4	Number of minutes for the temporary depersonalization (valid only for enabling a temporary unlock).

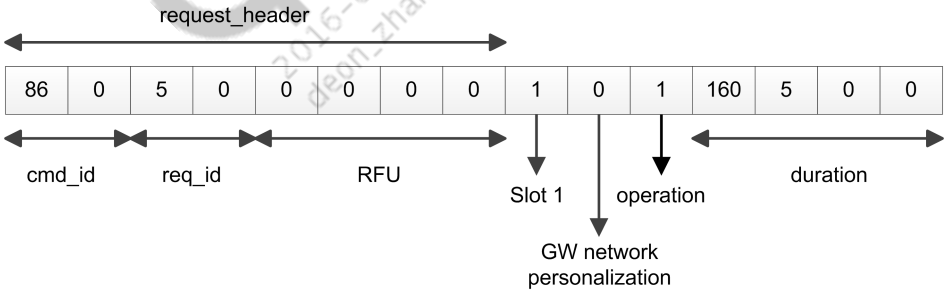
Example – GW network depersonalization request

Payload buffer (before encryption) for a depersonalization request of a GW network category on slot 1 without the control keys:



Example – Temporary depersonalization of a GW network category

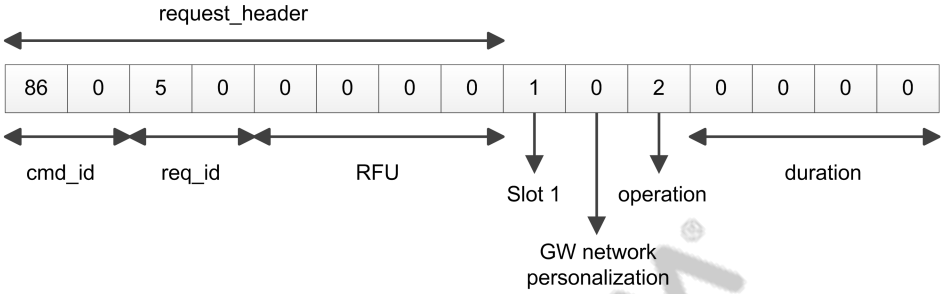
Payload buffer (before encryption) for enabling the temporary depersonalization of a GW network category on slot 1 for a duration of one day (1440 min):



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Example – Disabling a temporary depersonalization

Payload buffer (before encryption) for disabling the temporary depersonalization created in the previous example:



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D.2 QMI_UIM_PERSONALIZATION_SECURE_REQ

Table D-2 describes QMI_UIM_PERSONALIZATION_SECURE_REQ payload content before encryption.

Table D-2 Personalization secure payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0057) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – RFU (set to 0)
slot	1	Slot to be used. Valid values: • 1 – Slot 1 • 2 – Slot 2 • 3 – Slot 3 • 4 – Slot 4 • 5 – Slot 5
feature	1	Personalization feature to deactivate: • 0 – GW network personalization • 1 – GW network subset personalization • 2 – GW service provider personalization • 3 – GW corporate personalization • 4 – GW UIM personalization • 5 – 1X network type 1 personalization • 6 – 1X network type 2 personalization • 7 – 1X HRPD personalization • 8 – 1X service provider personalization • 9 – 1X corporate personalization • 10 – 1X RUIM personalization
num_codes	1	Number of sets of code_data.
code_data	var	Contains the personalization code data. See Table D-3 for various personalization codes.

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Table D-3 describes the lock codes associated with personalization features.

Table D-3 Personalization lock codes

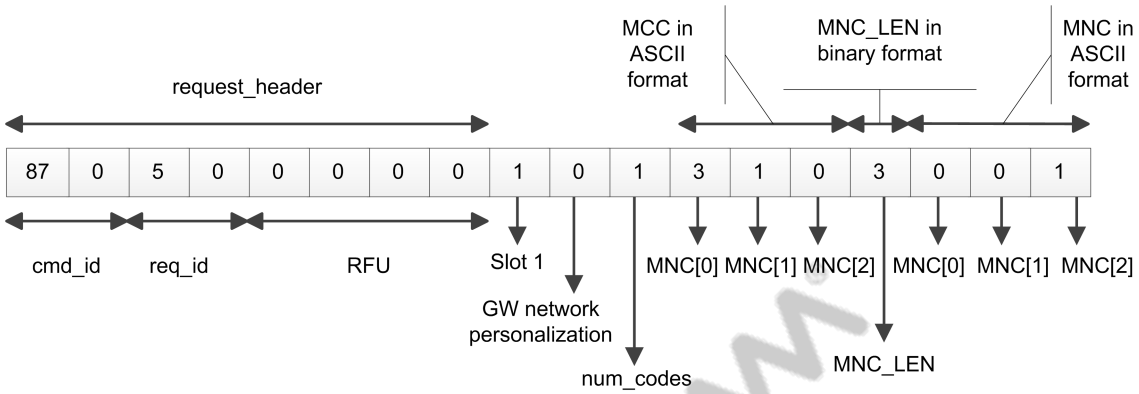
Feature	Lock codes
0 – GW network personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII)
1 – GW network subset personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII) IMSI digit 6 – 1 byte (ASCII) IMSI digit 7 – 1 byte (ASCII)
2 – GW service provider personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII) Gid1 – 1 byte (binary)
3 – GW corporate personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII) Gid1 – 1 byte (binary) Gid2 – 1 byte (binary)
4 – GW UIM personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII) MSIN length – 1 byte (binary) MSIN – bytes indicated by length (ASCII)
5 – 1X network type 1 personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII)
6 – 1X network type 2 personalization	IRM_CODE – 4 bytes (ASCII)
7 – 1X HRPD personalization	Not supported.
8 – 1X service provider personalization	Not supported.
9 – 1X corporate personalization	Not supported.
10 – 1X RUIM personalization	MCC – 3 bytes (ASCII) MNC length – 1 byte (binary) MNC – 2 or 3 bytes as indicated by length (ASCII) MSIN length – 1 byte (binary) MSIN – bytes indicated by length (ASCII)

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Example – GW network personalization request

Payload buffer (before encryption) for a GW network personalization request with MCC 310 and MNC 001:



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D.3 QMI_UIM_EMERGENCY_ONLY

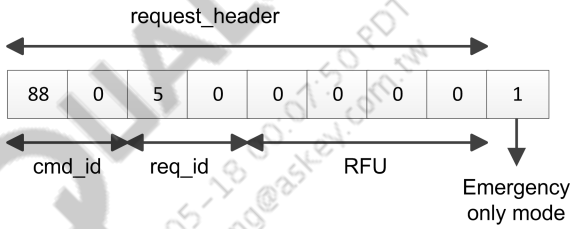
Table D-4 describes the payload content of QMI_UIM_EMERGENCY_ONLY before encryption.

Table D-4 Emergency only request payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0058) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – RFU (set to 0)
emergency_only	0	Whether to set or unset the device in emergency only mode: • 0 – Normal mode • 1 – Emergency-only mode

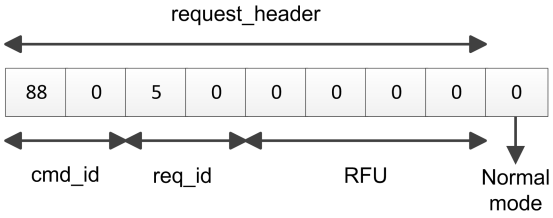
Example – Setting emergency-only mode

Payload buffer (before encryption) for setting the terminal in emergency-only mode:



Example – Setting normal mode

Payload buffer before encryption for setting the normal mode:



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D.4 QMI_UIM_SIMLOCK_CONFIGURATION

D.4.1 QMI_UIM_SIMLOCK_CONFIGURATION_REQ

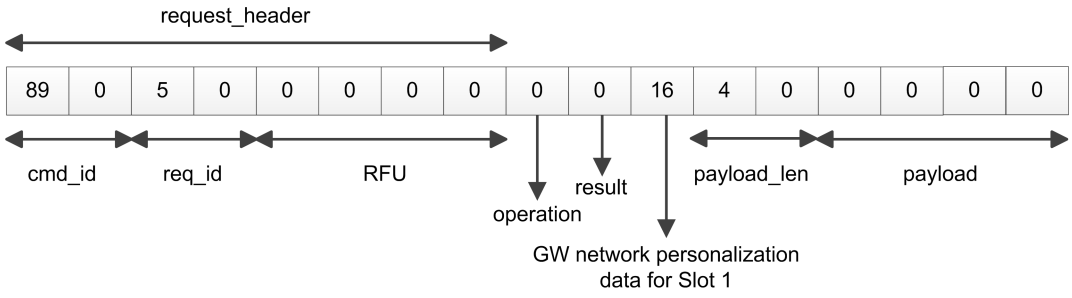
Table D-5 describes the payload content of QMI_UIM_SIMLOCK_CONFIGURATION request before encryption.

Table D-5 Simlock configuration request payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0059) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – RFU (set to 0)
operation	1	Simlock configuration request type. Valid values: • 0 – Read file • 1 – Write file • 2 – Enable remote storage • 3 – Disable remote storage
result	1	Result of the operation. Valid only for read/write operations. Valid values: • 0 – Success • 2 – Failure • 3 – File was not found
configuration_file_id	1	File ID of the simlock configuration data. Valid only for read/write operations. See Table D-7 for valid values.
payload_len	2	Length of the payload. Valid only for Read file operations.
payload	var	Contains the data read from the file. Valid only for Read file operations.

Example – Read simlock configuration

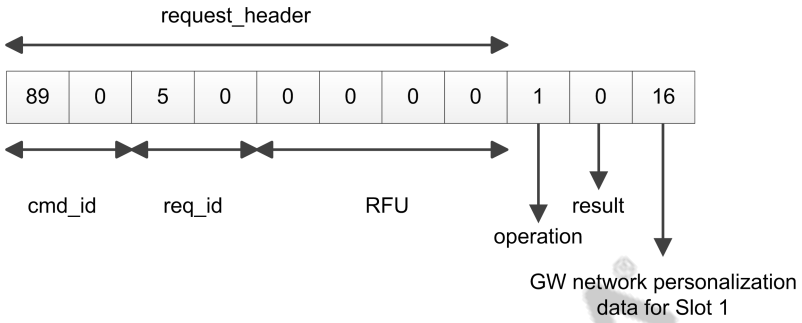
Payload buffer (before encryption) for a read simlock configuration request:



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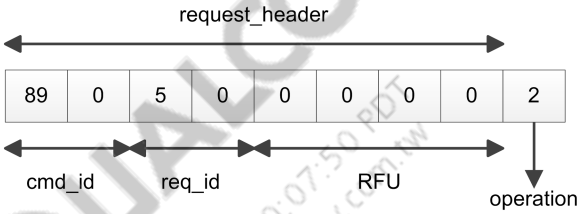
Example – Write simlock configuration

Payload buffer (before encryption) for a write simlock configuration request:



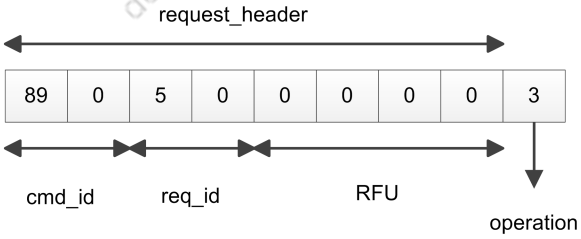
Example – Enable remote SFS storage

Payload buffer (before encryption) for an enable remote SFS storage request:



Example – Disable remote SFS storage

Payload buffer (before encryption) for a disable remote SFS storage request:



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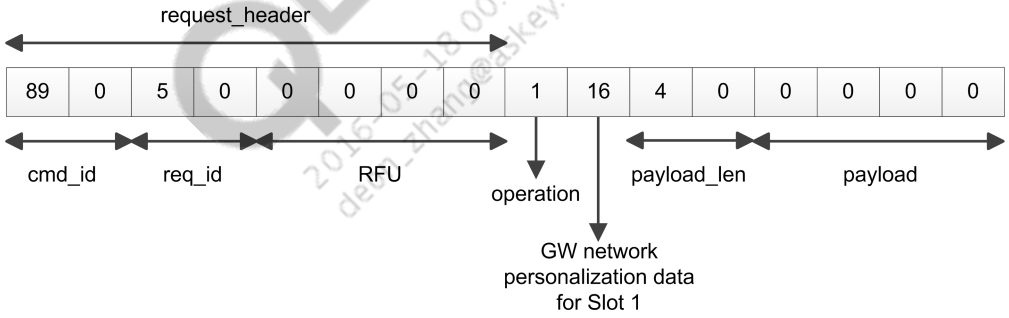
D.4.2 QMI_UIM_SIMLOCK_CONFIGURATION_IND

Table D-6 Simlock configuration indication payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0059) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – RFU (set to 0)
operation	1	Simlock configuration request type. Valid values: • 0 – Read file • 1 – Write file
configuration_file_id	1	File ID of the simlock configuration data. See Table D-7 for valid values.
payload_len	2	Length of the payload.
payload	var	Contains the data to be written to the file. Valid only for Write file operations.

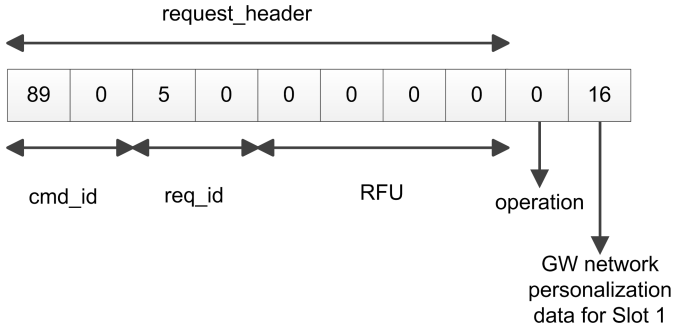
Example – Writing simlock configuration data

Payload buffer in the indication (before encryption) for writing the simlock configuration data:



Example – Reading simlock configuration data

Payload buffer in the indication (before encryption) for reading the simlock configuration data:



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Simlock configuration file IDs are coded in the 0xAB format as follows:

- Nibble A represents the corresponding slot value starting from 1
- Nibble B represents the corresponding feature value

Table D-7 Valid simlock configuration file IDs

File ID	Description
0 (0x00)	Lock status and emergency-only status data file
16 (0x10)	GW network personalization data file for slot 1
17 (0x11)	GW network subset personalization data file for slot 1
18 (0x12)	GW service provider personalization data file for slot 1
19 (0x13)	GW corporate personalization data file for slot 1
20 (0x14)	GW UIM personalization data file for slot 1
21 (0x15)	1X network type 1 personalization data file for slot 1
22 (0x16)	1X network type 2 personalization data file for slot 1
23 (0x17)	1X HRPD personalization data file for slot 1
24 (0x18)	1X service provider personalization data file for slot 1
25 (0x19)	1X corporate personalization data file for slot 1
26 (0x1A)	1X RUIM personalization data file for slot 1
32 (0x20)	GW network personalization data file for slot 2
33 (0x21)	GW network subset personalization data file for slot 2
34 (0x22)	GW service provider personalization data file for slot 2
35 (0x23)	GW corporate personalization data file for slot 2
36 (0x24)	GW UIM personalization data file for slot 2
37 (0x25)	1X network type 1 personalization data file for slot 2
38 (0x26)	1X network type 2 personalization data file for slot 2
39 (0x27)	1X HRPD personalization data file for slot 2
40 (0x28)	1X service provider personalization data file for slot 2
41 (0x29)	1X corporate personalization data file for slot 2
42 (0x2A)	1X RUIM personalization data file for slot 2
48 (0x30)	GW network personalization data file for slot 3
49 (0x31)	GW network subset personalization data file for slot 3
50 (0x32)	GW service provider personalization data file for slot 3
51 (0x33)	GW corporate personalization data file for slot 3
52 (0x34)	GW UIM personalization data file for slot 3
53 (0x35)	1X network type 1 personalization data file for slot 3
54 (0x36)	1X network type 2 personalization data file for slot 3
55 (0x37)	1X HRPD personalization data file for slot 3
56 (0x38)	1X service provider personalization data file for slot 3
57 (0x39)	1X corporate personalization data file for slot 3
58 (0x3A)	1X RUIM personalization data file for slot 3
64 (0x40)	GW network personalization data file for slot 4
65 (0x41)	GW network subset personalization data file for slot 4
66 (0x42)	GW service provider personalization data file for slot 4
67 (0x43)	GW corporate personalization data file for slot 4
68 (0x44)	GW UIM personalization data file for slot 4
69 (0x45)	1X network type 1 personalization data file for slot 4
70 (0x46)	1X network type 2 personalization data file for slot 4

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Table D-7 Valid simlock configuration file IDs (cont.)

File ID	Description
71 (0x47)	1X HRPD personalization data file for slot 4
72 (0x48)	1X service provider personalization data file for slot 4
73 (0x49)	1X corporate personalization data file for slot 4
74 (0x4A)	1X RUIM personalization data file for slot 4
80 (0x50)	GW network personalization data file for slot 5
81 (0x51)	GW network subset personalization data file for slot 5
82 (0x52)	GW service provider personalization data file for slot 5
83 (0x53)	GW corporate personalization data file for slot 5
84 (0x54)	GW UIM personalization data file for slot 5
85 (0x55)	1X network type 1 personalization data file for slot 5
86 (0x56)	1X network type 2 personalization data file for slot 5
87 (0x57)	1X HRPD personalization data file for slot 5
88 (0x58)	1X service provider personalization data file for slot 5
89 (0x59)	1X corporate personalization data file for slot 5
90 (0x5A)	1X RUIM personalization data file for slot 5

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D.5 QMI_UIM_READ_TRANSPARENT_RESP

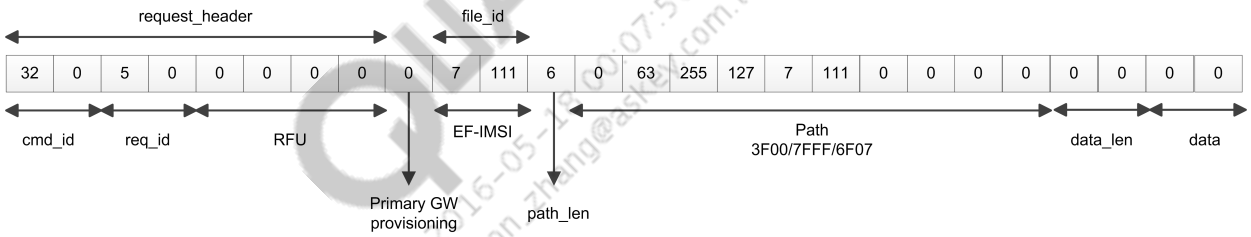
Table D-8 describes the result format for the read_result_ext field.

Table D-8 Read result response payload

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x0020) • 6 bytes – RFU (set to 0)
session_type	1	Session type. Values: 0 to 24. See Table A-1 for details.
file_id	2	File ID.
path_len	1	Valid path length.
path	10	Complete path; the size is fixed. Unused bytes are set to 0.
data_len	2	Data length as read from the card.
data	var	Data as read from the card.

Example – Read transparent response

Payload buffer in the response (before encryption) for a read transparent request:



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D.6 Encrypted Response Message Result

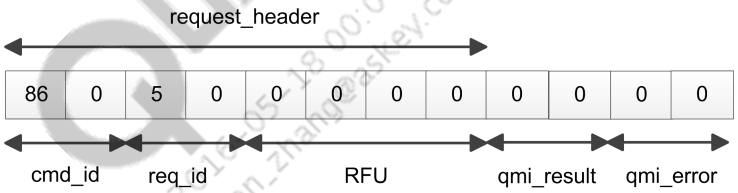
In addition to the mandatory Result TLV (see Section 2.3.1), an encrypted Result TLV is present in all the encrypted requests.

Table D-9 Decrypted response message result

Parameter	Size (byte)	Description
request_header	8	Request header: • 2 bytes – Command ID (0x00XX) • 2 bytes – Request ID (any number decided by the client) • 4 bytes – RFU (set to 0)
qmi_result	2	Result code: • QMI_RESULT_SUCCESS • QMI_RESULT_FAILURE
qmi_error	2	Error code – Described in the Error codes section of each message definition.

Example – Successful encrypted Result TLV

The Result TLV in the response (before encryption) for a successful operation:



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E References

E.1 Related Documents

Title	Number
Qualcomm Technologies	
QMI Client API Interface Specification	80-N1123-1
QMI Common Service Interface API Interface Specification	80-N1123-2
Qualcomm Messaging Interface (QMI) Architecture	80-VB816-1
Technical Memo: NAA Refresh High Level Guide	80-VM566-1
MMGSDI Session Enhancements Interface Specification and Operational Description	80-VT475-1
Standards	
Personalization of Mobile Equipment (ME)	3GPP TS 22.022
Characteristics of the Universal Subscriber Identity Module (USIM) Application	3GPP TS 31.102
IP Multimedia Services Identity Module (ISIM) application	3GPP TS 31.103
Generic Authentication Architecture (GAA); Generic Bootstrapping Architecture (GBA)	3GPP TS 33.220
Mobile Equipment (SIM - ME) interface	3GPP TS 51.011
SIM Access Profile	Bluetooth Ver 11r00
Removable User Identity Module for Spread Spectrum Systems	C.S0023-D
cdma2000® Application on UICC for Spread Spectrum Systems	C.S0065-0
Smart Cards: Card Application Toolkit (CAT) - Release 4	ETSI TS 102 223
Smart Cards: ETSI Numbering System for Telecommunication Application Providers	ETSI TS 101 220
Smart Cards: UICC-Terminal Interface; Physical and Logical Characteristics	ETSI TS 102 221
ISO 8601 - Time and date format	ISO 8601
Identification cards - Integrated circuit cards - Part 4: Organization, security and commands for interchange	ISO/IEC 7816-4

E.2 Acronyms and Terms

Acronym or term	Definition
ACL	access point name control list
AID	application identifier
AKA	authentication and key agreement
APDU	application protocol data unit
ARR	access rule reference

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Acronym or term	Definition
ATR	answer to reset
B-TID	Bootstrapping Transaction Identifier
BSF	Bootstrapping Server Function
CAVE	cellular authentication and voice encryption
CHAP	Challenge-Handshake Authentication Protocol
CSIM	cdma2000® SIM
DF	dedicated file
EF	elementary file
ESN	electronic serial number
FCI	file control information
FCN	file change notification
FCP	file control parameters
FDN	fixed dialing number
FQDN	fully qualified domain name
FMD	file management data
GBA	generic bootstrapping architecture
GID	group identifier
GW	GSM/WCDMA
HRPD	high rate packet data
ICC	integrated circuit card
IMPI	IMS Private User Identity
IMS	IP multimedia subsystem
IMSI	international mobile subscriber identity
ISIM	IMS SIM
Ks	key set
MBMS	multimedia broadcast/multicast services
MCC	mobile country code
ME	mobile equipment
MF	master file
MIP-RRQ	Mobile Interface Protocol registration request
MN-AAA	mobile network - authentication, authorization and accounting
MN-HA	mobile network home agent
MNC	mobile network code
MSK	MBMS service keys
MSIN	mobile station identification number
MUK	MBMS user key
NAF	network application function
PIN	personal identification number
PUK	PIN unlock key
QMI	Qualcomm messaging interface
RFU	reserved for future use
RUIM	removable user identity module
SAP	SIM access profile
SFS	secure file system
SIM	subscriber identify module
SW1/SW2	status word 1/status word 2
SWIO	single wire protocol input/output

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Acronym or term	Definition
SWP	single wire protocol
TLV	type-length-value
UE	user equipment
UICC	universal integrated circuit card
UIM	user identity module
UPIN	universal PIN
USIM	universal subscriber identity module
UTC	Coordinated Universal Time
VGCS	voice group call service
VSB	voice broadcast service

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